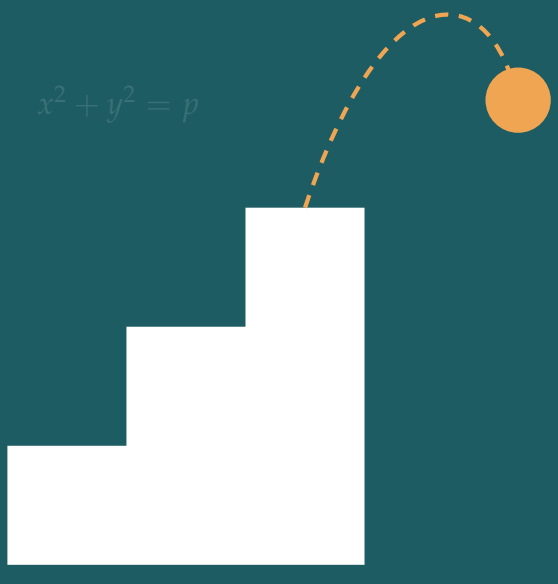
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2012 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2012 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2012 Worked Solutions

Worked solutions to all 39 questions of the 2012 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2012, Question 1

Pure • Optimisation with Calculus

STEP I 2012, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

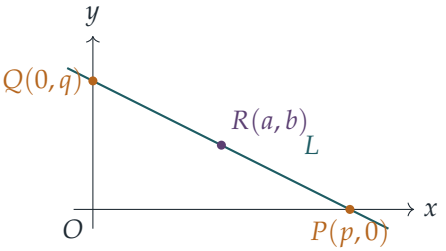
step.maths.org/questions/12-s1-q1

The key idea

Everything is controlled by one number, the gradient. Once L is forced through R, the intercepts p and q are simple functions of m, and both distances to be minimised become functions of m alone. The sum $p+q$ splits into a constant plus a term of the form $am + b/m$, which is a textbook job for AM–GM (or one derivative). For the length PQ it pays to notice that $q = mp$, which makes PQ^2 factorise nicely before we differentiate.

Solution

Finding p and q



Since $R(a,b)$ lies on $y = c - mx$, we have $b = c - ma$, so $c = b + ma$. The line meets the y-axis where $x = 0$, so $q = c$; it meets the x-axis where $y = 0$, so $p = c/m$ (allowed, since $m > 0$). Hence

$$p = a + \frac{b}{m}, \qquad q = b + ma.$$

Both are positive, as required, for every $m > 0$. Notice also that

$$q = b + ma = m\left(a + \frac{b}{m}\right) = mp,$$

which is just the statement that the gradient of L is $-q/p$. We shall use this below.

Minimising $OP + OQ$

Since $p, q > 0$, the distances of P and Q from O are p and q themselves, so as R stays fixed and m varies over $m > 0$ we must minimise

$$S(m) = p + q = a + b + \left(am + \frac{b}{m}\right).$$

The bracket is the sum of two positive numbers whose product, ab, does not depend on m. By the AM–GM inequality,

$$am + \frac{b}{m} \geq 2\sqrt{am \cdot \frac{b}{m}} = 2\sqrt{ab},$$

with equality exactly when $am = b/m$, that is, when $m = \sqrt{b/a}$ (which is positive, so this line is allowed). Therefore

$$S(m) \geq a + b + 2\sqrt{ab} = (a^{\frac{1}{2}} + b^{\frac{1}{2}})^2,$$

and the bound is attained. So the minimum value of the sum of the distances is $(a^{\frac{1}{2}} + b^{\frac{1}{2}})^2$, as required.

(If you prefer calculus: $S'(m) = a - b/m^2$, which vanishes only at $m = \sqrt{b/a}$, and substituting gives the same value.)

Minimising PQ

By Pythagoras, and using $q = mp$,

$$PQ^2 = p^2 + q^2 = p^2(1 + m^2) = \left(a + \frac{b}{m}\right)^2 (1 + m^2).$$

Minimising PQ is the same as minimising $f(m) = PQ^2$, since $PQ > 0$. Differentiating with the product and chain rules,

$$\begin{aligned} f'(m) &= 2\left(a + \frac{b}{m}\right)\left(-\frac{b}{m^2}\right)(1 + m^2) + 2m\left(a + \frac{b}{m}\right)^2 \\ &= 2\left(a + \frac{b}{m}\right)\left[-\frac{b}{m^2} - b + am + b\right] \\ &= 2\left(a + \frac{b}{m}\right)\left(am - \frac{b}{m^2}\right). \end{aligned}$$

Ask yourself Where does the $-b$ cancel?

Inside the square bracket, $-\frac{b}{m^2}(1 + m^2) = -\frac{b}{m^2} - b$, while $m(a + \frac{b}{m}) = am + b$; the two b's cancel. This cancellation is the reason for writing PQ^2 as $p^2(1 + m^2)$ rather than expanding $p^2 + q^2$ directly.

The first factor is positive, so $f'(m) = 0$ exactly when $am = b/m^2$, that is $m^3 = b/a$, giving the single stationary point

$$m = \left(\frac{b}{a}\right)^{\frac{1}{3}} = \frac{b^{\frac{1}{3}}}{a^{\frac{1}{3}}}.$$

(We are told we may assume it is a minimum; in fact $f' < 0$ for smaller m and $f' > 0$ for larger m, so it is the global minimum on $m > 0$.)

Now we substitute. With this m,

$$a + \frac{b}{m} = a + b \cdot \frac{a^{\frac{1}{3}}}{b^{\frac{1}{3}}} = a + a^{\frac{1}{3}} b^{\frac{2}{3}} = a^{\frac{1}{3}}(a^{\frac{2}{3}} + b^{\frac{2}{3}}), \qquad 1 + m^2 = 1 + \frac{b^{\frac{2}{3}}}{a^{\frac{2}{3}}} = \frac{a^{\frac{2}{3}} + b^{\frac{2}{3}}}{a^{\frac{2}{3}}}.$$

Hence

$$PQ^2 = a^{\frac{2}{3}}(a^{\frac{2}{3}} + b^{\frac{2}{3}})^2 \cdot \frac{a^{\frac{2}{3}} + b^{\frac{2}{3}}}{a^{\frac{2}{3}}} = (a^{\frac{2}{3}} + b^{\frac{2}{3}})^3,$$

and so the minimum distance between P and Q is

$$PQ_{\min} = (a^{\frac{2}{3}} + b^{\frac{2}{3}})^{\frac{3}{2}}.$$

This is the "similar form" asked for: the first answer has exponents $\frac{1}{2}$ inside and 2 outside, this one $\frac{2}{3}$ inside and $\frac{3}{2}$ outside.

Checks

- Symmetry: swapping a and b (reflecting in $y = x$) must leave both minima unchanged, and it does.
- A numerical case: take $a = 1$, $b = 8$. The second result predicts $m = 2$, giving $p = 5$, $q = 10$ and $PQ = \sqrt{125} = 5\sqrt{5} \approx 11.180$, and indeed $(1 + 4)^{3/2} = 5\sqrt{5}$. For the sum, $m = \sqrt{8}$ gives $p + q = 9 + 4\sqrt{2} \approx 14.657 = (1 + 2\sqrt{2})^2$.
- We checked both minima against a brute-force search over m for 2000 random pairs (a, b); they agreed to about 10^{-8}.

Common slips

- Expanding $p^2 + q^2$ in full and differentiating: correct but messy, and the factor $am - b/m^2$ is easy to miss. Spotting $q = mp$ first saves a lot of work.
- Forgetting that minimising PQ and PQ^2 are equivalent only because $PQ > 0$; and losing track of fractional powers when simplifying $a + b/m$.

Beyond STEP

The second result is the classic "ladder round a corner" problem in disguise: $(a^{2/3} + b^{2/3})^{3/2}$ is the longest ladder that can be carried horizontally round a corner between corridors of widths a and b. The family of segments PQ of fixed length envelopes an astroid, $x^{2/3} + y^{2/3} = \ell^{2/3}$, and R lies on the envelope exactly when ℓ is this minimum. Envelopes and constrained optimisation (later, Lagrange multipliers) appear in first-year calculus courses; Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

WORKED SOLUTION

STEP I 2012, Question 2

Pure • Curve Sketching and Roots

STEP I 2012, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q2

The key idea

Counting the real roots of $p(x) = 0$ is the same as counting where the graph of $y = p(x)$ meets the x-axis, or, after moving the constant across, where a fixed curve meets a horizontal line. So the whole question is about knowing the shape of a quartic: where it turns, and how high or low it is when it does. In part (i) the quartic is a perfect square; in part (ii) the condition on the derivatives makes $\frac{dy}{dx}$ factorise with a repeated root, which leaves only one turning point.

Solution

Part (i)

Notice first that

$$y = x^4 - 6x^2 + 9 = (x^2 - 3)^2,$$

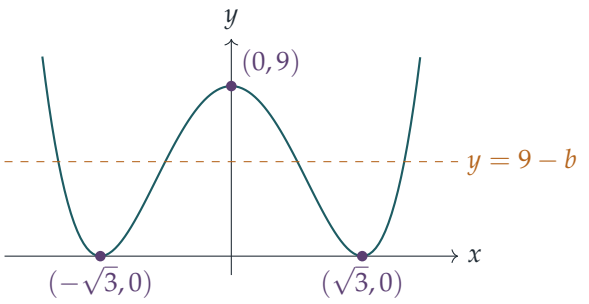
so $y \geq 0$ everywhere, with $y = 0$ exactly at $x = \pm\sqrt{3}$. The curve is even (symmetric in the y-axis), and it crosses the y-axis at $(0, 9)$. Differentiating,

$$\frac{dy}{dx} = 4x^3 - 12x = 4x(x^2 - 3),$$

which vanishes at $x = 0$ and $x = \pm\sqrt{3}$. The stationary points are therefore

$$(0,\, 9) \text{ (a local maximum)}, \qquad (\sqrt{3},\, 0) \text{ and } (-\sqrt{3},\, 0) \text{ (minima)}.$$

The minima are clear because $y \geq 0$ and $y = 0$ there; and $\frac{d^2y}{dx^2} = 12x^2 - 12$ is $-12 < 0$ at $x = 0$, so that point is a maximum. As $x \to \pm\infty$, $y \to +\infty$. The curve is a "W" that touches the x-axis at $x = \pm\sqrt{3}$.



For the root count, add 9 to both sides:

$$x^4 - 6x^2 + b = 0 \iff (x^2 - 3)^2 = 9 - b.$$

So n is the number of times the horizontal line $y = 9 - b$ meets the curve we have just sketched. Reading off the picture as the line moves from high to low:

- $9 - b > 9$, i.e. $b < 0$: the line is above the maximum and meets only the two outer arms, so $n = 2$.
- $9 - b = 9$, i.e. $b = 0$: the line passes through the maximum, giving $x = 0$ and two outer points; indeed $x^2(x^2 - 6) = 0$ gives $x = 0, \pm\sqrt{6}$. So $n = 3$.
- $0 < 9 - b < 9$, i.e. $0 < b < 9$: the line cuts each of the four monotonic pieces once, so $n = 4$.
- $9 - b = 0$, i.e. $b = 9$: the line touches at the two minima only, $x = \pm\sqrt{3}$, so $n = 2$.
- $9 - b < 0$, i.e. $b > 9$: the line lies below the curve, so $n = 0$.

Collecting these:

$$\text{(a) } n = 0 \iff b > 9; \quad \text{(b) } n = 1 \text{ for no } b; \quad \text{(c) } n = 2 \iff b < 0 \text{ or } b = 9;$$

$$\text{(d) } n = 3 \iff b = 0; \quad \text{(e) } n = 4 \iff 0 < b < 9.$$

That $n = 1$ is impossible also follows from symmetry: if x is a root then so is $-x$, so a single root would have to be $x = 0$, which forces $b = 0$, and then there are three roots.

Part (ii)

Now $y = x^4 - 6x^2 + ax + b$, so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3 - 12x + a, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2 - 12.$$

The second derivative vanishes only at $x = \pm 1$. At $x = 1$ the first derivative is $a - 8$, and at $x = -1$ it is $a + 8$. So both vanish at the same point exactly when

$$a = 8 \text{ (at } x = 1) \qquad \text{or} \qquad a = -8 \text{ (at } x = -1).$$

Case $a = 8$. Here

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 4(x^3 - 3x + 2) = 4(x-1)^2(x+2),$$

where we found the factor $(x-1)^2$ because we built $x = 1$ to be a root of both y' and y'', i.e. a repeated root of y'.

> **Ask yourself** How many turning points does this curve actually have?

Since $(x-1)^2 \geq 0$, the sign of $\frac{\mathrm{d}y}{\mathrm{d}x}$ is the sign of $x+2$. So y is strictly decreasing for $x < -2$ and strictly increasing for $x > -2$ (the gradient is zero at $x = 1$ but does not change sign: that is a stationary point of inflection, not a turning point). There is just one turning point, a global minimum at $x = -2$, where

$$y(-2) = 16 - 24 - 16 + b = b - 24.$$

A curve that falls to a single minimum and then rises meets the x-axis twice if the minimum is below the axis, once (touching) if it is on the axis, and not at all if it is above. Hence the number of distinct real roots is

$$\begin{cases} 2 & \text{if } b < 24, \\ 1 & \text{if } b = 24, \\ 0 & \text{if } b > 24. \end{cases}$$

(For $b = 24$ we can see the double root directly: $x^4 - 6x^2 + 8x + 24 = (x+2)^2(x^2 - 4x + 6)$, and the quadratic has discriminant $16 - 24 < 0$.)

Case $a = -8$. Replacing x by $-x$ turns $x^4 - 6x^2 - 8x + b$ into $x^4 - 6x^2 + 8x + b$, so the curve is the reflection of the previous one in the y-axis. The single minimum is now at $x = 2$, with value $b - 24$, and the number of roots is the same: 2 if $b < 24$, 1 if $b = 24$, 0 if $b > 24$.

Part (iii)

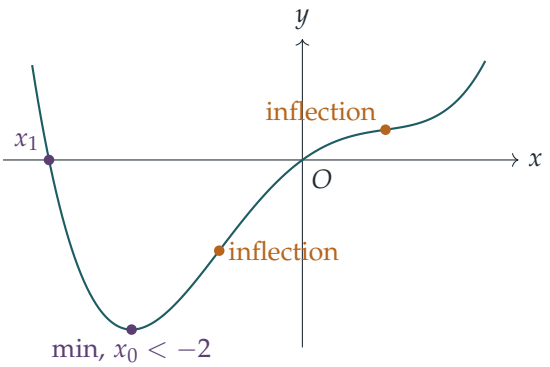
Take $y = x^4 - 6x^2 + ax$ with $a > 8$.

Stationary points. $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ means $x^3 - 3x = -\frac{a}{4}$. The cubic $g(x) = x^3 - 3x$ has a local maximum $g(-1) = 2$ and a local minimum $g(1) = -2$, and $g(-2) = -2$. Since $-\frac{a}{4} < -2$, the horizontal line at height $-\frac{a}{4}$ lies below the local minimum of g, so it meets g exactly once, on the left-hand branch where $x < -2$. So there is exactly one stationary point, at some $x_0 < -2$. The gradient is negative to its left and positive to its right, so it is the minimum.

Crossings of the x-axis. $y = x(x^3 - 6x + a)$, so $x = 0$ is one crossing, and the others are roots of $h(x) = x^3 - 6x + a$. This cubic has its local minimum at $x = \sqrt{2}$, where $h(\sqrt{2}) = a - 4\sqrt{2} > 0$ because $a > 8 > 4\sqrt{2}$. So h has only one real root, and it is negative because $h(0) = a > 0$. In fact $h(-2) = a + 4 > 0$, and h is increasing for $x < -\sqrt{2}$, so this root x_1 satisfies $x_1 < -2$. The curve crosses the axis at x_1 and at the origin, and is negative between them; the minimum lies in this interval.

Other features. At the origin the gradient is $a > 0$. The second derivative $12(x^2 - 1)$ changes sign at $x = \pm 1$, so there are points of inflection at $(-1, -a - 5)$ and $(1, a - 5)$; the gradient there is $a + 8$ and $a - 8$, both positive, so the curve is still rising through them. It is concave down for $-1 < x < 1$ and concave up outside. As $x \to \pm\infty$, $y \to +\infty$.

The sketch shows the case $a = 10$, where $x_0 \approx -2.05$, the minimum value is about -28, and $x_1 \approx -3.05$.



Checks

- In part (i), $b = 5$ should give four roots: $x^2 = 3 \pm 2$, so $x = \pm 1, \pm\sqrt{5}$. It does.
- In part (ii), $b = 24$ gives the factorisation $(x + 2)^2(x^2 - 4x + 6)$, confirming the single (double) root at the minimum.
- Numerically, we found all complex roots of the quartics for many values of b either side of 0, 9 and 24, and counted the distinct real ones; every count matched. For random a between 8 and 200, the curve in (iii) always had one stationary point and two axis crossings, with $x_1 < x_0 < -2$. For $a = 10$, $x_0 \approx -2.054$ and $y(x_0) \approx -28.05$.

Common slips

- Counting the double roots at $x = \pm\sqrt{3}$ (when $b = 9$) as four roots: the question asks for *distinct* values of x.
- Treating the point where $y' = y'' = 0$ in part (ii) as a turning point. Because y' has a squared factor there, its sign does not change, so it is a stationary point of inflection.
- Forgetting $a = -8$: $y'' = 0$ has two solutions, and each gives a value of a.

Beyond STEP

The way the number of real roots jumps as b passes through 0, 9 or 24 is a small example of a *bifurcation*: the answer changes only when a horizontal line passes through a stationary value. The special case $y' = y'' = 0$ in part (ii) is where two stationary points merge and vanish, which is the simplest "catastrophe" (the fold) and the reason the quartic in (iii) has only one turning point once $a > 8$. The same reasoning, made rigorous by the intermediate value theorem and monotonicity, is the start of real analysis; Abbott's *Understanding Analysis* is a good place to see it done carefully.

WORKED SOLUTION

STEP I 2012, Question 3

Pure • Inequalities from Areas

STEP I 2012, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q3

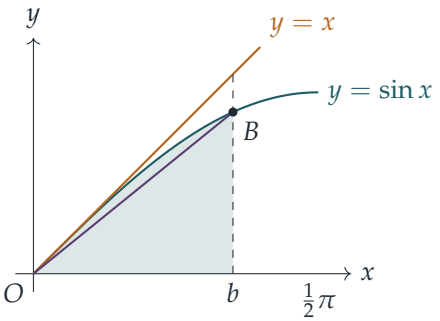
The key idea

A curve that bends one way all the time lies on one side of each of its tangents and on the other side of each of its chords. So the area under the curve is trapped between the area under a tangent and the area under a chord, and both of those are just triangles or trapezia. Write each area down exactly and the inequalities fall out; the only real work is deciding which tangent and which chord to use.

Solution

Part (i)

On $0 < x < \frac{1}{2}\pi$ we have $\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}\sin x = -\sin x < 0$, so the curve $y = \sin x$ is concave there: it bends downwards. The tangent at the origin has gradient $\cos 0 = 1$, so it is the line $y = x$. The chord from O to $B = (b, \sin b)$ is the line $y = \dfrac{\sin b}{b}\,x$.



Concavity means that between O and B the curve lies strictly *above* the chord, and for

$x > 0$ it lies strictly *below* the tangent at O. So, for $0 < x < b$,

$$\frac{\sin b}{b}\, x \;<\; \sin x \;<\; x.$$

Now compare the three areas over $0 \le x \le b$.

- Under the curve (shaded): $\displaystyle\int_0^b \sin x \,\mathrm{d}x = 1 - \cos b$.
- Under the tangent: a right-angled triangle with base b and height b, area $\frac{1}{2}b^2$.
- Under the chord: a triangle with base b and height $\sin b$, area $\frac{1}{2}b\sin b$.

Because the curve is strictly between the two lines on $0 < x < b$ (and all three are continuous), the areas satisfy strict inequalities:

$$\tfrac{1}{2}b\sin b \;<\; 1-\cos b \;<\; \tfrac{1}{2}b^2.$$

The right-hand inequality rearranges to $\cos b > 1 - \frac{1}{2}b^2$ and the left-hand one to $\cos b < 1 - \frac{1}{2}b\sin b$. Together,

$$1 - \tfrac{1}{2}b^2 < \cos b < 1 - \tfrac{1}{2}b\sin b,$$

as required.

Part (ii)

Write $k = \ln a$; since $a > 1$, $k > 0$, and $a^x = \mathrm{e}^{kx}$. Then

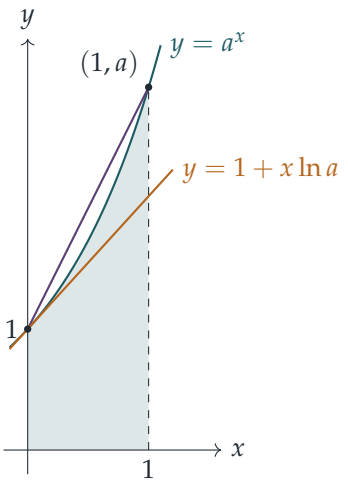
$$\frac{\mathrm{d}}{\mathrm{d}x}a^x = k\,\mathrm{e}^{kx}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}x^2}a^x = k^2\mathrm{e}^{kx} > 0,$$

so this curve is convex: it bends upwards. Now the roles swap: the curve lies strictly *below* its chords and strictly *above* its tangents (away from the point of contact).

? **Ask yourself** Over which interval should we compare areas?

The answer is to involve $\ln a$ and a, but nothing else. The interval $0 \le x \le 1$ does exactly that: the endpoints of the curve are $(0,1)$ and $(1,a)$, and the area under the curve is

$$\int_0^1 \mathrm{e}^{kx}\,\mathrm{d}x = \frac{\mathrm{e}^k - 1}{k} = \frac{a-1}{\ln a}.$$



Lower bound (chord). The chord from $(0,1)$ to $(1,a)$ lies above the curve on $0<x<1$, and the region under it is a trapezium of width 1 with parallel sides 1 and a, of area $\frac{1}{2}(1+a)$. Hence

$$\frac{a-1}{\ln a} < \frac{a+1}{2} \quad\Longrightarrow\quad \ln a > \frac{2(a-1)}{a+1},$$

where we multiplied through by the positive quantity $2\ln a/(a+1)$, so the inequality keeps its direction.

Upper bound (tangent). The tangent at $(0,1)$ has gradient $k=\ln a$, so it is $y=1+kx$. It lies below the curve for $x>0$, and the area under it on $[0,1]$ is a trapezium with parallel sides 1 and $1+k$, of area $1+\frac{1}{2}k$. Hence

$$1+\tfrac{1}{2}k < \frac{a-1}{k}.$$

Multiplying by $2k>0$:

$$2k+k^2 < 2a-2 \quad\Longleftrightarrow\quad k^2+2k+1 < 2a-1 \quad\Longleftrightarrow\quad (k+1)^2 < 2a-1.$$

Ask yourself Can we just take square roots?

Yes, but only because both sides are positive: $2a-1>1>0$, and $k+1>0$ since $k>0$. For positive numbers, $u^2<v^2$ gives $u<v$, so $k+1<\sqrt{2a-1}$, that is,

$$\frac{2(a-1)}{a+1} < \ln a < -1+\sqrt{2a-1},$$

which completes the proof.

Checks

- Part (i) with $b=1$: $1-\frac{1}{2}=0.5$, $\cos 1\approx 0.5403$, and $1-\frac{1}{2}\sin 1\approx 0.5793$. In order, as claimed.

- Part (ii) with $a = 2$: $\frac{2}{3} \approx 0.6667$, $\ln 2 \approx 0.6931$, and $\sqrt{3} - 1 \approx 0.7321$. Again in order.
- As $a \to 1$ all three expressions in part (ii) tend to 0, and they squeeze together: at $a = 1.01$ they agree to five decimal places. That makes sense, because over a short range a smooth curve is almost straight.
- We checked both chains of inequalities at thousands of random values, and the area formulas by Simpson's rule.

Common slips

- Getting the sides the wrong way round: the sine curve is concave (below its tangent, above its chord), while a^x is convex (the opposite). Justify this with the second derivative rather than the sketch.
- Forgetting that $\ln a > 0$ when multiplying or dividing an inequality by it.
- Square-rooting $(k+1)^2 < 2a - 1$ without checking that $k + 1 > 0$.

Beyond STEP

The chord bounds are a case of a general principle: for a convex function, the average value over an interval lies between the value at the midpoint and the average of the endpoint values (the Hermite–Hadamard inequality; using the tangent at the midpoint instead of at an end gives an even sharper lower bound). The lower bound in part (ii) says that the logarithmic mean $(a-1)/\ln a$ of a and 1 is less than their arithmetic mean, a fact that turns up throughout analysis and in physics. Bartle and Sherbert's *Introduction to Real Analysis* treats convex functions and integral estimates carefully.

WORKED SOLUTION

STEP I 2012, Question 4

Pure • Tangents and Normals

STEP I 2012, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q4

The key idea

Everything follows from the gradient of $y = \frac{1}{2x}$ and some careful elimination. When we intersect two lines that differ only by swapping p and q, subtracting the equations always produces a factor $p - q$, which we may cancel because the points are distinct. In the special case, it pays to notice what the condition does to Q: it makes Q the mirror image of P in the line $y = x$, and that symmetry explains why T and N land on that line.

Solution

The point T

On C we have $y = \dfrac{1}{2x}$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{1}{2x^2}$. The tangent at P therefore has gradient $-\frac{1}{2p^2}$ and equation

$$y - \frac{1}{2p} = -\frac{1}{2p^2}(x - p).$$

Multiplying by $2p^2$ gives $2p^2y - p = -x + p$, that is,

$$x + 2p^2y = 2p. \tag{1}$$

In the same way the tangent at Q is

$$x + 2q^2y = 2q. \tag{2}$$

Subtracting (2) from (1),

$$2(p^2 - q^2)\,y = 2(p - q).$$

The points are distinct, so $p \neq q$, and we may divide by $2(p - q)$ to get $(p + q)y = 1$. Since $p, q > 0$, $p + q \neq 0$, so

$$y = \frac{1}{p + q}.$$

Substituting into (1),

$$x = 2p - \frac{2p^2}{p+q} = \frac{2p(p+q) - 2p^2}{p+q} = \frac{2pq}{p+q}.$$

So T is the point $\left(\frac{2pq}{p+q}, \frac{1}{p+q}\right)$, as required.

The point N when $pq = \frac{1}{2}$

The normal at P is perpendicular to the tangent, so its gradient is $2p^2$, and its equation is

$$y = 2p^2x - 2p^3 + \frac{1}{2p}. \tag{3}$$

Likewise the normal at Q is $y = 2q^2x - 2q^3 + \frac{1}{2q}$. Equating the two expressions for y:

$$2(p^2 - q^2)\,x = 2(p^3 - q^3) - \left(\frac{1}{2q} - \frac{1}{2p}\right) = 2(p^3 - q^3) + \frac{p-q}{2pq}.$$

Dividing by $2(p-q) \neq 0$,

$$(p+q)\,x = p^2 + pq + q^2 + \frac{1}{4pq}.$$

This holds for any distinct p and q. Now we use $pq = \frac{1}{2}$, so $\frac{1}{4pq} = \frac{1}{2} = pq$, and the right-hand side becomes

$$p^2 + 2pq + q^2 = (p+q)^2.$$

Hence $x = p + q$. Putting this into (3),

$$y = 2p^2(p+q) - 2p^3 + \frac{1}{2p} = 2p^2q + \frac{1}{2p} = p\,(2pq) + q = p + q,$$

where we used $2pq = 1$ twice: $2p^2q = p(2pq) = p$, and $\frac{1}{2p} = q$. Therefore

$$N = (p+q,\ p+q).$$

(Note that $p \neq q$ together with $pq = \frac{1}{2}$ means $p \neq \frac{1}{\sqrt{2}}$, which is the only restriction on p.)

Ask yourself Why should T and N lie on $y = x$ at all?

When $pq = \frac{1}{2}$ we have $\frac{1}{2q} = p$ and $\frac{1}{2p} = q$, so $P = (p, q)$ and $Q = (q, p)$. The curve $xy = \frac{1}{2}$ is symmetric under reflection in $y = x$, and this reflection swaps P and Q. It therefore swaps the two tangents and swaps the two normals, so each intersection point is mapped to itself, and a point fixed by this reflection lies on $y = x$. The algebra confirms it.

T and N on $y = x$, and the product of distances

With $2pq = 1$, the result for T gives

$$T = \left(\frac{1}{p+q}, \frac{1}{p+q} \right),$$

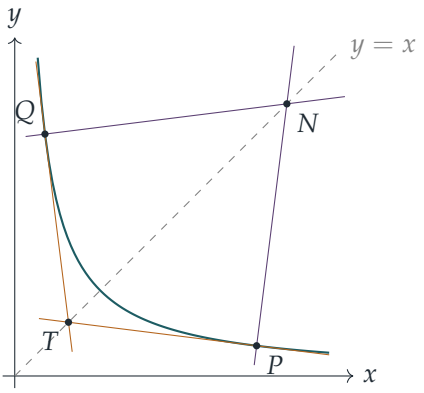
which lies on $y = x$, and we found $N = (p+q,\ p+q)$, which also lies on $y = x$.
A point (a, a) with $a > 0$ is at distance $\sqrt{2}\,a$ from the origin. Since $p + q > 0$,

$$OT = \frac{\sqrt{2}}{p+q}, \qquad ON = \sqrt{2}\,(p+q),$$

so

$$OT \cdot ON = 2,$$

which is constant (independent of p), as required.



The figure shows the case $p = 2$, $q = \frac{1}{4}$: tangents in amber, normals in plum.

Checks

Take $p = 2$, $q = \frac{1}{4}$, so $pq = \frac{1}{2}$. The tangents are $x + 8y = 4$ and $8x + y = 4$, which meet at $\left(\frac{4}{9}, \frac{4}{9}\right)$, matching $\frac{1}{p+q} = \frac{4}{9}$. The normals are $y = 8x - \frac{63}{4}$ and $y = \frac{x}{8} + \frac{63}{32}$, which meet at $\left(\frac{9}{4}, \frac{9}{4}\right)$, matching $p + q = \frac{9}{4}$. Then $OT \cdot ON = \frac{4\sqrt{2}}{9} \cdot \frac{9\sqrt{2}}{4} = 2$. We also confirmed the general formula for T, and both results in the special case, with exact rational arithmetic for 2000 random choices of p and q.

Common slips

- Dividing by $p - q$ without saying why it's allowed: this is exactly where "distinct points" is used.

- Sign errors in $\frac{1}{2q} - \frac{1}{2p} = \frac{p-q}{2pq}$ when finding N.
- Forgetting to use $pq = \frac{1}{2}$ to simplify T before claiming it lies on $y = x$.

Beyond STEP

The rectangular hyperbola $xy = c^2$ is a conic, and the fact that $OT \cdot ON$ is constant says that T and N are *inverse points* with respect to the circle $x^2 + y^2 = 2$. Inversion in a circle is a powerful transformation in geometry and in complex analysis, where it appears as the map $z \mapsto 1/\bar{z}$. Needham's *Visual Complex Analysis* has a beautiful account of inversion and its geometric properties.

WORKED SOLUTION

STEP I 2012, Question 5

Pure • Integration by Substitution

STEP I 2012, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q5

The key idea

Both of the first two integrals give way to standard tools. Writing $\sin 2x = 2\sin x \cos x$ shows that $\sin x$ is (minus) the derivative of $\cos x$, so the substitution $u = \cos x$ works. For the second, the logarithm is the awkward factor, so we integrate by parts to differentiate it away. For the final integral, $\cos x + \sin x$ is a cosine in disguise: $\sqrt{2}\cos(x - \frac{1}{4}\pi)$. Shifting the variable by $\frac{1}{4}\pi$ turns the range into $[0, \frac{1}{4}\pi]$ and the integrand into a combination of the two integrals we have just found.

Solution

The first integral

Write I_1 for the first integral. Since $\sin 2x = 2\sin x \cos x$, substitute $u = \cos x$, so $\mathrm{d}u = -\sin x \,\mathrm{d}x$. When $x = 0$, $u = 1$, and when $x = \frac{1}{4}\pi$, $u = \frac{1}{\sqrt{2}}$. So

$$I_1 = \int_0^{\frac{1}{4}\pi} 2\cos x \ln(\cos x)\, \sin x \,\mathrm{d}x = \int_{1/\sqrt{2}}^{1} 2u \ln u \,\mathrm{d}u,$$

where the minus sign from $\mathrm{d}u$ has been used to swap the limits. Integrating by parts (differentiate $\ln u$, integrate $2u$),

$$\int 2u \ln u \,\mathrm{d}u = u^2 \ln u - \int u^2 \cdot \frac{1}{u} \,\mathrm{d}u = u^2 \ln u - \tfrac{1}{2}u^2.$$

At $u = 1$ this is $-\frac{1}{2}$. At $u = \frac{1}{\sqrt{2}}$ we have $u^2 = \frac{1}{2}$ and $\ln u = -\frac{1}{2}\ln 2$, so it is $-\frac{1}{4}\ln 2 - \frac{1}{4}$. Hence

$$I_1 = -\tfrac{1}{2} - \left(-\tfrac{1}{4}\ln 2 - \tfrac{1}{4}\right) = \tfrac{1}{4}\ln 2 - \tfrac{1}{4} = \tfrac{1}{4}(\ln 2 - 1),$$

as required.

The second integral

Write I_2 for the second integral. We integrate by parts, differentiating $\ln(\cos x)$ and integrating $\cos 2x$. A convenient antiderivative of $\cos 2x$ is $\frac{1}{2}\sin 2x = \sin x \cos x$, and the derivative of $\ln(\cos x)$ is $-\sin x / \cos x$. So

$$\begin{aligned} I_2 &= \Big[\sin x \cos x \, \ln(\cos x)\Big]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} \sin x \cos x \cdot \left(-\frac{\sin x}{\cos x}\right) \mathrm{d}x \\ &= \tfrac{1}{2}\ln\left(\tfrac{1}{\sqrt{2}}\right) - 0 + \int_0^{\frac{1}{4}\pi} \sin^2 x \, \mathrm{d}x. \end{aligned}$$

Ask yourself Why choose $\sin x \cos x$ rather than just writing $\frac{1}{2}\sin 2x$?

They are the same function, but in the form $\sin x \cos x$ the factor $\cos x$ cancels with the $\cos x$ in the denominator of the derivative of $\ln(\cos x)$, leaving the easy integral of $\sin^2 x$.

The boundary term is $\frac{1}{2} \cdot (-\frac{1}{2}\ln 2) = -\frac{1}{4}\ln 2$. Using $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$,

$$\int_0^{\frac{1}{4}\pi} \sin^2 x \, \mathrm{d}x = \Big[\tfrac{1}{2}x - \tfrac{1}{4}\sin 2x\Big]_0^{\frac{1}{4}\pi} = \tfrac{\pi}{8} - \tfrac{1}{4}.$$

So

$$I_2 = -\tfrac{1}{4}\ln 2 + \tfrac{\pi}{8} - \tfrac{1}{4} = \tfrac{1}{8}\left(\pi - 2\ln 2 - 2\right) = \tfrac{1}{8}(\pi - \ln 4 - 2),$$

as required.

The final integral

Call the final integral I. By the compound-angle formula,

$$\cos x + \sin x = \sqrt{2}\left(\cos x \cos \tfrac{1}{4}\pi + \sin x \sin \tfrac{1}{4}\pi\right) = \sqrt{2}\cos\left(x - \tfrac{1}{4}\pi\right).$$

This suggests the substitution $t = x - \frac{1}{4}\pi$, which sends the range $[\frac{1}{4}\pi, \frac{1}{2}\pi]$ to $[0, \frac{1}{4}\pi]$, the range of I_1 and I_2. With $2x = 2t + \frac{1}{2}\pi$,

$$\cos 2x = \cos\left(2t + \tfrac{1}{2}\pi\right) = -\sin 2t, \qquad \sin 2x = \sin\left(2t + \tfrac{1}{2}\pi\right) = \cos 2t,$$

and

$$\ln(\cos x + \sin x) = \ln\left(\sqrt{2}\cos t\right) = \tfrac{1}{2}\ln 2 + \ln(\cos t),$$

which is valid because $\cos t > 0$ on $[0, \frac{1}{4}\pi]$. Therefore

$$\begin{aligned} I &= \int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\left(\tfrac{1}{2}\ln 2 + \ln(\cos t)\right) \mathrm{d}t \\ &= \tfrac{1}{2}\ln 2 \int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\, \mathrm{d}t + I_2 - I_1. \end{aligned}$$

The remaining integral vanishes:

$$\int_0^{\frac{1}{4}\pi} (\cos 2t - \sin 2t)\,\mathrm{d}t = \left[\tfrac{1}{2}\sin 2t + \tfrac{1}{2}\cos 2t\right]_0^{\frac{1}{4}\pi} = \tfrac{1}{2} - \tfrac{1}{2} = 0.$$

So, using the two results above,

$$I = I_2 - I_1 = \tfrac{1}{8}(\pi - 2\ln 2 - 2) - \tfrac{1}{8}(2\ln 2 - 2) = \frac{\pi - 4\ln 2}{8} = \frac{\pi}{8} - \frac{1}{2}\ln 2.$$

Checks

Simpson's rule gives $I_1 \approx -0.076713$, $I_2 \approx -0.030588$ and $I \approx 0.046125$, matching $\frac{1}{4}(\ln 2 - 1)$, $\frac{1}{8}(\pi - \ln 4 - 2)$ and $\frac{1}{8}(\pi - 4\ln 2)$ to ten decimal places. A quick sign check on the first integral: on $(0, \frac{1}{4}\pi)$ we have $\sin 2x > 0$ and $\ln(\cos x) < 0$, so I_1 must be negative, and indeed $\ln 2 - 1 < 0$.

Common slips

- Forgetting the boundary term in the integration by parts for I_2. It is not zero at $x = \frac{1}{4}\pi$.
- Getting the signs of $\cos(2t + \frac{1}{2}\pi)$ and $\sin(2t + \frac{1}{2}\pi)$ the wrong way round, which gives $I_1 - I_2$ or $I_1 + I_2$ instead of $I_2 - I_1$.
- Dropping the constant $\frac{1}{2}\ln 2$ from $\ln\sqrt{2}$ without checking that its coefficient integrates to zero.

Beyond STEP

Integrals of $\ln(\cos x)$ and $\ln(\sin x)$ are classics: the same symmetry tricks give the famous result $\int_0^{\pi/2} \ln(\sin x)\,\mathrm{d}x = -\frac{1}{2}\pi\ln 2$, and over a general range they lead to the Clausen function and Catalan's constant. In a first-year analysis course you'd also check that such integrals are well defined when the logarithm blows up at an endpoint, which makes them improper integrals. For lots more practice in this style, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2012, Question 6

Pure • Trigonometric Inequalities

STEP I 2012, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q6

The key idea

Every angle of elevation here has the same opposite side, the flagpole, so each cotangent is just a horizontal distance divided by the height. The first result then becomes a fact about lengths in the circle: $DC^2 = AD \cdot DB$, because the angle in a semicircle is a right angle. The trigonometric identity turns $\cot p \cot q - \cot^2 \frac{1}{2}(p+q)$ into something whose sign we can read off. Once we have that inequality, the geometry hands us the final result almost for free.

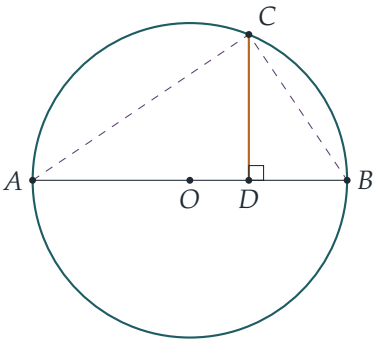
Solution

The geometric result

Let the flagpole have height $h > 0$ and top T, so DT is vertical. Each of A, B, C is on the ground, so the triangles TDA, TDB, TDC have right angles at D. Hence

$$\cot\alpha = \frac{AD}{h}, \qquad \cot\beta = \frac{DB}{h}, \qquad \cot\phi = \frac{DC}{h}.$$

Since α and β are acute, D is neither A nor B (from either of those points the elevation would be a right angle), so D lies strictly between A and B and $DC > 0$.



Now look at the ground plan. Let O be the centre, r the radius and $x = OD$ (measured towards B, so $-r < x < r$). Then $AD = r + x$ and $DB = r - x$, and by Pythagoras in triangle ODC, $DC^2 = OC^2 - OD^2 = r^2 - x^2$. Therefore

$$AD \cdot DB = (r + x)(r - x) = r^2 - x^2 = DC^2.$$

(Equivalently: $\angle ACB$ is a right angle, being the angle in a semicircle, and CD is the altitude of the right-angled triangle ACB.) Dividing by h^2,

$$\cot\alpha \cot\beta = \frac{AD \cdot DB}{h^2} = \frac{DC^2}{h^2} = \cot^2\phi,$$

as required.

The identity

Write $s = \frac{1}{2}(p + q)$ and use the double-angle forms $\sin^2 s = \frac{1}{2}\big(1 - \cos(p + q)\big)$ and $\cos^2 s = \frac{1}{2}\big(1 + \cos(p + q)\big)$. The left-hand side becomes

$$\begin{aligned}
&\tfrac{1}{2}\cos p \cos q\big(1 - \cos(p + q)\big) - \tfrac{1}{2}\sin p \sin q\big(1 + \cos(p + q)\big) \\
&\quad = \tfrac{1}{2}\big(\cos p \cos q - \sin p \sin q\big) - \tfrac{1}{2}\cos(p + q)\big(\cos p \cos q + \sin p \sin q\big) \\
&\quad = \tfrac{1}{2}\cos(p + q) - \tfrac{1}{2}\cos(p + q)\cos(p - q),
\end{aligned}$$

using the compound-angle formulae for $\cos(p + q)$ and $\cos(p - q)$. This holds for all p and q.

The inequality

Factorise the right-hand side of the identity:

$$\cos p \cos q \sin^2 s - \sin p \sin q \cos^2 s = \tfrac{1}{2}\cos(p + q)\big(1 - \cos(p - q)\big).$$

Now suppose $p, q > 0$ and $p + q \le \frac{1}{2}\pi$. Then $0 < p + q \le \frac{1}{2}\pi$, so $\cos(p + q) \ge 0$; and $1 - \cos(p - q) \ge 0$ always. So the right-hand side is non-negative, and

$$\cos p \cos q \sin^2 s \;\ge\; \sin p \sin q \cos^2 s.$$

Ask yourself Can we safely divide through by $\sin p \sin q \sin^2 s$?

Yes: since p and q are positive with sum at most $\frac{1}{2}\pi$, each of p, q and s lies in $(0, \frac{1}{2}\pi)$, so $\sin p$, $\sin q$ and $\sin s$ are all strictly positive. Dividing by the positive quantity $\sin p \sin q \sin^2 s$ keeps the direction of the inequality:

$$\cot p \cot q \;\ge\; \cot^2 s = \cot^2 \tfrac{1}{2}(p + q).$$

Equality holds exactly when $p = q$ or $p + q = \frac{1}{2}\pi$.

The final result

Take $p = \alpha$, $q = \beta$. Both are positive (they are angles of elevation of a pole of positive height) and $\alpha + \beta \leq \frac{1}{2}\pi$, so the inequality applies. Combined with the first result,

$$\cot^2\phi = \cot\alpha\cot\beta \;\geq\; \cot^2\tfrac{1}{2}(\alpha+\beta).$$

Both ϕ and $\frac{1}{2}(\alpha+\beta)$ lie in $(0, \frac{1}{2}\pi)$ (ϕ because $DC > 0$ and $h > 0$), so both cotangents are positive and we may take square roots: $\cot\phi \geq \cot\frac{1}{2}(\alpha+\beta)$.

? **Ask yourself** Why does this give $\phi \leq \frac{1}{2}(\alpha+\beta)$ and not $\geq$?

Because cot is strictly *decreasing* on $(0, \frac{1}{2}\pi)$: a larger cotangent means a smaller angle. Hence

$$\phi \leq \tfrac{1}{2}(\alpha+\beta) \qquad \text{whenever } \alpha+\beta \leq \tfrac{1}{2}\pi.$$

Checks

- If $D = O$ then $AD = DB = DC = r$, so $\alpha = \beta = \phi$: both sides of the first result are equal, and the final inequality is an equality, matching the case $p = q$.
- A numerical test: $r = 1$, $OD = \frac{1}{2}$, $h = 1$ gives $\cot\alpha = 1.5$, $\cot\beta = 0.5$, $\cot^2\phi = 0.75$, as predicted. Here $\alpha+\beta \approx 0.588 + 1.107 = 1.695 > \frac{1}{2}\pi$, so the final result need not apply; and indeed $\phi \approx 0.857 > \frac{1}{2}(\alpha+\beta) \approx 0.848$, showing the condition matters.
- We checked the identity at 10^5 random (p, q), the inequality at 2×10^5 random pairs in the allowed range, and $\phi \leq \frac{1}{2}(\alpha+\beta)$ in about 8×10^4 random configurations with $\alpha+\beta \leq \frac{1}{2}\pi$, with no failures.

Common slips

- Forgetting that cot is decreasing, and so concluding $\phi \geq \frac{1}{2}(\alpha+\beta)$.
- Dividing the inequality by $\sin p \sin q \sin^2 s$ without saying why it is positive.
- Ignoring the condition $p + q \leq \frac{1}{2}\pi$: it is exactly what makes $\cos(p+q) \geq 0$, and without it the inequality can fail.

Beyond STEP

The inequality says that $\ln\cot t$ is a convex function on $(0, \frac{1}{4}\pi]$, since it states that the average of $\ln\cot p$ and $\ln\cot q$ is at least $\ln\cot$ of the average. Convexity and Jensen's inequality are a powerful source of inequalities in first-year analysis. The relation $DC^2 = AD \cdot DB$ is Euclid's construction of a geometric mean: $\cot\phi$ is the geometric mean of $\cot\alpha$ and $\cot\beta$. Bartle and Sherbert's *Introduction to Real Analysis* treats convex

functions, and Siklos's *Advanced Problems in Mathematics* has more questions in this style.

WORKED SOLUTION

STEP I 2012, Question 7

Pure • Periodic Recurrence Relations

STEP I 2012, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q7

The key idea

A sequence that repeats with period 1, 2 or 3 is pinned down by its first few terms, so the recurrence only has to hold for the few consecutive triples inside one period (plus the wrap-around). Writing these out gives a small set of simultaneous linear equations in x, y, z. The equations have a solution that isn't all zeros, and that is what forces the condition on p and q. Throughout, the fact that $x \neq 0$ is what lets us cancel.

Solution

Part (i)

If $t_n = x$ for every n, the recurrence says $x = px + qx$, that is $(p+q-1)x = 0$. Since $x \neq 0$, we get $p + q = 1$.

Conversely, if $p + q = 1$ and $t_0 = t_1 = x$ then $t_2 = (p+q)x = x$, and by induction every term equals x. Nothing restricts x, so it can be any non-zero real number.

Part (ii)

The pattern $x, y, x, y, \ldots$ satisfies the recurrence exactly when the two consecutive triples (x, y, x) and (y, x, y) do:

$$x = py + qx, \qquad \text{i.e.} \quad (1-q)x = py, \tag{1}$$

$$y = px + qy, \qquad \text{i.e.} \quad (1-q)y = px. \tag{2}$$

Adding and subtracting (1) and (2) gives

$$(1-q-p)(x+y) = 0, \qquad (1-q+p)(x-y) = 0.$$

The numbers $x + y$ and $x - y$ can't both be zero, since that would make $x = 0$. If $x + y \neq 0$ then $p + q = 1$; if $x - y \neq 0$ then $q - p = 1$. So at least one of $q + p = 1$ and $q - p = 1$ holds, as required.

Now we deduce the relation between x and y.

- If $q + p = 1$ then $1 - q = p$ and (1) becomes $px = py$. So $x = y$, *provided* $p \neq 0$.
- If $q - p = 1$ then $1 - q = -p$ and (1) becomes $-px = py$. So $x = -y$, again provided $p \neq 0$.

The only escape is $p = 0$, and then either case gives $q = 1$. With $p = 0$ and $q = 1$ the recurrence is $t_{n+2} = t_n$, which every pattern $x, y, x, y, \ldots$ satisfies, so x and y are unrelated. Hence $x = y$ or $x = -y$ unless $p = 0$ and $q = 1$.

Part (iii)

Now the pattern is $x, y, z, x, y, z, \ldots$, and we need the three consecutive triples (x, y, z), (y, z, x) and (z, x, y):

$$z = py + qx, \tag{3}$$
$$x = pz + qy, \tag{4}$$
$$y = px + qz. \tag{5}$$

We use (3) to eliminate z from the other two:

$$\text{(4):}\quad x = p(py + qx) + qy \quad\Longrightarrow\quad (pq - 1)x + (p^2 + q)y = 0,$$
$$\text{(5):}\quad y = px + q(py + qx) \quad\Longrightarrow\quad (p + q^2)x + (pq - 1)y = 0.$$

Multiply the first equation by $pq - 1$ and the second by $p^2 + q$, then subtract. The y terms cancel, leaving

$$\left[(pq - 1)^2 - (p^2 + q)(p + q^2)\right]x = 0.$$

Ask yourself Why must the bracket vanish? Couldn't the equations simply have no solution?

They do have a solution: the sequence exists, so x and y satisfy both equations. And $x \neq 0$, so we can divide by it. Expanding the bracket,

$$\begin{aligned}(pq - 1)^2 - (p^2 + q)(p + q^2) &= p^2q^2 - 2pq + 1 - \left(p^3 + p^2q^2 + pq + q^3\right)\\ &= 1 - 3pq - p^3 - q^3,\end{aligned}$$

so $p^3 + q^3 + 3pq - 1 = 0$, as required.

Factorising. The expression is $a^3 + b^3 + c^3 - 3abc$ with $a = p$, $b = q$, $c = -1$, and the standard identity

$$a^3 + b^3 + c^3 - 3abc = (a + b + c) \cdot \tfrac{1}{2}\left[(a - b)^2 + (b - c)^2 + (c - a)^2\right]$$

gives

$$p^3 + q^3 + 3pq - 1 = \tfrac{1}{2}(p + q - 1)\left[(p - q)^2 + (q + 1)^2 + (p + 1)^2\right].$$

(If you don't remember the identity, check it by expanding: the square bracket is $2(p^2+q^2+1-pq+p+q)$, and multiplying by $p+q-1$ gives back the cubic.) The product is zero, so either $p+q=1$ or $(p-q)^2+(p+1)^2+(q+1)^2=0$.

Case $(p-q)^2+(p+1)^2+(q+1)^2=0$**.** A sum of real squares is zero only if each square is zero, so $p=q=-1$. Then (3) reads $z=-y-x$, that is, $x+y+z=0$.

Case $p+q=1$**.** Substitute $q=1-p$ into (3), (4) and (5) and rearrange each as a difference:

$$z-x=p(y-x), \qquad x-y=p(z-y), \qquad y-z=p(x-z).$$

Write $a=y-x$, $b=z-y$, $c=x-z$, so that $a+b+c=0$. The first and third equations say $c=-pa$ and $b=-pc=p^2a$. Then

$$0=a+b+c=a(1-p+p^2).$$

Ask yourself Can $1-p+p^2$ be zero?

No: $1-p+p^2=\left(p-\frac{1}{2}\right)^2+\frac{3}{4}>0$ for every real p. So $a=0$, which gives $c=-pa=0$ and then $b=0$. That is $x=y=z$.

The two cases don't overlap, because $p=q=-1$ gives $p+q=-2\neq 1$. So either $x=y=z$ (when $p+q=1$) or $x+y+z=0$ (when $p=q=-1$). Both happen: in the second case the recurrence $t_{n+2}=-t_{n+1}-t_n$ gives $x,\ y,\ -x-y,\ x,\ y,\ \ldots$ for any starting values.

Checks

- With $p=q=-1$, $t_0=2$ and $t_1=5$, the sequence runs $2,5,-7,2,5,-7,\ldots$, and $2+5-7=0$.
- With $p=3$ and $q=4$ (so $q-p=1$), starting $1,-1$ gives $t_2=-3+4=1$ and $t_3=3-4=-1$, alternating as part (ii) predicts.
- At $p=2$, $q=-1$ the cubic gives $8-1-6-1=0$, consistent with $p+q=1$. We also checked the expansion, the factorisation and the cubic's agreement with the 3×3 determinant of (3)–(5) at 1000 random (p,q). Simulating the recurrence confirmed every case above.

Common slips

- Concluding $p+q=1$ *and* $q-p=1$ in part (ii). The argument shows only that at least one holds; both hold only in the exceptional case $p=0$, $q=1$.
- Dividing by p in part (ii) without noticing that $p=0$ is the exception the question is asking for.
- Forgetting the wrap-around equations (4) and (5). One equation per period isn't enough to force the condition.

Beyond STEP

The standard way to handle $t_{n+2} = pt_{n+1} + qt_n$ is through the *characteristic equation* $\lambda^2 = p\lambda + q$: in general $t_n = A\alpha^n + B\beta^n$, where α and β are its roots. A sequence repeats every k steps exactly when the roots it uses satisfy $\lambda^k = 1$. Constant sequences need the root 1 (so $p + q = 1$), and alternating ones need -1 (so $q - p = 1$). The case $p = q = -1$ gives $\lambda^2 + \lambda + 1 = 0$, whose roots are the complex cube roots of unity, and that is why their sum $1 + \omega + \omega^2 = 0$ reappears as $x + y + z = 0$. This is linear algebra in disguise (the shift map acting on sequences), and Strang's *Introduction to Linear Algebra* treats recurrences through eigenvalues in exactly this way.

WORKED SOLUTION

STEP I 2012, Question 8

Pure • Homogeneous Differential Equations

STEP I 2012, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q8

The key idea

Both equations are *homogeneous*: every term has the same total degree in x and y, apart from an overall power of x. If you put $y = xv$, that power of x cancels and the equation separates into a v-part and an x-part. Part (i) walks you through this. Part (ii) leaves it to you: dividing through by x shows it's the same kind of equation, and the v-integral then needs partial fractions.

Solution

Part (i)

If $y = xv$, the product rule gives $\dfrac{\mathrm{d}y}{\mathrm{d}x} = v + x\dfrac{\mathrm{d}v}{\mathrm{d}x}$. Substituting,

$$x(xv)\left(v + x\frac{\mathrm{d}v}{\mathrm{d}x}\right) + x^2v^2 - 2x^2 = x^3v\frac{\mathrm{d}v}{\mathrm{d}x} + 2x^2v^2 - 2x^2 = 0.$$

Since $x \neq 0$, we can divide by x^2, which gives

$$xv\frac{\mathrm{d}v}{\mathrm{d}x} + 2v^2 - 2 = 0,$$

as required.

Now we separate the variables. The equation says $xv\dfrac{\mathrm{d}v}{\mathrm{d}x} = 2(1 - v^2)$. Provided $v^2 \neq 1$,

$$\int \frac{v}{1 - v^2}\,\mathrm{d}v = \int \frac{2}{x}\,\mathrm{d}x \quad \Longrightarrow \quad -\tfrac{1}{2}\ln|1 - v^2| = 2\ln|x| + c.$$

Multiplying by -2 and taking exponentials gives $|1 - v^2| = Ax^{-4}$ for some constant $A > 0$, so

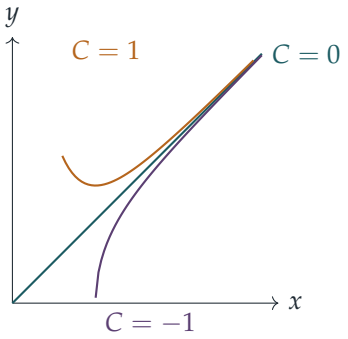
$$x^4(v^2 - 1) = C,$$

where $C = \pm A$ is a nonzero constant. The cases we set aside, $v = \pm 1$ (that is, $y = \pm x$), are also solutions: then $\mathrm{d}v/\mathrm{d}x = 0$ and $2v^2 - 2 = 0$. They are exactly the case $C = 0$, so C can be any constant.

Finally, put $v = y/x$:

$$x^4\left(\frac{y^2}{x^2} - 1\right) = x^2(y^2 - x^2) = C,$$

which is the required form.



The sketch shows a few solution curves in the first quadrant. They approach the line $y = x$ as $x \to \infty$, as $y^2 = x^2 + C/x^2$ suggests.

Part (ii)

Every term in $6x + 5y$ has degree 1, and so does $y\,\mathrm{d}y/\mathrm{d}x$. This suggests the same substitution, $y = xv$:

$$xv\left(v + x\frac{\mathrm{d}v}{\mathrm{d}x}\right) + 6x + 5xv = 0.$$

Dividing by $x \neq 0$ and collecting terms,

$$xv\frac{\mathrm{d}v}{\mathrm{d}x} = -(v^2 + 5v + 6) = -(v+2)(v+3).$$

Ask yourself How do we integrate $\dfrac{v}{(v+2)(v+3)}$?

Use partial fractions. Write $v = P(v+3) + Q(v+2)$. Putting $v = -2$ gives $P = -2$, and putting $v = -3$ gives $Q = 3$. So

$$\frac{v}{(v+2)(v+3)} = \frac{3}{v+3} - \frac{2}{v+2}.$$

Provided $v \neq -2, -3$, separating the variables gives

$$\int\left(\frac{3}{v+3} - \frac{2}{v+2}\right)\mathrm{d}v = -\int\frac{\mathrm{d}x}{x} \quad\Longrightarrow\quad 3\ln|v+3| - 2\ln|v+2| = -\ln|x| + c.$$

So $|x|\,|v+3|^3/(v+2)^2$ is constant. Removing the modulus signs (the sign of $x(v+3)^3$ can't change along a solution curve on which $v \neq -3$), we get

$$\frac{x(v+3)^3}{(v+2)^2} = K.$$

Now substitute $v = y/x$. The numerator is $x(y+3x)^3/x^3$ and the denominator is $(y+2x)^2/x^2$, so the powers of x cancel completely:

$$\boxed{(y+3x)^3 = K(y+2x)^2},$$

where K is a constant.

The lines we excluded are also solutions. If $y = -3x$, then $y\,\mathrm{d}y/\mathrm{d}x = 9x$ and $6x + 5y = -9x$, so the equation holds. This line is the case $K = 0$ of the boxed form. If $y = -2x$, then $y\,\mathrm{d}y/\mathrm{d}x = 4x$ and $6x + 5y = -4x$, so this line is a solution too. It isn't covered by any finite K; it corresponds to $K \to \infty$. Writing the solution as $(y+2x)^2 = k(y+3x)^3$ would cover it instead, with $k = 0$.

Checks

- Part (i): differentiating $x^2y^2 - x^4 = C$ implicitly gives $2xy^2 + 2x^2y\,y' - 4x^3 = 0$. Dividing by $2x$ gives back the original equation.
- Part (ii): we stepped the equation numerically (with a Runge–Kutta method) from many random starting points. Along each solution, the quantity $(y+3x)^3/(y+2x)^2$ stayed constant to about 10^{-14}. For example, the curve through $(1,1)$ has $K = 64/9$.
- The straight lines through the origin that solve each equation, $y = \pm x$ and $y = -2x$, $-3x$, come from the roots of $2v^2 - 2 = 0$ and $v^2 + 5v + 6 = 0$. That is a good sign that the v-equations are right.

Common slips

- Forgetting the product rule: $\mathrm{d}y/\mathrm{d}x$ is $v + x\,\mathrm{d}v/\mathrm{d}x$, not $x\,\mathrm{d}v/\mathrm{d}x$.
- Getting the partial fractions wrong (say, as $2/(v+2) - 3/(v+3)$). Test the decomposition at a value such as $v = 0$.
- Dividing by $1 - v^2$ or $(v+2)(v+3)$ without remarking that this loses some straight-line solutions.

Beyond STEP

An equation $\mathrm{d}y/\mathrm{d}x = F(y/x)$ is called homogeneous. Its solution curves are mapped to one another by the scalings $(x, y) \mapsto (\lambda x, \lambda y)$, which is why $v = y/x$ works. This is the simplest case of a deep idea, due to Sophus Lie: a symmetry of a differential

equation can be used to reduce its order. The straight-line solutions $y = mx$ are the curves that the scaling leaves fixed. Boyce and DiPrima, *Elementary Differential Equations*, covers homogeneous and other first-order equations.

WORKED SOLUTION

STEP I 2012, Question 9

Mechanics • Projectile Motion

STEP I 2012, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

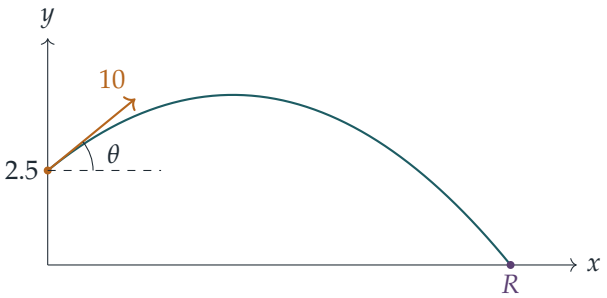
step.maths.org/questions/12-s1-q9

The key idea

This is a projectile launched from a height, so the time of flight comes from a quadratic in t, and the question is really about rewriting its root in terms of $c = \cos 2\theta$. The double-angle formulae $\sin^2\theta = \frac{1}{2}(1-c)$ and $\cos^2\theta = \frac{1}{2}(1+c)$ do all the work. Once the range is a function of the single variable c, maximising it is ordinary calculus, with a little care over squaring.

Solution

We model the shot as a particle moving freely under gravity, with no air resistance. Take the origin on the ground directly below the point of projection, with x horizontal and y vertically up, and assume $0 \le \theta < 90^\circ$ (the shot is thrown forwards and upwards), so that $\sin\theta \ge 0$ and $\cos\theta > 0$.



The time of flight

The horizontal velocity stays at $10\cos\theta$, and the vertical motion has initial velocity $10\sin\theta$ and acceleration -10, so at time t

$$x = 10t\cos\theta, \qquad y = 2.5 + 10t\sin\theta - 5t^2.$$

The shot lands when $y = 0$, that is, when

$$5t^2 - 10t\sin\theta - 2.5 = 0 \quad\Longleftrightarrow\quad t^2 - 2t\sin\theta - \tfrac{1}{2} = 0.$$

The product of the roots is $-\frac{1}{2} < 0$, so there is exactly one positive root, and it is the one with the plus sign:

$$T = \sin\theta + \sqrt{\sin^2\theta + \tfrac{1}{2}}.$$

Now use $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta) = \frac{1}{2}(1 - c)$. Since $\sin\theta \geq 0$ we may write $\sin\theta = \sqrt{\frac{1}{2}(1-c)}$, and $\sin^2\theta + \frac{1}{2} = \frac{1}{2}(2 - c)$. Hence

$$T = \sqrt{\frac{1-c}{2}} + \sqrt{\frac{2-c}{2}} = \frac{1}{\sqrt{2}}\left(\sqrt{1-c} + \sqrt{2-c}\right),$$

which is the required time in seconds.

The range

Similarly $\cos^2\theta = \frac{1}{2}(1 + c)$, and $\cos\theta > 0$, so $\cos\theta = \sqrt{\frac{1}{2}(1+c)}$. The range is

$$\begin{aligned} R = 10T\cos\theta &= 10\sqrt{\frac{1+c}{2}} \cdot \frac{1}{\sqrt{2}}\left(\sqrt{1-c} + \sqrt{2-c}\right) \\ &= 5\left(\sqrt{1-c^2} + \sqrt{(1+c)(2-c)}\right) = 5\left(\sqrt{1-c^2} + \sqrt{2+c-c^2}\right) \text{ m}. \end{aligned}$$

As θ runs over $[0°, 90°)$, $c = \cos 2\theta$ runs over $(-1, 1]$, so we maximise R over this interval.

Maximising the range

For $-1 < c < 1$, differentiating,

$$\frac{\mathrm{d}R}{\mathrm{d}c} = 5\left(\frac{-c}{\sqrt{1-c^2}} + \frac{1-2c}{2\sqrt{2+c-c^2}}\right).$$

A stationary point needs

$$\frac{c}{\sqrt{1-c^2}} = \frac{1-2c}{2\sqrt{2+c-c^2}}. \qquad (*)$$

Ask yourself Can we just square $(*)$, and what might go wrong?

Squaring can introduce false roots, because it loses the signs. Both denominators are positive, so in $(*)$ the numbers c and $1 - 2c$ must have the same sign (or both be zero, which is impossible). That forces $0 < c < \frac{1}{2}$, and we keep this in mind. Squaring and

clearing denominators:

$$\begin{aligned} 4c^2(2+c-c^2) &= (1-2c)^2(1-c^2) \\ 8c^2+4c^3-4c^4 &= 1-4c+3c^2+4c^3-4c^4 \\ 5c^2+4c-1 &= 0 \\ (5c-1)(c+1) &= 0. \end{aligned}$$

The root $c=-1$ is outside $0<c<\frac{1}{2}$ (it is the vertical throw), so the only stationary point is $c=\frac{1}{5}$. It really does satisfy $(*)$: both sides equal $\frac{1}{2\sqrt{6}}$, since $\sqrt{1-\frac{1}{25}}=\frac{2\sqrt{6}}{5}$ and $\sqrt{2+\frac{1}{5}-\frac{1}{25}}=\frac{3\sqrt{6}}{5}$.

At this point

$$R\left(\tfrac{1}{5}\right)=5\left(\frac{2\sqrt{6}}{5}+\frac{3\sqrt{6}}{5}\right)=5\sqrt{6}\approx 12.25.$$

To see that this is the greatest value, compare with the ends of the interval: $R\to 0$ as $c\to -1$, and $R(1)=5\sqrt{2}\approx 7.07$. Since R is continuous on $[-1,1]$ and differentiable inside, its maximum is at an end or at a stationary point; the only stationary point gives the largest of these values. So the range is greatest when $c=\frac{1}{5}$, as required. (This is $\theta=\frac{1}{2}\arccos\frac{1}{5}\approx 39.2^\circ$, a little below 45°.)

The extra distance

At 45° we have $c=\cos 90^\circ=0$, so

$$R(0)=5\left(1+\sqrt{2}\right).$$

The extra distance is therefore

$$5\sqrt{6}-5\left(1+\sqrt{2}\right)=5\left(\sqrt{6}-\sqrt{2}-1\right)\text{ m},$$

as required.

Checks

- At $\theta=0$ ($c=1$) the time is $\frac{1}{\sqrt{2}}$, matching $2.5=5t^2$ for a ball rolled off a 2.5 m ledge.
- As $c\to -1$ (vertical throw) the range tends to 0, as it should.
- Numerically, the extra distance is $5(2.4495-1.4142-1)\approx 0.176$ m, only about 18 cm: the optimum is very flat. A step-by-step simulation of the flight, and a fine grid search of $R(c)$, both agree with the formulas and put the maximum at $c=0.2$.

Common slips

- Taking the minus sign in the quadratic formula, which gives a negative time.

- Squaring $(*)$ and keeping $c = -1$ without checking it, or forgetting to justify that $c = \frac{1}{5}$ is a maximum rather than a minimum.
- Assuming the best angle is $45°$: that is only true when launch and landing are at the same height.

Beyond STEP

Launching from a height always lowers the optimal angle below $45°$; in general the best angle satisfies $\tan\theta = u/\sqrt{u^2 + 2gh}$, which here gives $\tan\theta = \sqrt{2/3}$, the same as $\cos 2\theta = \frac{1}{5}$. Optimising a function of a parameter, and noticing how flat the maximum is, is a first taste of sensitivity analysis: near a maximum, a small error in the parameter costs only a second-order loss. For more on projectiles with and without air resistance, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 2012, Question 10

Mechanics • Free Fall and Sound

STEP I 2012, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q10

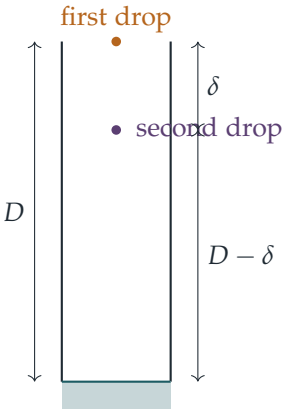
The key idea

Each measured time has two pieces: the stone falling freely from rest, and the sound of the splash travelling back up at constant speed. Writing both times down and subtracting gives t at once. For the "hence", the two square roots are the two fall times; we know their difference, and the difference of their squares is simple, so the factorisation $a^2 - b^2 = (a-b)(a+b)$ hands us their sum as well, and then each one separately.

Solution

The formula for t

We model the stone as a particle falling from rest under gravity alone (no air resistance), and assume the sound travels straight up the well at constant speed u.



A stone dropped from rest falls a distance h in time τ with $h = \frac{1}{2}g\tau^2$, so $\tau = \sqrt{2h/g}$. The sound then travels the same distance h back up to me, taking h/u. So the time I

measure when dropping from a height h above the water is

$$\sqrt{\frac{2h}{g}} + \frac{h}{u}.$$

From the top, $h = D$; after climbing down, I am (and the stone starts) a height $D - \delta$ above the water, and the sound only has to reach me there. Hence

$$\begin{aligned} t &= \left(\sqrt{\frac{2D}{g}} + \frac{D}{u}\right) - \left(\sqrt{\frac{2(D-\delta)}{g}} + \frac{D-\delta}{u}\right) \\ &= \sqrt{\frac{2D}{g}} - \sqrt{\frac{2(D-\delta)}{g}} + \frac{\delta}{u}, \end{aligned}$$

as required.

The formula for D

Write $a = \sqrt{2D/g}$ and $b = \sqrt{2(D-\delta)/g}$ for the two fall times. The result just shown says

$$a - b = t - \frac{\delta}{u} = \beta.$$

Since $\delta > 0$ we have $a > b$, so $\beta > 0$; in particular $\beta \neq 0$. Also

$$a^2 - b^2 = \frac{2D}{g} - \frac{2(D-\delta)}{g} = \frac{2\delta}{g}.$$

? **Ask yourself** We know $a - b$. How can we find a on its own?

Factorise: $a^2 - b^2 = (a-b)(a+b) = \beta(a+b)$, and we may divide by β because it is positive. So

$$a + b = \frac{2\delta}{\beta g}.$$

Adding this to $a - b = \beta$ and halving,

$$a = \tfrac{1}{2}\beta + \frac{\delta}{\beta g} = T.$$

Finally $a = \sqrt{2D/g}$ means $D = \frac{1}{2}ga^2$, so

$$D = \tfrac{1}{2}gT^2,$$

as required. (So T is just the time for the stone to fall the whole depth from the top.)

The numerical case

With $u = 300\,\mathrm{m\,s^{-1}}$, $g = 10\,\mathrm{m\,s^{-2}}$, $t = \frac{1}{5}\,\mathrm{s}$ and $\delta = 10\,\mathrm{m}$,

$$\beta = \frac{1}{5} - \frac{10}{300} = \frac{1}{5} - \frac{1}{30} = \frac{1}{6}\,\mathrm{s},$$
$$T = \frac{1}{12} + \frac{10}{\frac{1}{6}\cdot 10} = \frac{1}{12} + 6 = \frac{73}{12}\,\mathrm{s},$$
$$D = \tfrac{1}{2}\cdot 10\cdot\left(\frac{73}{12}\right)^2 = \frac{5\times 5329}{144} = \frac{26645}{144} \approx 185.03\ \mathrm{m}.$$

So the well is approximately 185 m deep.

Checks

- Put $D = 26645/144$ back into the formula for t with $\delta = 10$: the fall times are $73/12 \approx 6.083\,\mathrm{s}$ and $\sqrt{2(175.03)/10} \approx 5.917\,\mathrm{s}$, which differ by $1/6\,\mathrm{s}$, and adding $\delta/u = 1/30\,\mathrm{s}$ gives $t = 0.2\,\mathrm{s}$ exactly.
- If sound were instantaneous ($u \to \infty$), then $\beta = t$, as it should be: only the fall times would matter.
- A numerical time-step of the falling stone, plus the sound delay, reproduces the formula for t to $10^{-4}\,\mathrm{s}$, and the inversion $D = \frac{1}{2}gT^2$ recovers D in 10^4 random cases.

Common slips

- Forgetting that the sound only travels $D - \delta$ after I have climbed down, which loses the δ/u term.
- Trying to square the equation $a - b = \beta$ directly; this leads to messy algebra, while $a^2 - b^2 = (a-b)(a+b)$ gives $a + b$ in one line.
- Ignoring the sound altogether: with $\beta = t = 0.2$ the same method gives $T = 5.1\,\mathrm{s}$ and $D \approx 130\,\mathrm{m}$, badly wrong.

Beyond STEP

The method here is a *differential measurement*: rather than trusting one timing, we compare two experiments that differ by a known amount, so that unknowns cancel. The last slip shows how sensitive the answer is to small changes in the data, since t is a small difference of two large times; physicists and engineers quantify this with error propagation and condition numbers, studied in first-year experimental methods and numerical analysis. For more on modelling simple motion carefully, read Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 2012, Question 11

Mechanics • Pulleys and Friction

STEP I 2012, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q11

The key idea

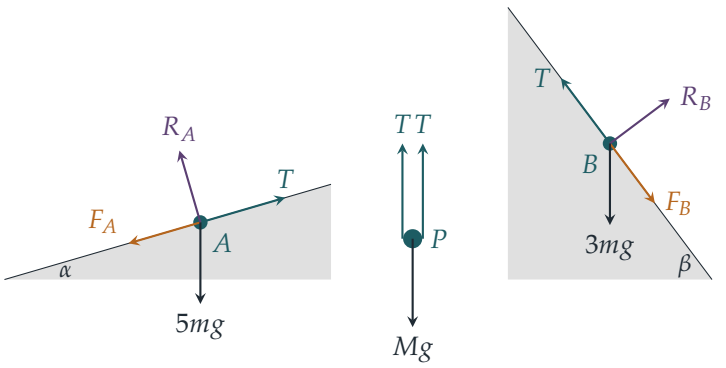
Because every pulley is smooth and the string is light, the tension is the same all the way along it. So each particle gives one equation linking the tension to friction, and the pulley P gives a third. The angles are chosen from the $7, 24, 25$ and $3, 4, 5$ triangles, so all the sines and cosines are exact fractions. For the moving case we need one more ingredient: how the accelerations of A, B and P are tied together by the fixed length of the string.

Solution

We model A, B as particles and the string as light and inextensible. Pulleys Q and R are smooth, light and fixed, and P is smooth with mass M. Let $\alpha = \arctan\frac{7}{24}$ and $\beta = \arctan\frac{4}{3}$. From the right-angled triangles with sides $7, 24, 25$ and $3, 4, 5$,

$$\sin\alpha = \tfrac{7}{25}, \quad \cos\alpha = \tfrac{24}{25}, \qquad \sin\beta = \tfrac{4}{5}, \quad \cos\beta = \tfrac{3}{5}.$$

Let T be the tension. Since all the pulleys are smooth, T is the same in every segment.



In both parts the particles are on the point of moving, or moving, *up* their planes, so

friction acts *down* each plane, as drawn. Resolving perpendicular to the planes,

$$R_A = 5mg\cos\alpha = \tfrac{24}{5}mg, \qquad R_B = 3mg\cos\beta = \tfrac{9}{5}mg.$$

Part (i)

Both particles are on the point of slipping, so friction is limiting: $F_A = \mu R_A$ and $F_B = \mu R_B$. Resolving along each plane (equilibrium):

$$\begin{aligned} A: &\quad T = 5mg\sin\alpha + \mu R_A = \tfrac{mg}{5}(7 + 24\mu), \\ B: &\quad T = 3mg\sin\beta + \mu R_B = \tfrac{mg}{5}(12 + 9\mu). \end{aligned}$$

Equating, $7 + 24\mu = 12 + 9\mu$, so

$$\mu = \tfrac{1}{3},$$

and then $T = \frac{mg}{5}(7 + 8) = 3mg$.

The pulley P is in equilibrium with two vertical string segments pulling it up and its weight pulling it down, so

$$2T = Mg \quad \Longrightarrow \quad M = \frac{2 \cdot 3mg}{g} = 6m,$$

as required.

Part (ii)

Now $M = 9m$, which is heavier than the $6m$ that could just be held in limiting equilibrium, so we expect P to descend and drag both particles up their planes.

Ask yourself How are the three accelerations related?

Let A and B have accelerations a and b up their planes, and let P have acceleration p downwards. The string length $AQ + QP + PR + RB$ is fixed. If A moves a distance x up its plane and B a distance y up its plane, then AQ and RB shorten by x and y; if P drops by z, then QP and PR each lengthen by z. Hence $2z = x + y$ at all times, and differentiating twice,

$$2p = a + b.$$

With $\mu = \frac{1}{3}$, the friction while sliding up is $\frac{1}{3}R_A = \frac{8}{5}mg$ on A and $\frac{1}{3}R_B = \frac{3}{5}mg$ on B. Newton's second law along each plane and vertically for P gives

$$\begin{aligned} A: &\quad T - \tfrac{7}{5}mg - \tfrac{8}{5}mg = 5ma, && \text{so } T - 3mg = 5ma, \\ B: &\quad T - \tfrac{12}{5}mg - \tfrac{3}{5}mg = 3mb, && \text{so } T - 3mg = 3mb, \\ P: &\quad 9mg - 2T = 9mp. \end{aligned}$$

Put $T = 3mg + X$. Then $a = \dfrac{X}{5m}$, $b = \dfrac{X}{3m}$, and the string condition gives

$$p = \tfrac{1}{2}(a + b) = \frac{4X}{15m}.$$

Substituting into the equation for P:

$$3mg - 2X = \frac{12X}{5} \quad \Longrightarrow \quad X = \frac{15mg}{22}.$$

Therefore

$$a = \frac{3g}{22} \text{ up the plane}, \qquad b = \frac{5g}{22} \text{ up the plane}, \qquad p = \frac{2g}{11} \text{ vertically downwards},$$

with tension $T = \frac{81}{22}mg$.

Ask yourself How do we know both particles really do move up, rather than one staying put?

From the equations above, a particle slides up its plane only if $T > 3mg$, and stays at rest (or would slide down) only if $T \leq 3mg$. The value $3mg$ is the same for both particles, so either both move up or neither does. If neither moved up, then $a + b \leq 0$, so $p \leq 0$; but $T \leq 3mg$ gives $9mg - 2T \geq 3mg > 0$, which forces $p > 0$. That is a contradiction, so both move up, and our solution (with $a, b > 0$) is consistent.

Checks

- In part (i), substituting $\mu = \frac{1}{3}$ into both particle equations gives the same tension, $3mg$.
- In part (ii), $2p = \frac{4g}{11} = \frac{3g}{22} + \frac{5g}{22} = a + b$, as the string requires, and $T = \frac{81}{22}mg \approx 3.68mg$ lies between $3mg$ (the threshold for slipping) and $4.5mg$ (the value at which P would have no acceleration).
- A numerical simulation stepping the three equations of motion keeps the string length fixed up to rounding error, and $p \approx 0.182g$.
- If M were $6m$ again, the same method gives $X = 0$, so all accelerations vanish, matching part (i).

Common slips

- Taking $a = b = p$: the pulley P moves at half the combined rate at which the two particles move, because two segments of string share its motion.
- Putting friction up the plane: both particles are being pulled *up*, so friction opposes that and acts down the plane.
- Forgetting that both strings on P carry tension T, so the upward force on it is $2T$.

Beyond STEP

The relation $2p = a + b$ is a *constraint*: it cuts the number of independent coordinates from three to two. In Lagrangian mechanics, met in the first or second year, you pick

coordinates that satisfy the constraints automatically and never have to find the tension at all. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many pulley systems treated carefully through constraint equations, a good bridge to that viewpoint.

WORKED SOLUTION

STEP I 2012, Question 12

Probability and Statistics • Conditional Probability

STEP I 2012, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q12

The key idea

Everything depends on which year the extinguisher becomes faulty in, so first turn the density into a cumulative distribution function and read off the probabilities of the fault arriving in the first, second and third year. Once we know the year, the tests are just a run of independent passes followed by a fail. The final part is a conditional probability, and the event "faulty at 18 months" is again a statement about which interval T lies in.

Solution

The distribution of T

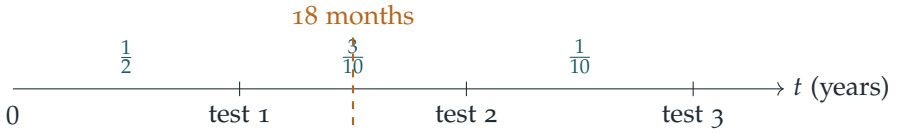
For $t \geq 0$, substituting $u = 1 + s^2$ (so $\mathrm{d}u = 2s\,\mathrm{d}s$),

$$F(t) = \mathrm{P}(T \leq t) = \int_0^t \frac{2s}{(1+s^2)^2}\,\mathrm{d}s = \left[-\frac{1}{1+s^2}\right]_0^t = 1 - \frac{1}{1+t^2} = \frac{t^2}{1+t^2}.$$

So $F(1) = \frac{1}{2}$, $F(2) = \frac{4}{5}$, $F(3) = \frac{9}{10}$, and

$$\mathrm{P}(0 < T < 1) = \tfrac{1}{2}, \qquad \mathrm{P}(1 < T < 2) = \tfrac{4}{5} - \tfrac{1}{2} = \tfrac{3}{10}, \qquad \mathrm{P}(2 < T < 3) = \tfrac{9}{10} - \tfrac{4}{5} = \tfrac{1}{10}.$$

Since T is continuous, the probability that the fault arrives exactly on a test day is zero, so we needn't worry about whether $<$ or $\leq$ is used at the endpoints.



Destroyed exactly three years after manufacture

Let D be the event that the extinguisher is destroyed at the test three years after manufacture. For this to happen it must be faulty by the third test (so $T < 3$), it must

pass every test it takes while faulty before then, and it must fail the third test. A non-faulty extinguisher always passes, so only the tests after time T matter, and they are independent with failure probability p.

- If $0 < T < 1$: it is faulty at tests 1, 2 and 3, so it must pass, pass, fail: probability $(1-p)^2 p$.
- If $1 < T < 2$: it passes test 1 automatically, then must pass test 2 and fail test 3: probability $(1-p)p$.
- If $2 < T < 3$: it passes tests 1 and 2 automatically and must fail test 3: probability p.

If $T > 3$ it is not faulty at test 3 and so cannot be destroyed then. By the law of total probability,

$$\begin{aligned} \mathrm{P}(D) &= \tfrac{1}{2}(1-p)^2 p + \tfrac{3}{10}(1-p)p + \tfrac{1}{10}p \\ &= \frac{p}{10}\Big(5(1-2p+p^2) + 3(1-p) + 1\Big) \\ &= \frac{p\,(5p^2 - 13p + 9)}{10}, \end{aligned}$$

as required.

Faulty at 18 months, given destroyed at three years

Let E be the event that the extinguisher was faulty 18 months after manufacture, that is, $T < \frac{3}{2}$. We want

$$\mathrm{P}(E \mid D) = \frac{\mathrm{P}(E \cap D)}{\mathrm{P}(D)}.$$

Ask yourself The interval $(0, \frac{3}{2})$ straddles a test. How do we split it?

What matters for the tests is the year in which the fault arrives, so split E into $0 < T < 1$ and $1 < T < \frac{3}{2}$. We have $F(\frac{3}{2}) = \dfrac{9/4}{13/4} = \dfrac{9}{13}$, so

$$\mathrm{P}\big(1 < T < \tfrac{3}{2}\big) = \tfrac{9}{13} - \tfrac{1}{2} = \tfrac{5}{26}.$$

Given $0 < T < 1$, the chance of D is $(1-p)^2 p$ as before; given $1 < T < \frac{3}{2}$, the fault still arrives in the second year, so the chance of D is $(1-p)p$. Hence

$$\begin{aligned} \mathrm{P}(E \cap D) &= \tfrac{1}{2}(1-p)^2 p + \tfrac{5}{26}(1-p)p \\ &= \frac{p(1-p)}{26}\big(13(1-p) + 5\big) = \frac{p(1-p)(18-13p)}{26}. \end{aligned}$$

Dividing by $\mathrm{P}(D)$ (which is positive for $0 < p \le 1$, because $5p^2 - 13p + 9$ has discriminant $169 - 180 < 0$ and so is always positive), the factor p cancels and

$$\mathrm{P}(E \mid D) = \frac{10(1-p)(18-13p)}{26(5p^2 - 13p + 9)} = \frac{5(1-p)(18-13p)}{13(5p^2 - 13p + 9)}.$$

Checks

- When $p = 1$ every faulty extinguisher is destroyed at its first test after the fault, so destruction at year 3 means the fault arrived in $(2, 3)$: $\mathrm{P}(D) = \frac{1}{10}$ and $\mathrm{P}(E \mid D) = 0$. Both formulae agree.
- As $p \to 0$, $\mathrm{P}(E \mid D) \to \dfrac{5 \cdot 18}{13 \cdot 9} = \dfrac{10}{13}$, which is $\mathrm{P}(T < \frac{3}{2} \mid T < 3) = \dfrac{9/13}{9/10}$: when the test barely detects faults, the destruction tells us almost nothing beyond $T < 3$.
- A simulation of 10^6 extinguishers with $p = 0.4$ gave $\mathrm{P}(D) \approx 0.1838$ (exact 0.184) and $\mathrm{P}(E \mid D) \approx 0.642$ (exact $\frac{192}{299} \approx 0.642$).

Common slips

- Forgetting that a faulty extinguisher has to *pass* the earlier tests before failing the third one.
- Using $\mathrm{P}(T < \frac{3}{2}) = \frac{9}{13}$ as a single block in the final part: faults in the first year and in the first half of the second year face different numbers of tests.

Beyond STEP

Here T is a continuous lifetime, but we only observe it through discrete, imperfect inspections. This is the setting of *interval-censored* data in survival analysis and reliability theory, and the last part is a small Bayesian update: the prior distribution of T is revised in the light of an observed failure. Grimmett and Welsh, *Probability: An Introduction*, covers continuous distributions and conditioning of this kind.

WORKED SOLUTION

STEP I 2012, Question 13

Probability and Statistics • Expectation via Indicators

STEP I 2012, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q13

The key idea

First get the conditional sample space straight: once we are told which digits are excluded, the number is just a string of five digits, each chosen from $\{1,2,3,4,5\}$, and every such string is equally likely. Then, rather than finding the whole distribution of the number of different digits, write that number as a sum of indicator variables, one for each possible digit. Linearity of expectation turns the problem into one probability, which is easy to find by looking at its complement.

Solution

The conditional sample space

Every integer from 10000 to 99999 is a string of five digits $d_1d_2d_3d_4d_5$ with $d_1 \neq 0$, and each such string appears exactly once. The condition says that every digit lies in $\{1,2,3,4,5\}$. Since 0 is excluded anyway, the restriction $d_1 \neq 0$ adds nothing, so the integers satisfying the condition correspond exactly to the

$$5^5 = 3125$$

strings of five digits from $\{1,2,3,4,5\}$.

The original choice was uniform. Conditioning on an event keeps the outcomes inside it equally likely, so given the condition each of these 3125 strings has probability $\frac{1}{3125}$. Put another way, the five digits are independent, and each is uniform on $\{1,2,3,4,5\}$: each of the 3125 outcomes is a product of five factors of $\frac{1}{5}$.

Indicator variables

Let N be the number of different digits in the choice. For $j = 1,2,3,4,5$ define

$$I_j = \begin{cases} 1 & \text{if the digit } j \text{ appears at least once,} \\ 0 & \text{otherwise.} \end{cases}$$

Each digit that appears contributes exactly 1 to the sum, so

$$N = I_1 + I_2 + I_3 + I_4 + I_5.$$

Ask yourself The I_j are not independent. Can we still add their expectations?

Yes. Linearity of expectation, $\mathrm{E}(X+Y) = \mathrm{E}(X) + \mathrm{E}(Y)$, holds for any random variables, independent or not. So

$$\mathrm{E}(N) = \sum_{j=1}^{5} \mathrm{E}(I_j) = \sum_{j=1}^{5} \mathrm{P}(\text{digit } j \text{ appears}).$$

The digit j fails to appear exactly when each of the five independent digits is one of the other four values, which has probability $\frac{4}{5}$ for each digit. Hence

$$\mathrm{P}(\text{digit } j \text{ appears}) = 1 - \left(\tfrac{4}{5}\right)^5 = 1 - \frac{1024}{3125} = \frac{2101}{3125}.$$

This is the same for all five values of j by symmetry, so

$$\mathrm{E}(N) = 5 \cdot \frac{2101}{3125} = \frac{2101}{625} = 3.3616,$$

as required.

Or otherwise: the full distribution

If you prefer to count directly, the number of strings using exactly k different digits is $\binom{5}{k}$ (choose the digits) times the number of ways to fill five places so that all k chosen digits appear. That second count comes from inclusion–exclusion. For example, with $k = 3$ it is $3^5 - 3 \cdot 2^5 + 3 \cdot 1^5 = 150$.

k	1	2	3	4	5	total
strings	$5 \cdot 1$	$10 \cdot 30$	$10 \cdot 150$	$5 \cdot 240$	$1 \cdot 120$	
	5	300	1500	1200	120	3125

Then

$$\mathrm{E}(N) = \frac{1 \cdot 5 + 2 \cdot 300 + 3 \cdot 1500 + 4 \cdot 1200 + 5 \cdot 120}{3125} = \frac{10505}{3125} = 3.3616.$$

This method works, but it is longer and has more places to go wrong. The indicator method needs only one probability.

Checks

- The counts in the table add up to $3125 = 5^5$, which confirms that every string has been counted exactly once.

- The answer is sensible: N lies between 1 and 5, and with five draws from five values we expect a fair number of repeats, so a value a little above 3 is plausible. As a general check, if we draw n times from m values the same argument gives $m\big(1-(1-\frac{1}{m})^n\big)$, which is 1 when $n=1$, as it should be.
- Enumerating all integers from 10000 to 99999 by computer, keeping the 3125 that use only the digits 1–5, and averaging the number of distinct digits gives exactly $\frac{2101}{625}=3.3616$.

Common slips

- Treating the sample space as all 90000 integers. The condition reduces it to 5^5 equally likely strings.
- Thinking that linearity of expectation needs independence. It doesn't, and that is exactly why indicators are so useful.
- Computing $\mathrm{P}(\text{digit } j \text{ appears})$ by adding $\frac{1}{5}$ for each of the five places, which gives 1 because it counts strings where j appears more than once several times over. Use the complement.

Beyond STEP

This is the classic "occupancy" or coupon-collector problem: n balls thrown at random into m boxes, and N is the number of boxes that end up occupied. The indicator trick, $\mathrm{E}(N)=m\big(1-(1-\frac{1}{m})^n\big)$, is one of the most useful tools in first-year probability and in the analysis of algorithms such as hashing. The exact distribution involves Stirling numbers of the second kind, which appeared in the inclusion–exclusion count above. Blitzstein and Hwang, *Introduction to Probability*, uses indicator random variables throughout and is an excellent next step.

WORKED SOLUTION

STEP II 2012, Question 1

Pure • Binomial Series Coefficients

STEP II 2012, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q1

The key idea

Each factor $(1-x^k)^{-1}$ is a geometric series in x^k, so multiplying such factors together and reading off the coefficient of x^n amounts to *counting* the ways of writing n as a sum of the allowed pieces. Once you see the coefficients as counts, the case analysis organises itself, and multiplying by $(1-x)^{-1}$ simply turns a list of coefficients into its running totals.

Solution

The general term

By the binomial expansion, $(1-y)^{-2} = \sum_{r\geq 0}(r+1)y^r$ for $|y|<1$ (the coefficient is $\binom{-2}{r}(-1)^r = r+1$). Putting $y = x^6$,

$$(1-x^6)^{-2} = \sum_{r=0}^{\infty}(r+1)\,x^{6r},$$

so the general term is $(r+1)x^{6r}$, and every power of x that is not a multiple of 6 has coefficient 0.

Part (i)

Also $(1-x^3)^{-1} = \sum_{s\geq 0} x^{3s}$. Multiplying the two series, a term x^n arises from each pair (r,s) of non-negative integers with $6r+3s=n$, contributing $(r+1)\cdot 1$. So

$$\text{coefficient of } x^n = \sum_{\substack{r,s\geq 0\\ 6r+3s=n}} (r+1).$$

For $n=24$ we need $2r+s=8$, so r can be $0,1,2,3,4$ (each with exactly one $s=8-2r\geq 0$). The coefficient of x^{24} is

$$1+2+3+4+5 = 15.$$

General n. Since $6r+3s$ is always a multiple of 3, the coefficient is 0 unless $3 \mid n$. If $n = 3m$, we need $2r + s = m$, and for each r with $0 \le 2r \le m$ there is exactly one valid s. So r runs from 0 to $\lfloor m/2 \rfloor$, and the coefficient is

$$1 + 2 + \cdots + (\lfloor m/2 \rfloor + 1).$$

Writing n according to its remainder on division by 6:

- $n = 6k$ (so $m = 2k$, $\lfloor m/2 \rfloor = k$): coefficient $\frac{1}{2}(k+1)(k+2)$;
- $n = 6k+3$ (so $m = 2k+1$, $\lfloor m/2 \rfloor = k$): coefficient $\frac{1}{2}(k+1)(k+2)$ as well;
- $n \not\equiv 0 \pmod 3$: coefficient 0.

As a check, $n = 24 = 6 \times 4$ gives $\frac{1}{2} \cdot 5 \cdot 6 = 15$, as found above.

Part (ii)

Write b_n for the coefficient of x^n found in part (i), and c_n for the coefficient we now want. Since $(1-x)^{-1} = 1 + x + x^2 + \cdots$,

$$\Big(\sum_j b_j x^j\Big)(1 + x + x^2 + \cdots) = \sum_n \Big(\sum_{j=0}^{n} b_j\Big) x^n,$$

so $c_n = b_0 + b_1 + \cdots + b_n$: the running total of the part (i) coefficients.

Ask yourself Why does multiplying by $(1-x)^{-1}$ give running totals?

To get x^n in the product we take x^j from the first series and x^{n-j} from $1 + x + x^2 + \cdots$, whose coefficient is always 1. So every b_j with $0 \le j \le n$ appears exactly once in c_n, with weight 1.

Only b_j with $3 \mid j$ are non-zero, and by part (i) they come in equal pairs: $b_{6k} = b_{6k+3} = T_{k+1}$, where $T_p = \frac{1}{2}p(p+1)$ is the pth triangular number. So for $n = 24$,

$$\begin{aligned} c_{24} &= (b_0 + b_3) + (b_6 + b_9) + (b_{12} + b_{15}) + (b_{18} + b_{21}) + b_{24} \\ &= 2(1 + 3 + 6 + 10) + 15 = 40 + 15 = 55, \end{aligned}$$

as required.

For $n = 25$, the extra term b_{25} is 0 because $3 \nmid 25$, so $c_{25} = c_{24} = 55$.

For $n = 66 = 6 \times 11$, the same pattern gives

$$c_{66} = 2\,(T_1 + T_2 + \cdots + T_{11}) + T_{12}.$$

The sum of the first p triangular numbers is the tetrahedral number $\frac{1}{6}p(p+1)(p+2)$ (by induction, or because $T_1 + \cdots + T_p = \binom{p+2}{3}$ by the hockey-stick identity). So $T_1 + \cdots + T_{11} = \frac{1}{6} \cdot 11 \cdot 12 \cdot 13 = 286$, and $T_{12} = 78$. Hence

$$c_{66} = 2 \times 286 + 78 = 650.$$

The same argument gives a general formula: if $n = 6K + r$ with $0 \leq r \leq 2$ then $c_n = 2\binom{K+2}{3} + \binom{K+2}{2}$, and if $3 \leq r \leq 5$ then $c_n = 2\binom{K+3}{3}$. With $K = 4$ the first gives $40 + 15 = 55$; with $K = 11$ it gives $572 + 78 = 650$.

Checks

The coefficient of x^n in part (ii) counts the non-negative solutions of $6a + 6b + 3c + d = n$ (one geometric series for each factor, with $(1 - x^6)^{-2}$ counted as two). A direct computer count of these solutions gives 55, 55 and 650 for $n = 24, 25, 66$. Multiplying out the series term by term up to x^{80} also confirms the formulas for both parts. By hand, small cases are quick: in part (ii), $c_0 = c_1 = c_2 = 1$ and $c_3 = c_4 = c_5 = 2$, matching the general formula with $K = 0$.

Common slips

- Forgetting the case $n = 6k + 3$ in part (i): odd multiples of 3 do appear, from an odd number of factors of x^3.
- Writing the general term of $(1 - x^6)^{-2}$ as $r\,x^{6r}$ instead of $(r + 1)x^{6r}$; the constant term is 1, not 0.
- Treating x^{25} as a fresh calculation: it equals the x^{24} coefficient, because no term of the part (i) series has exponent 25.

Beyond STEP

Power series used as counting devices like this are called *generating functions*; the product here counts the ways to pay n pence with coins worth 6, 6 (of a different design), 3 and 1. The theory of partitions grows from exactly this idea, and the fact that the answers are piecewise polynomials, depending on n modulo 6, is a general phenomenon ("quasi-polynomials"). For a lively introduction to the method, read Wilf, *generatingfunctionology*.

WORKED SOLUTION

STEP II 2012, Question 2

Pure • Composition of Polynomials

STEP II 2012, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q2

The key idea

Degrees multiply under composition, so a few iterations of p quickly produce a degree far too large to match the right-hand side. Comparing degrees pins down the degree of p; after that, writing p out with unknown coefficients turns each part into a short piece of algebra. In (ii) there is a neat shortcut: $\mathrm{p}(\mathrm{p}(x))$ is itself a quadratic expression *in* $\mathrm{p}(x)$, so the whole left-hand side is a quadratic in $\mathrm{p}(x)$, and completing the square finishes the job.

Solution

The degree of a composition

Write $\mathrm{p}(x) = a_m x^m + \dots$ and $\mathrm{q}(x) = b_n x^n + \dots$ with $a_m, b_n \neq 0$. In $\mathrm{p}(\mathrm{q}(x)) = a_m[\mathrm{q}(x)]^m + a_{m-1}[\mathrm{q}(x)]^{m-1} + \dots$, the term $a_m[\mathrm{q}(x)]^m$ has degree mn and leading coefficient $a_m b_n^m \neq 0$, while every other term has degree at most $(m-1)n < mn$ (when $n \geq 1$). So nothing cancels the top term, and

$$\deg \mathrm{p}(\mathrm{q}(x)) = mn.$$

(If $n = 0$ then $\mathrm{p}(\mathrm{q}(x))$ is a constant, and $mn = 0$ again.)

Part (i)

Let p have degree m.

If $m \geq 2$, then by the result above $\mathrm{p}(\mathrm{p}(\mathrm{p}(x)))$ has degree m^3, while $3\mathrm{p}(x)$ has degree $m < m^3$. So the left-hand side has degree $m^3 \geq 8$, but the right-hand side $-2x$ has degree 1. Impossible.

If $m = 0$, then $\mathrm{p}(x) = c$ for a constant c, and the left-hand side is the constant $c - 3c = -2c$, which cannot equal $-2x$ for all x. Impossible.

So $m = 1$. Write $\mathrm{p}(x) = ax + b$ with $a \neq 0$. Then

$$\mathrm{p}(\mathrm{p}(x)) = a(ax + b) + b = a^2x + (a + 1)b,$$
$$\mathrm{p}(\mathrm{p}(\mathrm{p}(x))) = a^3x + (a^2 + a + 1)b,$$

and the equation becomes

$$(a^3 - 3a)x + (a^2 + a + 1 - 3)b = -2x \quad \text{for all } x.$$

Two polynomials agree for all x only if their coefficients agree, so

$$a^3 - 3a + 2 = 0 \qquad \text{and} \qquad (a^2 + a - 2)\, b = 0.$$

Since $a = 1$ is a root of the first, it factorises as $(a - 1)^2(a + 2) = 0$, giving $a = 1$ or $a = -2$. The second condition is $(a - 1)(a + 2)b = 0$, which holds automatically for both of these values, whatever b is. Hence the solutions are exactly

$$\mathrm{p}(x) = x + b \qquad \text{and} \qquad \mathrm{p}(x) = -2x + b, \qquad b \text{ any constant.}$$

As a check on the second family: $\mathrm{p}(\mathrm{p}(x)) = 4x - b$ and $\mathrm{p}(\mathrm{p}(\mathrm{p}(x))) = -8x + 3b$, so the left-hand side is $-8x + 3b + 6x - 3b = -2x$.

Part (ii)

Let p have degree m. The three terms on the left have degrees m^2, $2m$ and m.

- If $m \geq 3$, then $m^2 > 2m > m$, so the left-hand side has degree $m^2 \geq 9 \neq 4$.
- If $m = 1$, the left-hand side has degree at most 2; if $m = 0$ it is constant. Neither can equal x^4.

So $m = 2$. (Here $m^2 = 2m = 4$, so the top terms of $2\mathrm{p}(\mathrm{p}(x))$ and $3[\mathrm{p}(x)]^2$ have the same degree, and we must check their coefficients.)

Write $\mathrm{p}(x) = ax^2 + bx + c$ with $a \neq 0$, and to save writing let P stand for $\mathrm{p}(x)$.

> **Ask yourself** Do we really have to expand $\mathrm{p}(\mathrm{p}(x))$ in full?

No. By definition $\mathrm{p}(\mathrm{p}(x)) = aP^2 + bP + c$, so the left-hand side is

$$2(aP^2 + bP + c) + 3P^2 - 4P = (2a + 3)P^2 + (2b - 4)P + 2c,$$

a quadratic in P. Its x^4 coefficient is $(2a + 3)a^2$, since P^2 has leading term a^2x^4 and P has degree only 2. Matching with x^4 gives

$$2a^3 + 3a^2 - 1 = 0, \qquad \text{that is,} \qquad (a + 1)^2(2a - 1) = 0,$$

(spot the root $a = -1$ and divide), so $a = -1$ or $a = \frac{1}{2}$.

We now use a simple fact: if $S(x)$ is a polynomial and $S(x)^2 = x^4 + K$ for a constant K, then $(S - x^2)(S + x^2) = K$. If neither factor were the zero polynomial, at least one of

them would have degree 2 when S has leading term $\pm x^2$, and the product could not be constant. So, for S with leading term $\pm x^2$, one factor is zero and $S = \pm x^2$ exactly.

Case $a = -1$. Then $2a + 3 = 1$, and completing the square,

$$P^2 + (2b-4)P + 2c = \big(P + b - 2\big)^2 - (b-2)^2 + 2c = x^4.$$

So $S = P + b - 2 = -x^2 + bx + (c + b - 2)$ satisfies $S^2 = x^4 + K$ with K constant, and has leading term $-x^2$. Then $S + x^2 = bx + (c + b - 2)$ has degree less than 2 while $S - x^2$ has degree 2, so the fact forces $S + x^2 = 0$: that is, $b = 0$ and $c + b - 2 = 0$. So $b = 0$, $c = 2$, and then $K = (b-2)^2 - 2c = 0$, consistent. This gives

$$\mathrm{p}(x) = 2 - x^2.$$

Case $a = \frac{1}{2}$. Then $2a + 3 = 4$, and

$$4P^2 + (2b-4)P + 2c = \left(2P + \tfrac{b-2}{2}\right)^2 - \tfrac{(b-2)^2}{4} + 2c = x^4.$$

Now $S = 2P + \frac{b-2}{2} = x^2 + 2bx + 2c + \frac{b-2}{2}$ has leading term x^2, so in the same way $S - x^2 = 0$: $b = 0$ and $2c - 1 = 0$. So $b = 0$, $c = \frac{1}{2}$, and $K = \frac{(b-2)^2}{4} - 2c = 1 - 1 = 0$, consistent. This gives

$$\mathrm{p}(x) = \tfrac{1}{2}(x^2 + 1).$$

Both really are solutions, since every step can be reversed, but it is reassuring to check directly. For $\mathrm{p}(x) = 2 - x^2$, $\mathrm{p}(\mathrm{p}(x)) = 2 - (2 - x^2)^2 = -x^4 + 4x^2 - 2$, and

$$2(-x^4 + 4x^2 - 2) + 3(x^4 - 4x^2 + 4) - 4(2 - x^2) = x^4.$$

For $\mathrm{p}(x) = \frac{1}{2}(x^2 + 1)$, write $u = x^2 + 1$; then $2\mathrm{p}(\mathrm{p}(x)) = \frac{1}{4}u^2 + 1$, and the left-hand side is $\frac{1}{4}u^2 + 1 + \frac{3}{4}u^2 - 2u = (u-1)^2 = x^4$.

So the polynomials satisfying the equation in (ii) are exactly $\mathrm{p}(x) = 2 - x^2$ and $\mathrm{p}(x) = \frac{1}{2}(x^2 + 1)$.

Checks

- In (i), try $b = 1$: $\mathrm{p}(x) = x + 1$ gives $(x + 3) - 3(x + 1) = -2x$.
- In (ii), put $x = 1$: for $\mathrm{p}(x) = 2 - x^2$ we get $\mathrm{p}(1) = 1$, $\mathrm{p}(\mathrm{p}(1)) = 1$, and $2 + 3 - 4 = 1 = 1^4$. For $\mathrm{p}(x) = \frac{1}{2}(x^2 + 1)$ at $x = 3$: $\mathrm{p}(3) = 5$, $\mathrm{p}(5) = 13$, and $26 + 75 - 20 = 81 = 3^4$.
- We also checked with exact arithmetic that both families in (i) work for many random b, and that a search over a grid of quadratics with b, c in steps of $\frac{1}{4}$ finds only the two solutions in (ii).

Common slips

- In (ii), assuming the degree of the left-hand side is simply m^2: when $m = 2$ the terms $2\mathrm{p}(\mathrm{p}(x))$ and $3[\mathrm{p}(x)]^2$ both have degree 4, and the possibility that they cancel is exactly why a must be found from $2a^3 + 3a^2 = 1$.
- Forgetting the case $m = 0$ when explaining why the degree must be 1 in (i).
- In (i), stopping at $a = 1$ or $a = -2$ and fixing a particular b: the constant term condition holds for every b.

Beyond STEP

The rule $\deg(\mathrm{p} \circ \mathrm{q}) = \deg \mathrm{p} \cdot \deg \mathrm{q}$, like $\deg(\mathrm{pq}) = \deg \mathrm{p} + \deg \mathrm{q}$, relies on there being no zero divisors among the coefficients: in a ring where two non-zero numbers can multiply to zero, leading terms can vanish and both rules fail. Functional equations such as $\mathrm{p}(\mathrm{p}(x)) = \ldots$ also lead into the study of iterated maps and dynamics. For polynomial rings and when degree arguments work, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 2012, Question 3

Pure • Integration by Substitution

STEP II 2012, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q3

The key idea

The awkward expression $x + \sqrt{1+x^2}$ is crying out to be called t. The trick that makes the substitution work is that its reciprocal is $\sqrt{1+x^2} - x$, so adding and subtracting t and $1/t$ gives both x and $\sqrt{1+x^2}$ as simple functions of t. After that, each of the two integrals is a matter of spotting how to write the integrand as a function of $x + \sqrt{1+x^2}$ alone.

Solution

The identity

Let $t = x + \sqrt{1+x^2}$. Multiplying by the conjugate,

$$\frac{1}{t} = \frac{1}{\sqrt{1+x^2}+x} \cdot \frac{\sqrt{1+x^2}-x}{\sqrt{1+x^2}-x} = \frac{\sqrt{1+x^2}-x}{(1+x^2)-x^2} = \sqrt{1+x^2} - x.$$

Subtracting and adding these two equations gives

$$x = \tfrac{1}{2}\left(t - \frac{1}{t}\right), \qquad \sqrt{1+x^2} = \tfrac{1}{2}\left(t + \frac{1}{t}\right).$$

The first of these gives x directly as a function of t, so

$$\mathrm{d}x = \tfrac{1}{2}\left(1 + \frac{1}{t^2}\right)\mathrm{d}t.$$

For the limits: when $x = 0$, $t = 1$; and as $x \to \infty$, $t \to \infty$. Also t is a strictly increasing function of x (both x and $\sqrt{1+x^2}$ increase for $x \geq 0$), so the substitution is one-to-one on $[0, \infty)$. Therefore

$$\int_0^\infty f\left(x + \sqrt{1+x^2}\right)\mathrm{d}x = \frac{1}{2}\int_1^\infty \left(1 + \frac{1}{t^2}\right) f(t)\,\mathrm{d}t,$$

as required.

The first integral

Ask yourself How can the denominator $2x^2+1+2x\sqrt{x^2+1}$ be a function of $x+\sqrt{1+x^2}$?

Try squaring: $\left(x+\sqrt{1+x^2}\right)^2 = x^2+2x\sqrt{1+x^2}+(1+x^2) = 2x^2+1+2x\sqrt{x^2+1}$, which is exactly the denominator. So the integrand is $f\left(x+\sqrt{1+x^2}\right)$ with $f(t)=1/t^2$, and the identity gives

$$\begin{aligned}\int_0^\infty \frac{\mathrm{d}x}{2x^2+1+2x\sqrt{x^2+1}} &= \frac{1}{2}\int_1^\infty \left(1+\frac{1}{t^2}\right)\frac{1}{t^2}\,\mathrm{d}t = \frac{1}{2}\int_1^\infty \left(t^{-2}+t^{-4}\right)\mathrm{d}t \\ &= \frac{1}{2}\left[-\frac{1}{t}-\frac{1}{3t^3}\right]_1^\infty = \frac{1}{2}\left(1+\frac{1}{3}\right) = \frac{2}{3}.\end{aligned}$$

The second integral

Put $x=\tan\theta$, so $\mathrm{d}x=\sec^2\theta\,\mathrm{d}\theta$, that is, $\mathrm{d}\theta = \dfrac{\mathrm{d}x}{1+x^2}$. As θ runs from 0 to $\frac{1}{2}\pi$, x runs from 0 to ∞. Since $0\le\theta<\frac{1}{2}\pi$, $\cos\theta>0$, so $\sec\theta=\sqrt{1+x^2}$ and

$$\sin\theta=\tan\theta\cos\theta=\frac{x}{\sqrt{1+x^2}}, \qquad 1+\sin\theta=\frac{\sqrt{1+x^2}+x}{\sqrt{1+x^2}}=\frac{t}{\sqrt{1+x^2}},$$

where again $t=x+\sqrt{1+x^2}$. Hence

$$\int_0^{\frac{1}{2}\pi}\frac{\mathrm{d}\theta}{(1+\sin\theta)^3} = \int_0^\infty \frac{(1+x^2)^{3/2}}{t^3}\cdot\frac{\mathrm{d}x}{1+x^2} = \int_0^\infty \frac{\sqrt{1+x^2}}{t^3}\,\mathrm{d}x.$$

Ask yourself There's still a $\sqrt{1+x^2}$ in the integrand. How do we write it in terms of t?

We found it in the first part: $\sqrt{1+x^2}=\frac{1}{2}(t+1/t)$. So the integrand is

$$\frac{t+t^{-1}}{2t^3} = \tfrac{1}{2}\left(t^{-2}+t^{-4}\right),$$

a function of $t=x+\sqrt{1+x^2}$ only. Applying the identity with $f(t)=\frac{1}{2}(t^{-2}+t^{-4})$,

$$\begin{aligned}\int_0^{\frac{1}{2}\pi}\frac{\mathrm{d}\theta}{(1+\sin\theta)^3} &= \frac{1}{2}\int_1^\infty\left(1+\frac{1}{t^2}\right)\cdot\frac{1}{2}\left(\frac{1}{t^2}+\frac{1}{t^4}\right)\mathrm{d}t = \frac{1}{4}\int_1^\infty\left(t^{-2}+2t^{-4}+t^{-6}\right)\mathrm{d}t \\ &= \frac{1}{4}\left(1+\frac{2}{3}+\frac{1}{5}\right) = \frac{1}{4}\cdot\frac{28}{15} = \frac{7}{15}.\end{aligned}$$

Checks

- Test the identity on a simple f: with $f(t)=1/t^3$ the right-hand side is $\frac{1}{2}\int_1^\infty(t^{-3}+t^{-5})\,\mathrm{d}t = \frac{1}{2}\left(\frac{1}{2}+\frac{1}{4}\right)=\frac{3}{8}$, and numerical quadrature of the left-hand side also gives

0.375.

- Simpson's rule on the original integrals gives 0.666667 and 0.466667, matching $\frac{2}{3}$ and $\frac{7}{15}$.
- Sanity check on size: on $[0, \frac{1}{2}\pi]$ the integrand $1/(1+\sin\theta)^3$ falls from 1 to $\frac{1}{8}$, so the integral lies between $\frac{1}{8}\cdot\frac{1}{2}\pi \approx 0.20$ and $\frac{1}{2}\pi \approx 1.57$. The value $\frac{7}{15} \approx 0.47$ is comfortably inside.

Common slips

- Trying to find $\mathrm{d}t/\mathrm{d}x$ and then express it in terms of t the hard way. Writing x as a function of t first and differentiating that is much quicker.
- In the second integral, forgetting the factor $1/(1+x^2)$ from $\mathrm{d}\theta$, or leaving $\sqrt{1+x^2}$ in the integrand rather than converting it to $\frac{1}{2}(t+1/t)$.

Beyond STEP

The substitution $t = x + \sqrt{1+x^2}$ is really $t = \mathrm{e}^u$ where $x = \sinh u$: indeed $\sinh^{-1} x = \ln\left(x+\sqrt{1+x^2}\right)$, and the formulas $x = \frac{1}{2}(t-1/t)$, $\sqrt{1+x^2} = \frac{1}{2}(t+1/t)$ are just $\sinh u$ and $\cosh u$ written with exponentials. Hyperbolic substitutions like this are a standard first-year tool for integrals involving $\sqrt{1+x^2}$. Siklos, *Advanced Problems in Mathematics*, has more STEP integrals in this spirit.

WORKED SOLUTION

STEP II 2012, Question 4

Pure • Logarithmic Series Inequalities

STEP II 2012, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q4

The key idea

Everything comes from the series for $\ln(1+x)$ and the fact that e^t is increasing. To compare $(1+\frac{1}{n})^n$ with e, take logarithms: we need to compare $n\ln(1+\frac{1}{n})$ with 1. An alternating series with shrinking terms can be bracketed in pairs to show it is less than its first term; and the combination $\ln\frac{1+x}{1-x}$ kills all the negative terms, so it is bigger than its first term. Squeezing between the two bounds gives the limit.

Solution

Part (i)

Since $n > 1$ and $k \geq 1$, we have $(k+1)n > n > 1$, and certainly $(k+1)n > k$. Multiplying by $n^k > 0$,

$$(k+1)n^{k+1} > kn^k > 0,$$

and taking reciprocals of these positive numbers reverses the inequality:

$$\frac{1}{(k+1)n^{k+1}} < \frac{1}{kn^k}.$$

Now put $x = \frac{1}{n}$ in the series; this is allowed because $n > 1$ gives $0 < x < 1$. Grouping the terms in pairs after the first,

$$\ln\left(1+\frac{1}{n}\right) = \frac{1}{n} - \left(\frac{1}{2n^2} - \frac{1}{3n^3}\right) - \left(\frac{1}{4n^4} - \frac{1}{5n^5}\right) - \cdots.$$

Each bracket has the form $\frac{1}{kn^k} - \frac{1}{(k+1)n^{k+1}}$ with k even, and so is strictly positive by the inequality just proved.

? **Ask yourself** Are we allowed to bracket an infinite series like this?

Yes, because the series converges. The partial sums with an odd number of terms form a subsequence of all the partial sums, so they tend to the same limit, $\ln(1+\frac{1}{n})$. The bracketed series is exactly the sequence of these odd partial sums.

The partial sums of the bracketed series are $\frac{1}{n}-b_1$, $\frac{1}{n}-b_1-b_2$, …, where every $b_j>0$. They decrease, so all of them are at most $\frac{1}{n}-b_1$, and therefore so is their limit:

$$\ln\left(1+\frac{1}{n}\right)\le\frac{1}{n}-b_1<\frac{1}{n},\qquad b_1=\frac{1}{2n^2}-\frac{1}{3n^3}>0.$$

(Keeping b_1 aside is what makes the final inequality strict; a limit of numbers less than $\frac{1}{n}$ could otherwise equal $\frac{1}{n}$.)

Multiplying by $n>0$ gives $n\ln(1+\frac{1}{n})<1$, that is, $\ln\left((1+\frac{1}{n})^n\right)<1$. Since the exponential function is increasing,

$$\left(1+\frac{1}{n}\right)^n<\mathrm{e}.$$

Part (ii)

Let $x=\frac{1}{2y}$. Since $y>\frac{1}{2}$, we have $0<x<1$, so both x and $-x$ lie in the range where the series is valid:

$$\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots,$$
$$\ln(1-x)=-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}-\cdots.$$

Subtracting term by term (legitimate for two convergent series), the even powers cancel and the odd powers double:

$$\ln\left(\frac{1+x}{1-x}\right)=2\left(x+\frac{x^3}{3}+\frac{x^5}{5}+\cdots\right).$$

Now

$$\frac{1+x}{1-x}=\frac{1+\frac{1}{2y}}{1-\frac{1}{2y}}=\frac{2y+1}{2y-1},$$

so, with $2x=\frac{1}{y}$,

$$\ln\left(\frac{2y+1}{2y-1}\right)=\frac{1}{y}+\frac{2}{3(2y)^3}+\frac{2}{5(2y)^5}+\cdots=\frac{1}{y}\sum_{r=0}^{\infty}\frac{1}{(2r+1)(2y)^{2r}}.$$

Every term after the first is strictly positive (as $y>0$), so the partial sums from the second onwards are all at least $\frac{1}{y}+\frac{2}{3(2y)^3}$, and hence

$$\ln\left(\frac{2y+1}{2y-1}\right)\ge\frac{1}{y}+\frac{2}{3(2y)^3}>\frac{1}{y},$$

as required.

For the deduction, choose y so that the fraction becomes $1+\frac{1}{n}$. Taking $y = n+\frac{1}{2}$ (which exceeds $\frac{1}{2}$ for every positive integer n),

$$\frac{2y+1}{2y-1} = \frac{2n+2}{2n} = 1+\frac{1}{n},$$

so

$$\ln\left(1+\frac{1}{n}\right) > \frac{1}{n+\frac{1}{2}}.$$

Multiplying by $n+\frac{1}{2} > 0$ gives $\left(n+\frac{1}{2}\right)\ln(1+\frac{1}{n}) > 1$, and exponentiating (an increasing function again),

$$\mathrm{e} < \left(1+\frac{1}{n}\right)^{n+\frac{1}{2}}.$$

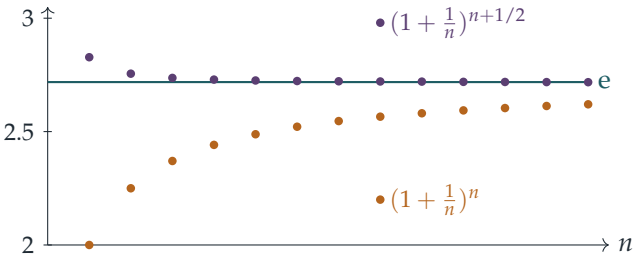
Part (iii)

Dividing the result of part (ii) by $(1+\frac{1}{n})^{1/2} > 0$ and combining it with part (i), for every integer $n \geq 2$

$$\frac{\mathrm{e}}{\sqrt{1+\frac{1}{n}}} < \left(1+\frac{1}{n}\right)^{n} < \mathrm{e}.$$

As $n \to \infty$, $1+\frac{1}{n} \to 1$, so $\sqrt{1+\frac{1}{n}} \to 1$ (the square root is continuous) and the left-hand side tends to e. The right-hand side is the constant e. By the squeeze (sandwich) theorem,

$$\left(1+\frac{1}{n}\right)^{n} \to \mathrm{e}.$$



Checks

For $n = 10$: $(1.1)^{10} \approx 2.5937$ and $(1.1)^{10.5} \approx 2.7203$, which do straddle $\mathrm{e} \approx 2.7183$. For $n = 1000$ the lower bound $\mathrm{e}/\sqrt{1.001} \approx 2.7169237$ is just below $(1.001)^{1000} \approx 2.7169239$, showing how tight the squeeze is. We also checked both strict inequalities for every n up to a few thousand, and checked numerically that the series in part (ii) sums to $\ln\frac{2y+1}{2y-1}$ for random $y > \frac{1}{2}$.

Common slips

- Putting $x = 1$ in part (i): the series is only given for $|x| < 1$, which is why $n > 1$ is needed there (part (iii) only uses large n, so nothing is lost).
- Concluding a strict inequality straight from a limit of strict inequalities. Limits only preserve $\leq$; keep one positive term aside to get $<$.
- Forgetting that exponentiating preserves the inequality only because e^t is increasing; say so.

Beyond STEP

The bounds show more than the limit: since $(1+\frac{1}{n})^{n+1/2}/(1+\frac{1}{n})^n = \sqrt{1+1/n} \approx 1 + \frac{1}{2n}$, the error $\mathrm{e} - (1+\frac{1}{n})^n$ is of size about $\mathrm{e}/(2n)$, so this sequence converges rather slowly. Rearranging convergent series, comparing partial sums and the squeeze theorem are bread and butter in a first analysis course, where you also meet the subtleties (rearranging conditionally convergent series can change the sum!). Abbott's *Understanding Analysis* treats all of this beautifully.

WORKED SOLUTION

STEP II 2012, Question 5

Pure • Sketching Reciprocal Functions

STEP II 2012, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q5

The key idea

Both curves are reciprocals, so we sketch the denominator in our heads and flip it: where the denominator is zero we get a vertical asymptote, where it is large y is near 0, and a turning point of the denominator is a turning point of the curve. In part (ii) the two quadratic factors are mirror images of each other about the midpoint of a and b, so shifting the origin to that midpoint makes g an even function of the new variable and the stationary points fall out of a quadratic in t^2.

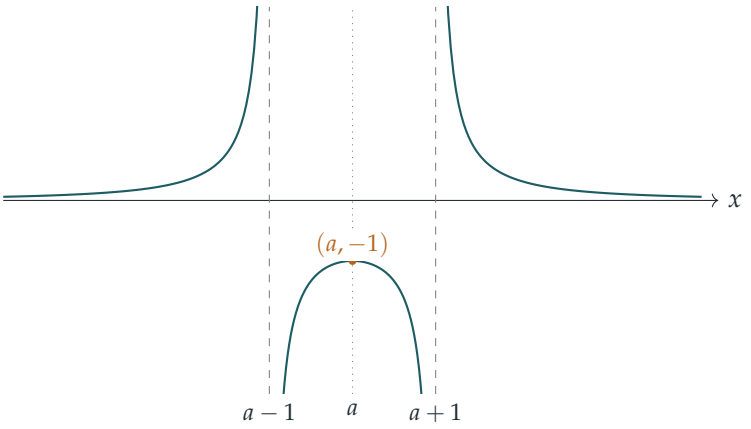
Solution

Part (i)

Write $D(x) = (x-a)^2 - 1$, so $f = 1/D$. The denominator D is a parabola, symmetric about $x = a$, with minimum value -1 at $x = a$ and zeros at $x = a \pm 1$. Reading off the reciprocal:

- The curve is symmetric about $x = a$, with vertical asymptotes $x = a \pm 1$.
- For $|x-a| < 1$ we have $-1 \leq D < 0$, so $f \leq -1$, with equality only at $x = a$. So $(a, -1)$ is a local maximum, and $f \to -\infty$ as x approaches $a \pm 1$ from inside.
- For $|x-a| > 1$ we have $D > 0$ and D increases as $|x-a|$ grows, so f is positive and decreases from $+\infty$ (next to the asymptotes) towards 0. The x-axis is a horizontal asymptote, with $f \approx 1/x^2$ for large $|x|$.

Indeed $f'(x) = -2(x-a)/D^2$ vanishes only at $x = a$. The curve crosses the y-axis at $1/(a^2-1)$ (when $a \neq \pm 1$), but where that lies depends on a, so we draw the picture relative to $x = a$.



Part (ii)

Let $p = (x-a)^2 - 1$ and $q = (x-b)^2 - 1$, so $g = 1/(pq)$. Put

$$c = \tfrac{1}{2}(a+b), \qquad d = \tfrac{1}{2}(b-a) > 0, \qquad t = x - c,$$

so that $x - a = t + d$ and $x - b = t - d$. Then, using $(t+d)(t-d) = t^2 - d^2$ and $(t+d)^2 + (t-d)^2 = 2t^2 + 2d^2$,

$$\begin{aligned} pq &= \big((t+d)^2 - 1\big)\big((t-d)^2 - 1\big) \\ &= (t^2 - d^2)^2 - (2t^2 + 2d^2) + 1. \end{aligned}$$

This depends only on t^2, so g is symmetric about $x = c$, the midpoint of a and b. Writing $u = t^2$ and $h(u) = (u - d^2)^2 - 2u - 2d^2 + 1$,

$$\frac{\mathrm{d}(pq)}{\mathrm{d}t} = 2t\,h'(u) = 2t\big(2(u - d^2) - 2\big) = 4t\,(t^2 - d^2 - 1).$$

Since $g' = -(pq)'/(pq)^2$, the stationary points of g are where $(pq)' = 0$:

$$t = 0 \qquad \text{or} \qquad t = \pm\sqrt{d^2 + 1},$$

provided these are not asymptotes. The asymptotes are at $t = \pm(d+1)$ and $t = \pm(d-1)$, and since $d < \sqrt{d^2+1} < d+1$, the outer two stationary points are always genuine. In terms of a and b the candidates are

$$x = \frac{a+b}{2} \qquad \text{and} \qquad x = \frac{a + b \pm \sqrt{(b-a)^2 + 4}}{2}.$$

The values there are $h(0) = (d^2-1)^2$ and $h(d^2+1) = 1 - 2(d^2+1) - 2d^2 + 1 = -4d^2$.

Ask yourself Which of these candidates actually survive, and are they maxima or minima?

It depends on whether $t = 0$ coincides with an asymptote, that is, whether $d = 1$; this is exactly the split between $b > a + 2$ (so $d > 1$) and $b = a + 2$ (so $d = 1$). For the nature of each point, we look at the sign of g on each interval between asymptotes and its behaviour at the ends, rather than differentiating again.

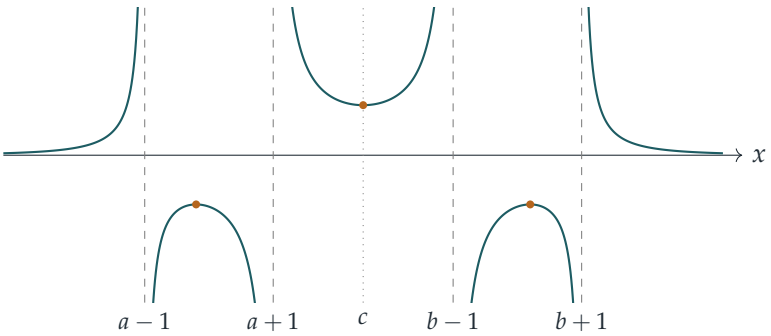
Case $b > a+2$. Now $d > 1$, and the four asymptotes, in order, are

$$t = -(d+1),\ -(d-1),\ d-1,\ d+1, \quad \text{i.e.} \quad x = a-1,\ a+1,\ b-1,\ b+1,$$

all distinct, with c strictly between $a+1$ and $b-1$.

- *Outside, $|t| > d+1$:* here $|x-a| > 1$ and $|x-b| > 1$, so $p, q > 0$ and $g > 0$. There are no stationary points, so g falls from $+\infty$ at the asymptote to 0 at infinity, behaving like $1/x^4$.
- *Middle, $|t| < d-1$, i.e. $a+1 < x < b-1$:* again $p, q > 0$, so $g > 0$, and $g \to +\infty$ at both ends. The only stationary point is $t = 0$, which must therefore be a minimum, with value
$$g(c) = \frac{1}{(d^2-1)^2} = \frac{16}{\big((b-a)^2-4\big)^2}.$$
- *The two gaps, $d-1 < |t| < d+1$:* for t in $(d-1, d+1)$, we have $|x-b| < 1$ but $x - a > 2d - 1 > 1$, so $q < 0 < p$ and $g < 0$, with $g \to -\infty$ at both ends. The only stationary point there is $t = \sqrt{d^2+1}$, which must be a maximum, with value
$$g = \frac{1}{-4d^2} = -\frac{1}{(b-a)^2}.$$
The gap $-(d+1) < t < -(d-1)$ is the mirror image.

So there are three stationary points: a local minimum at $x = \frac{1}{2}(a+b)$ and two local maxima, both with $y = -1/(b-a)^2$, at $x = \frac{1}{2}\big(a+b \pm \sqrt{(b-a)^2+4}\big)$. The sketch below has $b - a = 4.8$, with the y-direction stretched so that the turning points show.



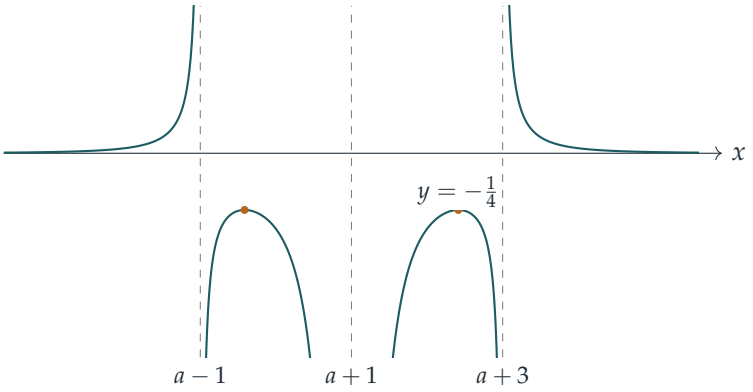
Case $b = a+2$. Now $d = 1$, so $a+1 = b-1 = c$ and the two middle asymptotes merge into one, at $x = a+1$. With $t = x - a - 1$,

$$pq = h(t^2) = t^4 - 4t^2 = t^2(t^2-4), \qquad g = \frac{1}{t^2(t^2-4)},$$

so there are three asymptotes, $x = a-1,\ a+1,\ a+3$. The candidate $t = 0$ is now an asymptote, not a stationary point, which leaves only $t = \pm\sqrt{2}$.

- For $|t| > 2$, $g > 0$ and g falls from $+\infty$ to 0, as before.
- For $0 < |t| < 2$, $g < 0$, and $g \to -\infty$ both as $t \to 0$ (where $g \approx -1/(4t^2)$, so both sides go down) and as $|t| \to 2$. The stationary point at $t = \sqrt{2}$ is therefore a maximum, with $g = 1/\big(2 \cdot (-2)\big) = -\frac{1}{4}$; likewise at $t = -\sqrt{2}$.

So the stationary points are at $x = a + 1 \pm \sqrt{2}$, both local maxima with $y = -\frac{1}{4}$. (This agrees with the general formula: with $b - a = 2$ it gives $x = \frac{1}{2}(2a + 2 \pm \sqrt{8})$, and $-1/(b-a)^2 = -\frac{1}{4}$.) The curve now never goes above the x-axis between $x = a - 1$ and $x = a + 3$.



Checks

- With $a = 0$, $b = 5$: $g(2.5) = 1/\big((6.25 - 1)^2\big) = 1/27.5625$, and the formula $16/(25 - 4)^2 = 16/441$ gives the same, about 0.0363. At $x = \frac{1}{2}(5 + \sqrt{29}) \approx 5.193$, direct evaluation gives $g \approx -0.04 = -1/25$.
- Letting $b \to a + 2$ in the first case, the minimum value $16/\big((b-a)^2 - 4\big)^2$ blows up, matching the middle branch disappearing into the merged asymptote.
- A numerical scan of the sign changes of g' for random a and b finds exactly three stationary points when $b > a + 2$ and two when $b = a + 2$, at the positions above.

Common slips

- Expanding pq as a quartic in x and trying to factorise the cubic $(pq)'$ directly: it works, but the symmetric substitution $t = x - c$ makes the factorisation immediate.
- Forgetting that in the case $b = a + 2$ the point $x = a + 1$ is an asymptote (a double zero of the denominator), not a turning point.
- Getting the sign of g wrong in the gaps: exactly one of p, q is negative there, so $g < 0$.

Beyond STEP

Sketching $1/D$ from D is a first taste of thinking about functions qualitatively: sign, zeros, symmetry and behaviour at infinity tell you almost everything before any calculus. The shift $t = x - \frac{1}{2}(a+b)$ is a use of symmetry to reduce a quartic to a quadratic in t^2, a trick that recurs throughout algebra and analysis. For the careful treatment of limits and asymptotes behind such sketches, see Abbott, *Understanding Analysis*, or for many more problems of this kind, Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2012, Question 6

Pure • Cyclic Quadrilateral Area

STEP II 2012, Question 6

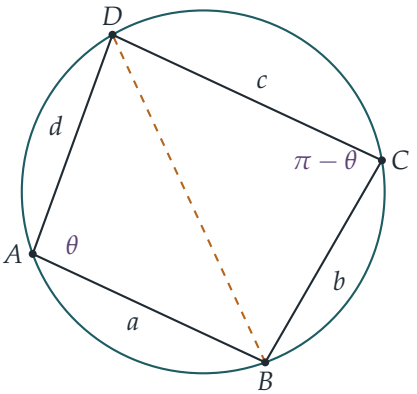
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q6

The key idea

The diagonal BD splits the quadrilateral into two triangles, and BD can be written in two ways with the cosine rule. What makes a cyclic quadrilateral special is that opposite angles add up to π, so the angle at C is $\pi - \theta$. Then $\cos C = -\cos\theta$ and $\sin C = \sin\theta$, and both the cosine rule and the area formula $\frac{1}{2}xy\sin(\text{angle})$ involve only the single angle θ. After that, $\sin^2\theta + \cos^2\theta = 1$ and some careful factorising do the rest.

Solution



Finding $\cos\theta$ and $\sin\theta$

Since $ABCD$ is cyclic, the opposite angles DAB and BCD sum to π, so $\angle BCD = \pi - \theta$. Hence

$$\cos\angle BCD = -\cos\theta, \qquad \sin\angle BCD = \sin\theta.$$

Apply the cosine rule to the diagonal BD, first in triangle ABD (sides a, d, included angle θ) and then in triangle CBD (sides b, c, included angle $\pi-\theta$):

$$BD^2 = a^2 + d^2 - 2ad\cos\theta = b^2 + c^2 + 2bc\cos\theta.$$

Collecting the $\cos\theta$ terms gives $2(ad+bc)\cos\theta = a^2+d^2-b^2-c^2$, so

$$\cos\theta = \frac{a^2+d^2-b^2-c^2}{2(ad+bc)}.$$

The diagonal BD divides the quadrilateral into these two triangles, so their areas add to Q:

$$Q = \tfrac{1}{2}ad\sin\theta + \tfrac{1}{2}bc\sin(\pi-\theta) = \tfrac{1}{2}(ad+bc)\sin\theta, \qquad \text{so} \qquad \sin\theta = \frac{2Q}{ad+bc}.$$

(We may divide by $ad+bc$ because all the side lengths are positive.)

The formula for $16Q^2$

Substitute both expressions into $\sin^2\theta + \cos^2\theta = 1$:

$$\frac{4Q^2}{(ad+bc)^2} + \frac{(a^2+d^2-b^2-c^2)^2}{4(ad+bc)^2} = 1.$$

Multiplying through by $4(ad+bc)^2$,

$$16Q^2 = 4(ad+bc)^2 - (a^2+d^2-b^2-c^2)^2,$$

as required.

Brahmagupta's formula

The right-hand side is a difference of two squares, $X^2 - Y^2$ with $X = 2(ad+bc)$ and $Y = a^2+d^2-b^2-c^2$.

? **Ask yourself** How do we turn $X-Y$ and $X+Y$ into products of linear factors?

Group the terms so that each is again a difference of two squares. Put the a,d terms together and the b,c terms together:

$$X - Y = (b^2+2bc+c^2) - (a^2-2ad+d^2) = (b+c)^2 - (a-d)^2,$$
$$X + Y = (a^2+2ad+d^2) - (b^2-2bc+c^2) = (a+d)^2 - (b-c)^2.$$

Factorising each of these again,

$$\begin{aligned} 16Q^2 &= (X-Y)(X+Y) \\ &= (b+c-a+d)(b+c+a-d)(a+d-b+c)(a+d+b-c). \end{aligned}$$

With $2s = a+b+c+d$, each bracket is $2s$ minus twice one side: for example $b+c+d-a = 2s-2a$. So the four brackets are $2(s-a)$, $2(s-d)$, $2(s-b)$ and $2(s-c)$, and

$$16Q^2 = 16(s-a)(s-b)(s-c)(s-d), \qquad \text{that is,} \qquad Q^2 = (s-a)(s-b)(s-c)(s-d).$$

The area of a triangle

Every triangle is cyclic, since its three vertices lie on its circumcircle. Now let the side DA shrink to nothing, with D sliding along the circle into A. Then $ABCD$ becomes triangle ABC with sides a, b, c, and $d = 0$. Both sides of the formula depend continuously on D, so it still holds in the limit. With $s = \frac{1}{2}(a+b+c)$, the area Δ of a triangle satisfies

$$\Delta^2 = s(s-a)(s-b)(s-c), \qquad \text{so} \qquad \Delta = \sqrt{s(s-a)(s-b)(s-c)}.$$

This is Heron's formula.

Checks

- A unit square is cyclic with $s = 2$, and $(2-1)^4 = 1$ is its area. For a 3–4–5 triangle, $s = 6$ and $\sqrt{6 \cdot 3 \cdot 2 \cdot 1} = 6 = \frac{1}{2} \cdot 3 \cdot 4$.
- For a rectangle, $a = c$ and $b = d$, so $\cos\theta = 0$, which is right because every angle of a rectangle is $\frac{\pi}{2}$.
- We placed 20 000 random cyclic quadrilaterals on circles and compared the formulas for $\cos\theta$, $\sin\theta$ and Q^2 with the actual angle and the shoelace area. They agreed to within 3×10^{-12}. Heron's formula matched 1000 random triangles.

Common slips

- Forgetting that $\cos(\pi - \theta) = -\cos\theta$. That turns the $+2bc\cos\theta$ into a minus and spoils everything that follows.
- Trying to expand $4(ad+bc)^2 - (a^2+d^2-b^2-c^2)^2$ in full. Factorising it as a difference of squares, twice, is far quicker and much less error-prone.

Beyond STEP

For a general (not necessarily cyclic) quadrilateral, Bretschneider's formula gives the area. It subtracts a term $abcd\cos^2\frac{\alpha+\gamma}{2}$, where α and γ are a pair of opposite angles, so for fixed side lengths the cyclic quadrilateral has the largest possible area. The factorised form of Heron's formula also shows at once that the area vanishes exactly when the triangle inequality becomes an equality. For more geometry in this spirit, see Coxeter and Greitzer, *Geometry Revisited*.

WORKED SOLUTION

STEP II 2012, Question 7

Pure • Vectors and Circles

STEP II 2012, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

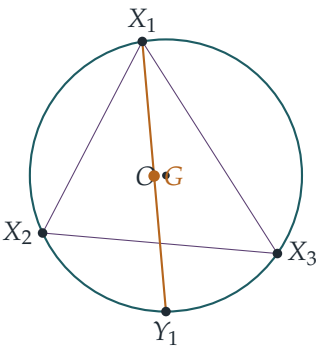
step.maths.org/questions/12-s2-q7

The key idea

The point Y_1 lies on the line through X_1 and G, so its position vector is a simple combination of $\mathbf{x}_1$ and $\mathbf{g}$, with one unknown scalar. The only other fact we know about Y_1 is that it is on the unit circle, so $|\overrightarrow{OY_1}|^2 = 1$ gives an equation for λ_1. It is a quadratic, but one root is obvious in advance: the line meets the circle at X_1 itself, so that root can be factored out.

Solution

Write $\mathbf{g} = \frac{1}{3}(\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3)$ and $\lambda = \lambda_1$. Since each $\mathbf{x}_i$ is a unit vector, $\mathbf{x}_i \cdot \mathbf{x}_i = 1$.



The position vector of Y_1

From $\overrightarrow{GY_1} = -\lambda\overrightarrow{GX_1}$ we get $\mathbf{y}_1 - \mathbf{g} = -\lambda(\mathbf{x}_1 - \mathbf{g})$, so

$$\begin{aligned}\overrightarrow{OY_1} = \mathbf{y}_1 &= (1+\lambda)\mathbf{g} - \lambda\mathbf{x}_1 \\ &= \tfrac{1}{3}(1+\lambda)(\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3) - \lambda\mathbf{x}_1 \\ &= \tfrac{1}{3}\big((1-2\lambda)\mathbf{x}_1 + (1+\lambda)(\mathbf{x}_2 + \mathbf{x}_3)\big),\end{aligned}$$

since $\frac{1}{3}(1+\lambda) - \lambda = \frac{1}{3}(1-2\lambda)$. This is the required expression.

Finding λ_1

Because Y_1 is on the circle, $|\mathbf{y}_1|^2 = 1$, so $|3\mathbf{y}_1|^2 = 9$. We need

$$|\mathbf{x}_2 + \mathbf{x}_3|^2 = 1 + 2\alpha + 1 = 2 + 2\alpha, \qquad \mathbf{x}_1 \cdot (\mathbf{x}_2 + \mathbf{x}_3) = \beta + \gamma.$$

Expanding $|3\mathbf{y}_1|^2$ as a dot product with itself therefore gives

$$(1-2\lambda)^2 + (1+\lambda)^2(2+2\alpha) + 2(1-2\lambda)(1+\lambda)(\beta+\gamma) = 9.$$

Collecting powers of λ and dividing by 2:

$$(3 + \alpha - 2\beta - 2\gamma)\lambda^2 + (2\alpha - \beta - \gamma)\lambda - (3 - \alpha - \beta - \gamma) = 0.$$

To keep this tidy, write

$$A = 3 + \alpha - 2\beta - 2\gamma, \qquad B = 3 - \alpha - \beta - \gamma,$$

and notice that the middle coefficient is exactly $A - B$. So the equation is $A\lambda^2 + (A - B)\lambda - B = 0$, which factorises:

$$(\lambda + 1)(A\lambda - B) = 0.$$

Ask yourself Why should $\lambda = -1$ be a root, and why can we discard it?

Putting $\lambda = -1$ into $\mathbf{y}_1 = (1+\lambda)\mathbf{g} - \lambda\mathbf{x}_1$ gives $\mathbf{y}_1 = \mathbf{x}_1$. Every point of the line through X_1 and G has the form $(1+\lambda)\mathbf{g} - \lambda\mathbf{x}_1$, and this line meets the circle at X_1 as well as at Y_1, so the quadratic was always going to have $\lambda = -1$ as a root. We are told $\lambda_1 > 0$, so $\lambda_1 \neq -1$ and hence $A\lambda_1 = B$.

It remains to check we may divide by A. If $A = 0$ then $A\lambda_1 = B$ forces $B = 0$. But

$$|\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3|^2 = 3 + 2(\alpha + \beta + \gamma), \qquad \text{so} \qquad B = \tfrac{1}{2}\big(9 - |\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3|^2\big) = \tfrac{9}{2}\big(1 - |\mathbf{g}|^2\big).$$

By the triangle inequality $|\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3| \leq 3$, with equality only if the three unit vectors are equal; the points are distinct, so $B > 0$. Hence $A \neq 0$ and

$$\lambda_1 = \frac{B}{A} = \frac{3 - \alpha - \beta - \gamma}{3 + \alpha - 2\beta - 2\gamma},$$

as required.

The final part

Since $\overrightarrow{GY_1} = -\lambda_1 \overrightarrow{GX_1}$ with $\lambda_1 > 0$, the lengths satisfy $GY_1 = \lambda_1 GX_1$, so

$$\frac{GX_1}{GY_1} = \frac{1}{\lambda_1} = \frac{3+\alpha-2\beta-2\gamma}{3-\alpha-\beta-\gamma}.$$

The same argument with the labels cycled ($1 \to 2 \to 3 \to 1$, which sends $\alpha \to \beta \to \gamma \to \alpha$) gives the other two ratios. The denominator $B = 3-\alpha-\beta-\gamma$ is symmetric, so it is the same for all three, while the numerators are

$$3+\alpha-2\beta-2\gamma, \qquad 3+\beta-2\gamma-2\alpha, \qquad 3+\gamma-2\alpha-2\beta.$$

These add up to $9+(1-2-2)(\alpha+\beta+\gamma) = 9-3(\alpha+\beta+\gamma) = 3B$. Therefore

$$\frac{GX_1}{GY_1} + \frac{GX_2}{GY_2} + \frac{GX_3}{GY_3} = \frac{3B}{B} = 3.$$

Checks

- Equilateral triangle: then $G = O$, each chord through G is a diameter, and every ratio is 1. In the formula, $\alpha = \beta = \gamma = -\frac{1}{2}$, giving $\lambda_1 = \frac{9/2}{9/2} = 1$.
- There is a geometric reading: $A = \frac{1}{2}|\mathbf{x}_2 + \mathbf{x}_3 - 2\mathbf{x}_1|^2 = \frac{9}{2} GX_1^2$ and $B = \frac{9}{2}(1 - OG^2)$, so $\lambda_1 = B/A$ says $GX_1 \cdot GY_1 = 1 - OG^2$, the intersecting chords (power of a point) theorem.
- Numerically: with X_1, X_2, X_3 at angles 0, 1 and 2.5 radians, the three ratios GX_i/GY_i come out as about 1.13, 0.31 and 1.57, and their sum is 3 to machine precision. We tested 20 000 random triangles the same way.

Common slips

- Forgetting that $|\mathbf{x}_2 + \mathbf{x}_3|^2$ contains the cross term 2α, not just $1+1$.
- Grinding through the quadratic formula instead of spotting the root $\lambda = -1$.
- Dividing by $3+\alpha-2\beta-2\gamma$ without saying why it is non-zero.

Beyond STEP

The identity $GX_1 \cdot GY_1 = 1 - OG^2$ hiding in this question is the *power of a point* with respect to a circle, and the dot-product calculation is the vector proof of it. Writing geometry in terms of inner products, where lengths and angles all come from $\mathbf{u} \cdot \mathbf{v}$, is the starting point for inner-product spaces in a first linear algebra course. Strang's *Introduction to Linear Algebra* develops this viewpoint.

WORKED SOLUTION

STEP II 2012, Question 8

Pure • Non-Linear Recurrence Relations

STEP II 2012, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q8

The key idea

The recurrence $u_{n+1}u_{n-1} = u_n^2 - q^2$ is non-linear, but it holds for *every* n, so writing it down for two consecutive values of n and subtracting kills the awkward constant q^2. What survives says that the ratio $(u_{n+1} + u_{n-1})/u_n$ never changes, and that turns the sequence into a *linear* recurrence with constant coefficient p. The opening inequality is exactly the statement $p > 2$, which is what drives the growth.

Solution

The inequality

Putting everything over the common denominator $\alpha\beta$,

$$\frac{\alpha^2 + \beta^2 - q^2}{\alpha\beta} - 2 = \frac{\alpha^2 - 2\alpha\beta + \beta^2 - q^2}{\alpha\beta} = \frac{(\beta - \alpha)^2 - q^2}{\alpha\beta}.$$

Since $\beta - \alpha > q > 0$, squaring (both sides positive) gives $(\beta - \alpha)^2 > q^2$, so the numerator is positive. The denominator $\alpha\beta$ is positive because $\alpha, \beta > 0$. Hence the expression is positive, as required.

The first identity

Multiply the recurrence through by u_{n-1}: for every $n \geq 1$,

$$u_{n+1}u_{n-1} = u_n^2 - q^2.$$

Replacing n by $n + 1$ (still allowed, since $n + 1 \geq 1$),

$$u_{n+2}u_n = u_{n+1}^2 - q^2.$$

Subtracting the first equation from the second removes q^2:

$$u_n u_{n+2} - u_{n-1}u_{n+1} = u_{n+1}^2 - u_n^2,$$

and rearranging,

$$u_n^2 + u_n u_{n+2} = u_{n+1}^2 + u_{n-1}u_{n+1}, \qquad \text{that is,} \qquad u_n(u_n + u_{n+2}) = u_{n+1}(u_{n-1} + u_{n+1})$$

for all $n \geq 1$, which is the required result.

The linear recurrence

Ask yourself The identity just proved has four different terms in it. How does it give a recurrence with a single constant p?

Divide it by $u_n u_{n+1}$, which is allowed because no term of the sequence is zero:

$$\frac{u_{n+2} + u_n}{u_{n+1}} = \frac{u_{n+1} + u_{n-1}}{u_n} \qquad (n \geq 1).$$

So the quantity $v_n = (u_{n+1} + u_{n-1})/u_n$ satisfies $v_{n+1} = v_n$ for all $n \geq 1$: it is constant. Call its common value p; then $u_{n+1} + u_{n-1} = pu_n$, i.e.

$$u_{n+1} - pu_n + u_{n-1} = 0 \qquad (n \geq 1).$$

To find p, use $n = 1$. From the recurrence, $u_2 = (\beta^2 - q^2)/\alpha$, so

$$p = \frac{u_2 + u_0}{u_1} = \frac{1}{\beta}\left(\frac{\beta^2 - q^2}{\alpha} + \alpha\right) = \frac{\alpha^2 + \beta^2 - q^2}{\alpha\beta}.$$

This is the expression from the first part, so when $\beta - \alpha > q$ we already know that $p > 2$.

The sequence is increasing when $\beta > \alpha + q$

Let $d_n = u_{n+1} - u_n$. Using the linear recurrence in the form $u_{n+1} = pu_n - u_{n-1}$,

$$d_n - d_{n-1} = u_{n+1} - 2u_n + u_{n-1} = (p-2)u_n.$$

Ask yourself We know $p - 2 > 0$, but why should $(p-2)u_n$ be positive? We don't yet know that $u_n > 0$.

Positivity and growth have to be proved together, by induction. Claim: for every $n \geq 1$, $u_n > 0$ and $d_{n-1} > 0$.

For $n = 1$: $u_1 = \beta > 0$ and $d_0 = \beta - \alpha > q > 0$.

Suppose $u_n > 0$ and $d_{n-1} > 0$ for some $n \geq 1$. Then, since $p > 2$,

$$d_n = d_{n-1} + (p-2)u_n > 0, \qquad u_{n+1} = u_n + d_n > 0,$$

which is the claim for $n + 1$. By induction, $d_n = u_{n+1} - u_n > 0$ for every $n \geq 0$, so the sequence is strictly increasing. (In fact the differences themselves increase, so the growth is faster than linear. Since every term is positive, the condition that no term is zero holds automatically in this case.)

The case $\beta = \alpha + q$

Now $(\beta - \alpha)^2 - q^2 = 0$, so $p = 2$ exactly and $d_n - d_{n-1} = 0$. The differences are all equal to $d_0 = \beta - \alpha = q$, so the sequence is the arithmetic progression

$$u_n = \alpha + nq.$$

It is still strictly increasing, since $q > 0$, but now only linearly. Direct check: $(\alpha + (n+1)q)(\alpha + (n-1)q) = (\alpha + nq)^2 - q^2$, so this sequence does satisfy the original recurrence.

Checks

- Take $\alpha = 1$, $\beta = 4$, $q = 2$ (so $\beta > \alpha + q$). The recurrence gives $1, 4, 12, 35, \frac{407}{4}, \ldots$, and $p = \frac{1+16-4}{4} = \frac{13}{4}$; indeed $\frac{13}{4} \cdot 4 - 1 = 12$ and $\frac{13}{4} \cdot 12 - 4 = 35$.
- With $\alpha = 1$, $q = 2$, $\beta = 3$ the terms are $1, 3, 5, 7, \ldots$, the arithmetic progression of the last part.
- We checked both identities in exact rational arithmetic for hundreds of random α, β, q, and checked that the sequence increases whenever $\beta > \alpha + q$.

Common slips

- Concluding $d_n > d_{n-1}$ from $p > 2$ without first knowing that $u_n > 0$: the two facts must go into the same induction.
- Dividing by $u_n u_{n+1}$ without remarking that it is non-zero.
- Forgetting that the subtraction step uses the recurrence at index $n + 1$, which needs $n + 1 \geq 1$.

Beyond STEP

A non-linear recurrence that secretly satisfies a linear one is said to have a hidden *invariant*: here $(u_{n+1} + u_{n-1})/u_n$, or equivalently $u_n^2 - u_{n+1}u_{n-1} = q^2$, which is a discrete analogue of a conserved quantity in mechanics. The linear recurrence $u_{n+1} - pu_n + u_{n-1} = 0$ is solved by $u_n = A\lambda^n + B\lambda^{-n}$ where $\lambda + \lambda^{-1} = p$; this is real and exponentially growing when $p > 2$, and becomes $A + Bn$ in the repeated-root case $p = 2$, exactly the dichotomy above. Siklos, *Advanced Problems in Mathematics*, has more STEP-style recurrences, and the parallel with differential equations is developed in Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP II 2012, Question 9

Mechanics • Projectile Motion

STEP II 2012, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

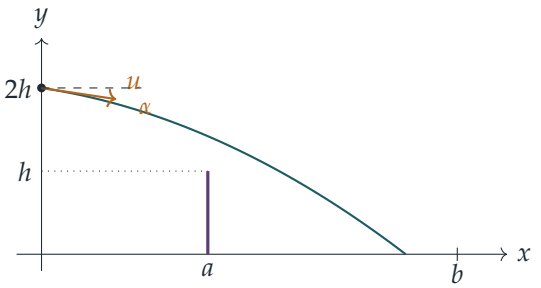
step.maths.org/questions/12-s2-q9

The key idea

Write the height of the ball as a function of its horizontal distance x from the point of projection: the trajectory equation. "Passes over the net" is then one inequality at $x = a$, and "lands before b" is an inequality on the time of flight. The final part works because both conditions turn out to be bounds on the same quantity $1/u^2$, one from above and one from below, so the lower bound must be smaller than the upper one, and u drops out.

Solution

We model the ball as a particle moving freely under gravity, ignoring air resistance and spin, with $0 \leq \alpha < \frac{1}{2}\pi$. Take the origin on the ground directly below the point of projection, x horizontal towards the net and y vertically upwards.



Passing over the net

The initial velocity has components $u\cos\alpha$ horizontally and $u\sin\alpha$ *downwards*. After time t,

$$x = ut\cos\alpha, \qquad y = 2h - ut\sin\alpha - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(u\cos\alpha)$ gives the trajectory

$$y = 2h - x\tan\alpha - \frac{gx^2\sec^2\alpha}{2u^2}.$$

The horizontal velocity is constant and positive, so the ball reaches $x = a$ exactly once, and it clears the net when $y > h$ there:

$$2h - a\tan\alpha - \frac{ga^2\sec^2\alpha}{2u^2} > h \iff \frac{ga^2\sec^2\alpha}{2u^2} < h - a\tan\alpha.$$

Multiplying by the positive quantity $2/(ga^2\sec^2\alpha)$,

$$\frac{1}{u^2} < \frac{2(h - a\tan\alpha)}{ga^2\sec^2\alpha},$$

as required.

Landing before b

The ball lands when it has fallen $2h$, that is when $\frac{1}{2}gt^2 + ut\sin\alpha - 2h = 0$. This quadratic in t has roots of opposite sign (their product is $-4h/g < 0$), and the landing time is the positive one:

$$T = \frac{-u\sin\alpha + \sqrt{u^2\sin^2\alpha + 4gh}}{g}.$$

It lands before horizontal distance b exactly when $uT\cos\alpha < b$, i.e. $T < \dfrac{b}{u\cos\alpha}$ (as $u\cos\alpha > 0$). Multiplying by g and adding $u\sin\alpha$,

$$\sqrt{u^2\sin^2\alpha + 4gh} < \frac{bg}{u\cos\alpha} + u\sin\alpha,$$

which is the second result.

The final inequality

Ask yourself How can two conditions involving u produce a condition with no u in it?

The first result is an upper bound on $1/u^2$. If we can turn the second into a *lower* bound on $1/u^2$, then the lower bound must be less than the upper bound, whatever u is.

Both sides of the second inequality are positive (the right-hand side because $\sin\alpha \geq 0$ and $\cos\alpha > 0$), so squaring preserves it:

$$u^2\sin^2\alpha + 4gh < \frac{b^2g^2}{u^2\cos^2\alpha} + 2bg\tan\alpha + u^2\sin^2\alpha.$$

The $u^2\sin^2\alpha$ terms cancel, leaving $2g(2h - b\tan\alpha) < b^2g^2\sec^2\alpha/u^2$, that is

$$\frac{2(2h - b\tan\alpha)}{gb^2\sec^2\alpha} < \frac{1}{u^2}.$$

Combining with the first result,

$$\frac{2(2h - b\tan\alpha)}{gb^2\sec^2\alpha} < \frac{1}{u^2} < \frac{2(h - a\tan\alpha)}{ga^2\sec^2\alpha}.$$

Cancelling the positive factor $2/(g\sec^2\alpha)$ and multiplying through by $a^2b^2 > 0$,

$$\begin{aligned} & a^2(2h - b\tan\alpha) < b^2(h - a\tan\alpha) \\ \iff \quad & (ab^2 - a^2b)\tan\alpha < h(b^2 - 2a^2) \\ \iff \quad & ab(b - a)\tan\alpha < h(b^2 - 2a^2). \end{aligned}$$

Ask yourself Can we divide by $b - a$?

Yes, and the sign matters. The ball is still above height $h > 0$ at $x = a$, so it lands at some distance greater than a; and it lands before b. Hence $b > a$, so $ab(b - a) > 0$ and dividing keeps the inequality the right way round:

$$\tan\alpha < \frac{h(b^2 - 2a^2)}{ab(b - a)}.$$

Checks

- Units: $a\tan\alpha$ is a length like h, and ga^2/u^2 is a length too, so each bound on $1/u^2$ has units $\mathrm{s}^2\,\mathrm{m}^{-2}$, as it should.
- With $\alpha = 0$ the final result says $b^2 > 2a^2$. Indeed, a ball fired horizontally has fallen less than h by $x = a$; since the drop is proportional to x^2, it has fallen less than $2h$ by $x = \sqrt{2}\,a$, so it lands beyond $\sqrt{2}\,a$ and $b > \sqrt{2}\,a$.
- Numerically, with $g = 9.8$, $u = 12$, $\alpha = 0.2$, $h = 1.5$, stepping the motion gives a landing distance of 6.7759, agreeing with $uT\cos\alpha$. In 20 000 random cases the first two inequalities matched "clears the net" and "lands before b" exactly, and the final inequality held whenever both did.

Common slips

- Taking the vertical component of the initial velocity as $+u\sin\alpha$: the ball is projected *below* the horizontal.
- Squaring an inequality without checking that both sides are positive.
- Dividing by $b - a$ without saying why it is positive.

Beyond STEP

The final part is a small instance of eliminating a parameter between inequalities: a system $L < 1/u^2 < U$, with L and U free of u, has a solution exactly when $L < U$.

The same idea, done systematically for many linear inequalities, is Fourier–Motzkin elimination, one of the foundations of linear programming. For the mechanics, including projectiles with air resistance where the trajectory is no longer a parabola, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 2012, Question 10

Mechanics • Friction and Moments

STEP II 2012, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q10

The key idea

The rod is a chord of the circular cross-section, so every normal reaction points straight at the centre O and the angle ϕ between the rod and each radius is fixed by the geometry: $\cos\phi = a/r$. The only way the rod can move is by turning about O, so both ends slip together, and we know which way friction acts at each end. Taking moments about the *midpoint* kills the weight and gives the first relation at once; moments about O and one resolution then give $\tan\theta$.

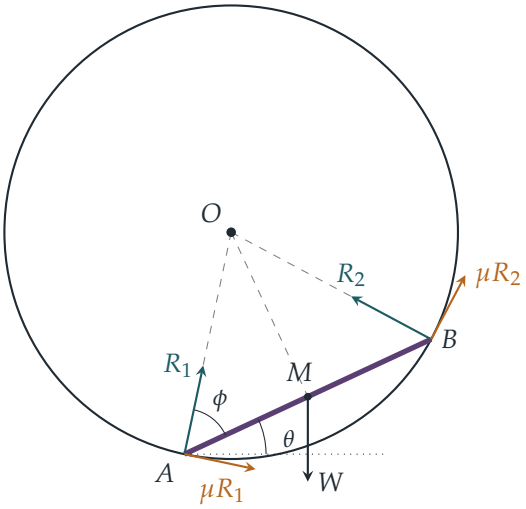
Solution

Setting up

Model the rod as a uniform rigid rod of weight W, touching the rough cylinder only at its ends A (lower) and B (higher). Let O be the centre of the circular cross-section and M the midpoint of AB. Since $OA = OB = r$, the triangle OAB is isosceles, so OM is perpendicular to AB and

$$\cos\phi = \frac{a}{r}, \qquad \sin\phi = \frac{\sqrt{r^2 - a^2}}{r}, \qquad OM = \sqrt{r^2 - a^2} = r\sin\phi,$$

with $0 < \phi < \frac{\pi}{2}$. The angle between the rod and the radius is ϕ at *both* ends.



Ask yourself Which way does friction act at each end?

The ends stay on the circle and AB stays the same length, so OM keeps its length and stays perpendicular to the rod: the only possible motion is a rotation about O. Since $\theta \neq 0$, M is not directly below O, and the weight has a moment about O turning the rod towards the horizontal (in the diagram, clockwise: A would move up the wall and B down it). The reactions pass through O, so only friction can resist this. Both ends would move together, so on the point of slipping friction is limiting at both: μR_1 and μR_2 act along the tangents at A and B, both in the anticlockwise sense about O, as shown.

The relation between R_1 and R_2

Take moments about M, which removes the weight. The line of action of R_1 is AO, which makes angle ϕ with AM, so its perpendicular distance from M is $a\sin\phi$. The friction at A acts along the tangent, which is perpendicular to OA, so its distance from M is $a\cos\phi$. The same holds at B. In the diagram R_1 turns the rod clockwise about M, while R_2 and both friction forces turn it anticlockwise, so

$$R_1\, a\sin\phi = R_2\, a\sin\phi + \mu R_1\, a\cos\phi + \mu R_2\, a\cos\phi.$$

Dividing by $a\cos\phi$ and rearranging,

$$\mu(R_1 + R_2) = (R_1 - R_2)\tan\phi,$$

as required. In particular $R_1 > R_2$: the lower end presses harder.

Finding $\tan\theta$

Moments about O. The reactions pass through O, and each friction force has moment arm r. The weight acts at M, and since OM is perpendicular to the rod it makes angle

θ with the vertical, so the horizontal distance of M from O is $OM\sin\theta = r\sin\phi\sin\theta$. Hence

$$\mu(R_1+R_2)\,r = Wr\sin\phi\sin\theta \quad\Longrightarrow\quad \mu(R_1+R_2) = W\sin\phi\sin\theta. \tag{1}$$

Resolving perpendicular to the rod (towards the side containing O). Each reaction makes angle ϕ with the rod, so contributes $R_i\sin\phi$. Each friction force makes angle ϕ with this perpendicular direction; at A it points away from O's side of the rod and at B towards it. The weight contributes $-W\cos\theta$. So

$$(R_1+R_2)\sin\phi - \mu(R_1-R_2)\cos\phi = W\cos\theta. \tag{2}$$

Write $S = R_1+R_2$. The first result gives $R_1-R_2 = \mu S\cot\phi$, so (2) becomes

$$S\,\frac{\sin^2\phi - \mu^2\cos^2\phi}{\sin\phi} = W\cos\theta,$$

while (1) says $\dfrac{\mu S}{\sin\phi} = W\sin\theta$. Dividing (the rod is inclined, not vertical, so $0<\theta<\frac{\pi}{2}$ and $\cos\theta\neq 0$),

$$\tan\theta = \frac{\mu}{\sin^2\phi - \mu^2\cos^2\phi}.$$

Multiply top and bottom by r^2 and use $r^2\sin^2\phi = r^2-a^2$, $r^2\cos^2\phi = a^2$:

$$\tan\theta = \frac{\mu r^2}{r^2 - a^2 - \mu^2 a^2} = \frac{\mu r^2}{r^2 - a^2(1+\mu^2)},$$

as required.

Deducing $\lambda < \phi$

> **Ask yourself** Where does an inequality come from, when all we have is an equation?

From the signs. The rod is inclined at θ to the horizontal with $\theta\neq 0$, so $0<\theta<\frac{\pi}{2}$ and $\tan\theta>0$. As $\mu>0$, the numerator μr^2 is positive, so the denominator must be positive too:

$$r^2\sin^2\phi - \mu^2 r^2\cos^2\phi > 0 \quad\Longleftrightarrow\quad \tan^2\phi > \mu^2 = \tan^2\lambda.$$

Both ϕ and λ lie in $(0,\frac{\pi}{2})$, where tan is positive and increasing, so $\tan\phi > \tan\lambda$ gives $\lambda<\phi$.

Checks

- Smooth limit: as $\mu\to 0$, $\tan\theta\to 0$, so a smooth rod can only rest horizontally, with M directly below O. Correct.

- Numbers: with $r = 5$, $a = 3$, $\mu = \frac{1}{2}$ we get $\tan\phi = \frac{4}{3} > \frac{1}{2}$ and $\tan\theta = \frac{12.5}{25-11.25} = \frac{10}{11}$. Solving the two force equations for R_1, R_2 in vector form at this angle (and at thousands of random r, a, μ) makes the moment about O vanish and satisfies $\mu(R_1 + R_2) = (R_1 - R_2)\tan\phi$ to within 10^{-12}.
- Physical sense of $\lambda < \phi$: λ is the angle of friction. If $\lambda \geq \phi$, friction is strong enough to hold the rod at every inclination, so it is never on the point of slipping and no such θ exists.

Common slips

- Taking the reactions perpendicular to the *rod* rather than along the radii: the contact is with the cylinder, so the normal is the cylinder's normal.
- Putting the two friction forces in opposite directions along the rod. Both ends move in the same rotational sense about O, so friction is anticlockwise about O at both.
- Resolving along and perpendicular to the rod for the first result: both equations contain W. Moments about M avoid it.

Beyond STEP

The deduction $\lambda < \phi$ is the friction-cone idea: the total reaction at a rough contact must lie within angle λ of the normal. Here the rod is in contact at two points, and the problem is about whether the two friction cones leave room for the three forces to meet at a point. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body statics more fully, including problems where the direction of impending motion has to be worked out.

WORKED SOLUTION

STEP II 2012, Question 11

Mechanics • Inelastic Collisions

STEP II 2012, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

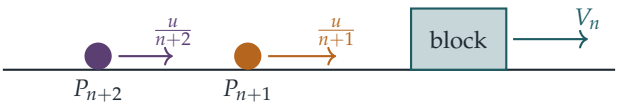
step.maths.org/questions/12-s2-q11

The key idea

Look at the momentum each particle carries: mass im times speed u/i is mu, the same for every particle. So after n particles have been absorbed the block's momentum is simply nmu, and conservation of momentum gives its speed at once. Whether the next particle ever arrives is then a comparison of two speeds, and the energy loss is "kinetic energy in minus kinetic energy out".

Solution

We model the block and the particles as particles moving along one straight line on a smooth surface, so no horizontal external force acts and momentum along the line is conserved in every collision. Each collision is perfectly inelastic, since the particle embeds in the block.



The speed after the nth collision

Particle P_i has momentum $im \cdot \frac{u}{i} = mu$, independent of i.

Suppose the nth particle collides with the block. Then particles $P_1, \ldots, P_{n-1}$ must have collided before it: the particles are fired in order and each is slower than the one before (u/i decreases), so a later particle can never overtake an earlier one, and no particle can pass the block without hitting it. So just after the nth collision the block contains $P_1, \ldots, P_n$, and its mass is

$$km + m(1 + 2 + \cdots + n) = m\left(k + \tfrac{1}{2}n(n+1)\right).$$

The block started at rest, and each absorbed particle brought momentum mu, so by

conservation of momentum the total momentum is now nmu. Hence the speed V_n of the block after the nth collision satisfies

$$m\Big(k+\tfrac{1}{2}n(n+1)\Big)V_n = nmu \quad\Longrightarrow\quad V_n = \frac{2nu}{2k+n(n+1)},$$

as required.

The number of collisions when $2k = N(N+1)$

After n collisions the block moves at the constant speed V_n (the surface is smooth). Particle P_{n+1}, which is behind the block, catches it if and only if it is strictly faster:

$$\frac{u}{n+1} > \frac{2nu}{2k+n(n+1)} \iff 2k+n(n+1) > 2n(n+1) \iff 2k > n(n+1),$$

where we multiplied up by the positive quantities $n+1$ and $2k+n(n+1)$. (For $n=0$ this reads $2k>0$, which is true: P_1 certainly hits the block at rest.)

> **?** **Ask yourself** Once one particle misses, could a later one still hit the block?

No. If P_{n+1} fails to catch the block, the block keeps the speed $V_n \geq \frac{u}{n+1}$ for ever, and every later particle is slower still, since $\frac{u}{i} < \frac{u}{n+1}$ for $i > n+1$. So the collisions are exactly $P_1, P_2, \ldots$ up to the first particle that fails.

Now take $2k = N(N+1)$. The condition for P_{n+1} to hit becomes $n(n+1) < N(N+1)$, that is $n < N$, because $n(n+1)$ is strictly increasing in $n \geq 0$. So $P_1, \ldots, P_N$ all hit the block, and P_{N+1} does not: with $n = N$ we have equality, and indeed

$$V_N = \frac{2Nu}{N(N+1)+N(N+1)} = \frac{u}{N+1},$$

exactly the speed of P_{N+1}, which therefore stays a fixed distance behind the block. Hence there are exactly N collisions.

The kinetic energy lost

Only $P_1, \ldots, P_N$ take part in collisions, so the total loss is the kinetic energy of those particles at firing (the block starts at rest) minus the final kinetic energy of the block.

Initially,

$$\sum_{i=1}^{N} \tfrac{1}{2}(im)\Big(\frac{u}{i}\Big)^2 = \tfrac{1}{2}mu^2 \sum_{i=1}^{N} \frac{1}{i}.$$

Finally the block has mass $m\big(k+\frac{1}{2}N(N+1)\big) = mN(N+1)$ and speed $\frac{u}{N+1}$, so its kinetic energy is

$$\tfrac{1}{2}\,mN(N+1)\,\frac{u^2}{(N+1)^2} = \tfrac{1}{2}mu^2\,\frac{N}{N+1} = \tfrac{1}{2}mu^2\Big(1-\frac{1}{N+1}\Big).$$

Subtracting,

$$\begin{aligned}\text{KE lost} &= \tfrac{1}{2}mu^2\left(\sum_{i=1}^{N}\frac{1}{i} - 1 + \frac{1}{N+1}\right)\\ &= \tfrac{1}{2}mu^2\left(\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{N}+\frac{1}{N+1}\right) = \tfrac{1}{2}mu^2\sum_{n=2}^{N+1}\frac{1}{n},\end{aligned}$$

which is the required result.

Checks

- $N = 1$ ($k = 1$): one particle of mass m at speed u hits a block of mass m, giving speed $\frac{1}{2}u$ and loss $\frac{1}{2}mu^2 - \frac{1}{4}mu^2 = \frac{1}{4}mu^2$, matching $\frac{1}{2}mu^2 \cdot \frac{1}{2}$. Then P_2 also moves at $\frac{1}{2}u$ and never arrives.
- $N = 2$ ($k = 3$): the formula gives loss $\frac{1}{2}mu^2(\frac{1}{2}+\frac{1}{3}) = \frac{5}{12}mu^2 \approx 0.4167\,mu^2$. An exact step-by-step simulation of the motion (positions and collision times in fractions) gave the same value, exactly two collisions, and speeds V_n matching the formula; it also confirmed N collisions and the loss formula for every N up to 11.
- Units: the loss is a mass times a speed squared, as an energy should be.

Common slips

- Using only the nth particle's momentum and forgetting that the block already contains $P_1, \ldots, P_{n-1}$ (or forgetting the block's own mass km).
- Counting $N + 1$ collisions: P_{N+1} has exactly the block's speed, so it never makes contact.
- Including the kinetic energy of $P_{N+1}, P_{N+2}, \ldots$ in the loss; they never collide, so they lose nothing.

Beyond STEP

The block is a body whose mass grows as it sweeps up material, a cousin of the rocket and the raindrop falling through a cloud. For such variable-mass systems one applies Newton's second law in the form "rate of change of momentum equals force" to the whole system, exactly as we did collision by collision. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats variable-mass systems and rocket motion carefully.

WORKED SOLUTION

STEP II 2012, Question 12

Probability and Statistics • Conditional Probability

STEP II 2012, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q12

The key idea

Pressing a switch *toggles* it, so the pool's hallway switch ends up on either because it started on and was left alone, or because it started off and was pressed. The kitchen half is a red herring in disguise: a switch that is on with probability $\frac{1}{2}$ stays on with probability $\frac{1}{2}$ whether or not it is toggled. Once part (i) gives a single conditional probability, the seven Mondays are a binomial experiment, and "most likely value is 3" becomes a pair of inequalities between neighbouring binomial probabilities.

Solution

Part (i)

Define the events

- H: the guest presses the pool's switch in the hallway, so $\mathrm{P}(H) = \frac{1}{4}$;
- A: the hallway switches are all on at the start, so $\mathrm{P}(A) = p$;
- L: the pool light is on at the end.

The guest's choice of switch is made at random, so H is independent of A.

Hallway. After the press, the pool's hallway switch is on exactly when either A happens and H does not (it was on and untouched), or A fails and H happens (it was off and got toggled on). So

$$\mathrm{P}(\text{hallway pool switch on at end}) = p \cdot \tfrac{3}{4} + (1-p) \cdot \tfrac{1}{4} = \frac{1+2p}{4}.$$

Kitchen. The pool's kitchen switch starts on with probability $\frac{1}{2}$. If it is not pressed it stays as it was; if it is pressed it flips, and a switch that is off with probability $\frac{1}{2}$ becomes on with probability $\frac{1}{2}$. Either way it ends on with probability $\frac{1}{2}$, whatever the guest does in the kitchen.

The kitchen switches and the guest's kitchen choice are independent of everything in the hallway, so

$$\mathrm{P}(L) = \frac{1+2p}{4} \cdot \frac{1}{2} = \frac{1+2p}{8}.$$

For the conditional probability, $H \cap L$ requires the guest to press the pool's hallway switch, that switch to have started off (so that pressing turns it on), and the kitchen switch to end on:

$$\mathrm{P}(H \cap L) = \frac{1}{4} \cdot (1-p) \cdot \frac{1}{2} = \frac{1-p}{8}.$$

Hence

$$\mathrm{P}(H \mid L) = \frac{(1-p)/8}{(1+2p)/8} = \frac{1-p}{1+2p},$$

as required.

Part (ii)

Write $q = \dfrac{1-p}{1+2p}$. On each Monday we are given that the light was on, so by part (i) the pool's hallway switch was pressed with probability q, independently from week to week (the guests act independently and the starting states are fresh each Monday). So the number X of days on which it was pressed satisfies

$$X \sim \mathrm{B}(7, q).$$

We need $1-q$ as well:

$$1 - q = \frac{(1+2p) - (1-p)}{1+2p} = \frac{3p}{1+2p}, \qquad \text{so} \qquad \frac{q}{1-q} = \frac{1-p}{3p}.$$

(Here $p > 0$; if $p = 0$ then $q = 1$ and $X = 7$ with certainty, so 3 could not be the most likely value.)

Ask yourself How do we turn "3 is the most likely value" into something we can calculate?

Compare neighbouring probabilities. For $1 \le k \le 7$,

$$\frac{\mathrm{P}(X=k)}{\mathrm{P}(X=k-1)} = \frac{\binom{7}{k} q^k (1-q)^{7-k}}{\binom{7}{k-1} q^{k-1} (1-q)^{8-k}} = \frac{8-k}{k} \cdot \frac{q}{1-q}.$$

The factor $\frac{8-k}{k}$ strictly decreases as k increases, so the probabilities rise while this ratio exceeds 1 and fall once it drops below 1. Thus 3 is the (unique) most likely value exactly when the probabilities rise from $k=2$ to $k=3$ and fall from $k=3$ to $k=4$:

$$\frac{\mathrm{P}(X=3)}{\mathrm{P}(X=2)} = \frac{5}{3} \cdot \frac{q}{1-q} > 1 \qquad \text{and} \qquad \frac{\mathrm{P}(X=4)}{\mathrm{P}(X=3)} = \frac{4}{4} \cdot \frac{q}{1-q} < 1.$$

(These two conditions alone are enough for the direction we need: if 3 is most likely then certainly $\mathrm{P}(X=3) > \mathrm{P}(X=2)$ and $\mathrm{P}(X=3) > \mathrm{P}(X=4)$.)

Substituting $\frac{q}{1-q} = \frac{1-p}{3p}$ and multiplying through by the positive quantity $3p$:

- the second condition gives $1 - p < 3p$, that is, $p > \frac{1}{4}$;
- the first gives $\frac{5}{3}(1 - p) > 3p$, that is, $5 - 5p > 9p$, so $p < \frac{5}{14}$.

Hence $\frac{1}{4} < p < \frac{5}{14}$, as required.

Checks

- At $p = 0$ the hallway switches are all off, so the light can only be on if the pool switch was pressed: part (i) gives $\mathrm{P}(H \mid L) = 1$, as it should. At $p = 1$ the light can only be on if that switch was *not* pressed, and the formula gives 0.
- The endpoints correspond to $q = \frac{1}{2}$ (at $p = \frac{1}{4}$) and $q = \frac{3}{8}$ (at $p = \frac{5}{14}$). At $q = \frac{1}{2}$, $\mathrm{B}(7, \frac{1}{2})$ is symmetric, so 3 and 4 tie; at $q = \frac{3}{8}$, $(n+1)q = 3$ and 2 and 3 tie. So the strict inequalities are exactly right.
- We checked part (i) by enumerating every starting state and every choice of switch exactly with fractions, and part (ii) by computing the $\mathrm{B}(7, q)$ probabilities for p on a grid of step $\frac{1}{2800}$: the unique mode is 3 precisely for $\frac{1}{4} < p < \frac{5}{14}$.

Common slips

- Forgetting that pressing a switch that is already on turns it *off*: the hallway calculation needs both routes to "on".
- Computing the kitchen by cases when the answer is simply $\frac{1}{2}$; the case analysis is harmless but an easy place for arithmetic errors.
- Using p instead of q as the binomial parameter in part (ii): the conditioning on the light being on is what changes the probability of pressing the switch from $\frac{1}{4}$ to q.

Beyond STEP

Part (i) is Bayes' theorem in action: an observation (the light is on) updates the probability that the guest pressed the switch from $\frac{1}{4}$ to $\frac{1-p}{1+2p}$. Part (ii) runs the idea in reverse, asking which values of an unknown parameter p are consistent with a given most likely outcome, which is the seed of statistical inference and maximum likelihood estimation. The ratio-of-consecutive-terms trick for locating a mode works for many discrete distributions (Poisson, hypergeometric). Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and these distributions thoroughly.

WORKED SOLUTION

STEP II 2012, Question 13

Probability and Statistics • Poisson Nearest Distance

STEP II 2012, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.19, *Random Points in the Plane*: https://beyondstep.keithmcnulty.org/ch12/spatial-poisson.html

The key idea

The event "the nearest supermarket is at least y away" is exactly the event "there are no supermarkets in the circle of radius y centred at our point", and the Poisson model tells us the probability of that directly. This gives the cumulative distribution function of Y, and differentiating gives the density. The moments are then Gaussian-type integrals: one integration by parts turns $\mathrm{E}(Y)$ into the integral we're allowed to assume, and a substitution $u = y^2$ makes $\mathrm{E}(Y^2)$ elementary.

Solution

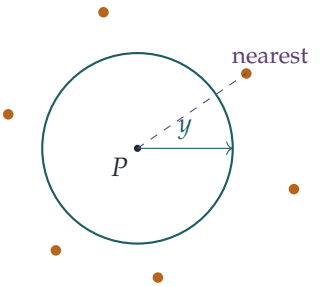
Throughout, we treat the region as so large that edge effects can be ignored, so a circle of radius y about the chosen point lies inside the region for every y that matters.

No supermarkets in a circle

Let N be the number of supermarkets inside a circle of radius y. The circle has area πy^2, so $N \sim \mathrm{Po}(\pi k y^2)$ and

$$\mathrm{P}(N = 0) = \mathrm{e}^{-\pi k y^2}.$$

The distribution of Y



Let P be the randomly chosen point. The nearest supermarket is at distance at least y from P precisely when the open circle of radius y centred at P contains no supermarkets. So, for $y \geq 0$,

$$\mathrm{P}(Y \geq y) = \mathrm{e}^{-\pi k y^2}, \qquad \text{hence} \qquad \mathrm{P}(Y < y) = 1 - \mathrm{e}^{-\pi k y^2}.$$

(Since Y is continuous, whether we write $<$ or $\leq$ makes no difference.) This is the cumulative distribution function $F(y)$ of Y, and $F(y) = 0$ for $y < 0$ because a distance can't be negative. Differentiating, for $y \geq 0$,

$$f(y) = F'(y) = \frac{\mathrm{d}}{\mathrm{d}y}\left(1 - \mathrm{e}^{-\pi k y^2}\right) = 2\pi k y\, \mathrm{e}^{-\pi k y^2},$$

which is the required density. As a quick sanity check, $F(y) \to 1$ as $y \to \infty$, so f integrates to 1.

The mean

$$\mathrm{E}(Y) = \int_0^\infty y \cdot 2\pi k y\, \mathrm{e}^{-\pi k y^2}\, \mathrm{d}y.$$

Ask yourself How can we use the given integral when there's a y^2 in front of the exponential?

Split the integrand as $y \times \left(2\pi k y\, \mathrm{e}^{-\pi k y^2}\right)$: the second factor is the derivative of $-\mathrm{e}^{-\pi k y^2}$. Integrating by parts with $u = y$,

$$\mathrm{E}(Y) = \left[-y\, \mathrm{e}^{-\pi k y^2}\right]_0^\infty + \int_0^\infty \mathrm{e}^{-\pi k y^2}\, \mathrm{d}y = \int_0^\infty \mathrm{e}^{-\pi k y^2}\, \mathrm{d}y,$$

since $y\, \mathrm{e}^{-\pi k y^2} \to 0$ as $y \to \infty$ (the exponential wins) and vanishes at $y = 0$. Now match the exponent to $-\frac{1}{2}x^2$: put $\pi k y^2 = \frac{1}{2}x^2$, that is, $y = x/\sqrt{2\pi k}$, $\mathrm{d}y = \mathrm{d}x/\sqrt{2\pi k}$. The limits stay 0 and ∞, so

$$\mathrm{E}(Y) = \frac{1}{\sqrt{2\pi k}} \int_0^\infty \mathrm{e}^{-x^2/2}\, \mathrm{d}x = \frac{1}{\sqrt{2\pi k}} \sqrt{\frac{\pi}{2}} = \frac{1}{2\sqrt{k}}.$$

The variance

For $\mathrm{E}(Y^2)$ we don't need the given integral at all, because the odd power of y lets us substitute $u = \pi k y^2$, $\mathrm{d}u = 2\pi k y \,\mathrm{d}y$:

$$\mathrm{E}(Y^2) = \int_0^\infty y^2 \cdot 2\pi k y \,\mathrm{e}^{-\pi k y^2} \,\mathrm{d}y = \int_0^\infty \frac{u}{\pi k} \,\mathrm{e}^{-u} \,\mathrm{d}u = \frac{1}{\pi k} \int_0^\infty u \,\mathrm{e}^{-u} \,\mathrm{d}u = \frac{1}{\pi k},$$

using $\int_0^\infty u \,\mathrm{e}^{-u} \,\mathrm{d}u = 1$ (integrate by parts, or recognise the mean of an exponential distribution with parameter 1). Therefore

$$\mathrm{Var}(Y) = \mathrm{E}(Y^2) - \big(\mathrm{E}(Y)\big)^2 = \frac{1}{\pi k} - \frac{1}{4k} = \frac{4-\pi}{4\pi k},$$

as required.

Checks

- *Scaling.* If supermarkets are k times as dense, lengths should shrink by a factor $\sqrt{k}$, so $\mathrm{E}(Y) \propto k^{-1/2}$ and $\mathrm{Var}(Y) \propto k^{-1}$. Both answers have this form.
- *Positivity.* $4 - \pi > 0$, so the variance is positive, as it must be.
- *Numerics.* With $k = 1$, midpoint quadrature of the density gives $\mathrm{E}(Y) = 0.5000$ and $\mathrm{Var}(Y) = 0.06831$, matching $(4-\pi)/(4\pi) \approx 0.06831$. A simulation of 10^5 random Poisson scatters of supermarkets around a point gave mean 0.4996 and variance 0.0681.

Common slips

- Writing $\mathrm{P}(Y < y) = \mathrm{e}^{-\pi k y^2}$: that is the probability of *no* supermarket within y, i.e. $\mathrm{P}(Y \geq y)$.
- Substituting $x = \sqrt{\pi k}\, y$ instead of $x = \sqrt{2\pi k}\, y$, which gives e^{-x^2} rather than the assumed $\mathrm{e}^{-x^2/2}$ and an answer out by a factor of $\sqrt{2}$.

Beyond STEP

The supermarkets here form a *Poisson point process* in the plane, the basic model of "complete spatial randomness" used for trees in a forest, stars in the sky and mobile-phone masts. The distribution of Y is a *Rayleigh distribution*, which also appears as the length of a vector whose two components are independent normal variables; the neat tail formula $\mathrm{E}(Y) = \int_0^\infty \mathrm{P}(Y \geq y) \,\mathrm{d}y$ that our integration by parts uncovered holds for every non-negative random variable. Grimmett and Welsh, *Probability: An Introduction*, covers Poisson processes and these tail-integral ideas.

WORKED SOLUTION

STEP III 2012, Question 1

Pure • Non-linear Differential Equations

STEP III 2012, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q1

The key idea

Both differential equations are second order and non-linear, so there is no standard recipe. The trick is that the bracket in $\mathrm{d}z/\mathrm{d}x$ can be made to match the left-hand side of each equation by choosing n well. The equation then says $\mathrm{d}z/\mathrm{d}x$ equals something we can integrate at once, which gives a first-order equation we can separate.

Solution

The preliminary result

Write $y' = \mathrm{d}y/\mathrm{d}x$ and $y'' = \mathrm{d}^2y/\mathrm{d}x^2$. By the product rule, using the chain rule on y^n and on $(y')^2$,

$$\begin{aligned}\frac{\mathrm{d}z}{\mathrm{d}x} &= ny^{n-1}y'\,(y')^2 + y^n \cdot 2y'y'' \\ &= y^{n-1}y'\big(n(y')^2 + 2yy''\big),\end{aligned}$$

which is the required result.

Part (i)

The left-hand side is $(y')^2 + 2yy''$, which is the bracket with $n = 1$. So take $z = y(y')^2$. Then

$$\frac{\mathrm{d}z}{\mathrm{d}x} = y'\big((y')^2 + 2yy''\big) = y'\sqrt{y} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\tfrac{2}{3}y^{3/2}\right).$$

Integrating, $y(y')^2 = \tfrac{2}{3}y^{3/2} + C$. At $x = 0$ we have $y = 1$ and $y' = 0$, so $0 = \tfrac{2}{3} + C$ and

$$y\,(y')^2 = \tfrac{2}{3}\big(y^{3/2} - 1\big). \qquad (*)$$

Ask yourself $(*)$ only tells us $(y')^2$. Which sign of the square root do we take, and is $y \equiv 1$ (which makes both sides zero) a possibility?

The constant $y \equiv 1$ is *not* a solution of the original equation: it would give $0 = 1$. In fact, putting $x = 0$ into the equation gives $0 + 2y''(0) = 1$, so $y''(0) = \frac{1}{2} > 0$. Hence y has a minimum at $x = 0$ and $y' > 0$ for small $x > 0$. Also the equation is unchanged by $x \mapsto -x$, so the solution is even and we can work with $x > 0$, $y' > 0$, $y > 1$.

Multiply $(*)$ by y, take the positive square root, and separate:

$$\frac{\sqrt{y}\,y'}{\sqrt{y^{3/2} - 1}} = \sqrt{\tfrac{2}{3}}.$$

Substitute $w = y^{3/2} - 1$, so $\mathrm{d}w = \frac{3}{2}y^{1/2}\,\mathrm{d}y$. The left-hand side becomes $\frac{2}{3}w^{-1/2}\,\mathrm{d}w/\mathrm{d}x$, and integrating from $x = 0$ (where $w = 0$),

$$\tfrac{4}{3}\sqrt{w} = \sqrt{\tfrac{2}{3}}\,x \quad \Longrightarrow \quad w = \tfrac{9}{16} \cdot \tfrac{2}{3}\,x^2 = \tfrac{3}{8}x^2.$$

So $y^{3/2} = \frac{3}{8}x^2 + 1$, which by evenness holds for $x < 0$ too. Therefore

$$y = \left(\tfrac{3}{8}x^2 + 1\right)^{2/3},$$

as required.

As a direct check, with $u = \frac{3}{8}x^2 + 1$ we get $y' = \frac{2}{3}u^{-1/3} \cdot \frac{3}{4}x = \frac{1}{2}xu^{-1/3}$, so $y(y')^2 = u^{2/3} \cdot \frac{1}{4}x^2u^{-2/3} = \frac{1}{4}x^2$. Meanwhile $\frac{2}{3}(y^{3/2} - 1) = \frac{2}{3} \cdot \frac{3}{8}x^2 = \frac{1}{4}x^2$, so $(*)$ holds.

Part (ii)

Now the left-hand side is $(y')^2 - yy''$. We need $n(y')^2 + 2yy''$ to be a multiple of this, so $n : 2 = 1 : -1$, giving $n = -2$. With $z = y^{-2}(y')^2$ (fine, since y is near 1 and so non-zero),

$$\frac{\mathrm{d}z}{\mathrm{d}x} = y^{-3}y'\left(-2(y')^2 + 2yy''\right) = -2y^{-3}y'\left((y')^2 - yy''\right) = -2y^{-3}y'\left(-y^2\right) = \frac{2y'}{y},$$

using the equation in the form $(y')^2 - yy'' = -y^2$. Integrating, $z = 2\ln y + C$, and $x = 0$ gives $0 = 0 + C$. So

$$\left(\frac{y'}{y}\right)^2 = 2\ln y.$$

Let $v = \ln y$, so $v' = y'/y$ and the equation becomes $(v')^2 = 2v$, with $v(0) = 0$.

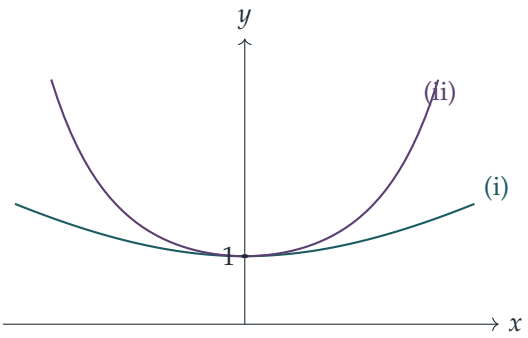
As in part (i), $v \equiv 0$ (that is, $y \equiv 1$) solves this first-order equation but not the original one, which would read $1 = 0$. At $x = 0$ the original equation gives $0 - y''(0) + 1 = 0$, so $y''(0) = 1 > 0$. The equation is again unchanged by $x \mapsto -x$, so we take $x > 0$, where $v' > 0$. Then

$$\frac{v'}{\sqrt{2v}} = 1 \quad \Longrightarrow \quad \sqrt{2v} = x \quad \Longrightarrow \quad v = \tfrac{1}{2}x^2,$$

and by evenness this holds for all x. Hence

$$y = \mathrm{e}^{x^2/2}.$$

To check: $y' = xy$ and $y'' = (1 + x^2)y$, so $(y')^2 - yy'' + y^2 = y^2\left(x^2 - 1 - x^2 + 1\right) = 0$. Also $y(0) = 1$ and $y'(0) = 0$.



Checks

- Both solutions are even, with a minimum value of 1 at $x = 0$, as the values of $y''(0)$ predict.
- For part (ii) there is a quicker route: putting $y = \mathrm{e}^v$ straight into the equation gives $y' = v'\mathrm{e}^v$ and $y'' = (v'' + (v')^2)\mathrm{e}^v$, and everything cancels down to $v'' = 1$. So $v = \frac{1}{2}x^2$ immediately.
- Numerically, a Runge–Kutta solution of (i) from $y(0) = 1$, $y'(0) = 0$ gives $y(2) \approx 1.842016$, matching $(\frac{5}{2})^{2/3}$. For (ii) it gives $y(1) \approx 1.648721 = \mathrm{e}^{1/2}$.

Common slips

- Forgetting the constant of integration in z. The initial conditions make it non-zero in part (i): $C = -\frac{2}{3}$.
- Choosing $n = 1$ in part (ii) by analogy. The ratio of the coefficients of $(y')^2$ and yy'' must match, which forces $n = -2$.
- Accepting $y \equiv 1$ from the first-order equation without checking it against the original equation.

Beyond STEP

The method here is an *integrating factor* for a second-order equation. Multiplying $(y')^2 + 2yy''$ by y' turns it into an exact derivative, just as multiplying $\ddot{x} = -\omega^2 x$ by $\dot{x}$ gives conservation of energy. Equations with no explicit x, like these, can always be

reduced to first order by taking $p = y'$ as a function of y. The worry about $y \equiv 1$ is a glimpse of non-uniqueness: $v' = \sqrt{2v}$ fails the Lipschitz condition at $v = 0$. Boyce and DiPrima, *Elementary Differential Equations*, covers both reduction of order and the existence and uniqueness theorem.

WORKED SOLUTION

STEP III 2012, Question 2

Pure • Infinite Products and Series

STEP III 2012, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q2

The key idea

The product in (i) collapses one bracket at a time through the difference of two squares, so it telescopes to something very simple. Once $\frac{1}{1-x}$ is written as an infinite product, taking logarithms turns the product into a sum, and differentiating that sum term by term gives the series of fractions. Part (ii) is the same trick in disguise: $1+x+x^2$ is a quotient of two expressions of the form $1-t$, so the result of (i) applies to each of them.

Solution

Throughout, $|x|<1$ and, as the question allows, we don't worry about convergence.

Part (i)

Combine the brackets from the left, using $(1-a)(1+a)=1-a^2$ each time:

$$\begin{aligned}(1-x)(1+x)&=1-x^2,\\(1-x^2)(1+x^2)&=1-x^4,\\(1-x^4)(1+x^4)&=1-x^8,\end{aligned}$$

and so on. Each step doubles the power, and the last bracket $1+x^{2^n}$ turns $1-x^{2^n}$ into $1-x^{2^{n+1}}$. So

$$(1-x)(1+x)(1+x^2)(1+x^4)\cdots(1+x^{2^n})=1-x^{2^{n+1}}.$$

Since $x\neq 1$ we can divide by $1-x$:

$$(1+x)(1+x^2)\cdots(1+x^{2^n})=\frac{1}{1-x}-\frac{x^{2^{n+1}}}{1-x},$$

which rearranges to the required expression for $\frac{1}{1-x}$.

Now let $n \to \infty$. Because $|x| < 1$, $x^{2^{n+1}} \to 0$, so the remainder term vanishes and

$$\frac{1}{1-x} = \prod_{r=0}^{\infty}(1 + x^{2^r}).$$

Every factor is positive (as $x^{2^r} > -1$), and so is $1 - x$, so we may take logarithms. The log of a product is the sum of the logs:

$$-\ln(1-x) = \sum_{r=0}^{\infty} \ln(1 + x^{2^r}),$$

which is the required result after multiplying by -1.

Ask yourself How do we get from a sum of logarithms to a sum of fractions?

Differentiate both sides with respect to x. The derivative of $\ln(1 + x^{2^r})$ is, by the chain rule,

$$\frac{2^r x^{2^r - 1}}{1 + x^{2^r}}.$$

On the left, $\frac{\mathrm{d}}{\mathrm{d}x}\big(-\ln(1-x)\big) = \frac{1}{1-x}$. Differentiating term by term (the convergence issue we are told to ignore) gives

$$\frac{1}{1-x} = \sum_{r=0}^{\infty} \frac{2^r x^{2^r-1}}{1 + x^{2^r}} = \frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \cdots,$$

where the terms shown are $r = 0, 1, 2$. This is the required series.

Part (ii)

The left-hand side is the derivative of $\ln(1 + x + x^2)$, since $\frac{\mathrm{d}}{\mathrm{d}x}(1 + x + x^2) = 1 + 2x$. So we look for a log-sum for $1 + x + x^2$, and the factorisation $1 - x^3 = (1 - x)(1 + x + x^2)$ gives one:

$$\ln(1 + x + x^2) = \ln(1 - x^3) - \ln(1 - x).$$

Since $|x^3| < 1$ too, the log result of (i) holds with x replaced by x^3. Writing $y_r = x^{2^r}$, so that $(x^3)^{2^r} = y_r^3$,

$$\ln(1 + x + x^2) = -\sum_{r=0}^{\infty} \ln(1 + y_r^3) + \sum_{r=0}^{\infty} \ln(1 + y_r) = -\sum_{r=0}^{\infty} \ln \frac{1 + y_r^3}{1 + y_r}.$$

The sum of cubes factorises as $1 + y^3 = (1 + y)(1 - y + y^2)$, so each fraction is $1 - y_r + y_r^2$ and

$$\ln(1 + x + x^2) = -\sum_{r=0}^{\infty} \ln(1 - x^{2^r} + x^{2^{r+1}}).$$

(Every $1 - y + y^2 = (y - \frac{1}{2})^2 + \frac{3}{4}$ is positive, so these logarithms make sense.)

Ask yourself Where does the minus sign go when we differentiate?

The chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln\left(1-x^{2^r}+x^{2^{r+1}}\right)=\frac{-2^r x^{2^r-1}+2^{r+1}x^{2^{r+1}-1}}{1-x^{2^r}+x^{2^{r+1}}}=-\frac{2^r x^{2^r-1}\left(1-2x^{2^r}\right)}{1-x^{2^r}+x^{2^{r+1}}},$$

and this minus sign cancels the one in front of the sum. Differentiating both sides term by term,

$$\frac{1+2x}{1+x+x^2}=\sum_{r=0}^{\infty}\frac{2^r x^{2^r-1}\left(1-2x^{2^r}\right)}{1-x^{2^r}+x^{2^{r+1}}}.$$

The first three terms are

$$\begin{aligned} r=0:&\quad \frac{1-2x}{1-x+x^2},\\ r=1:&\quad \frac{2x(1-2x^2)}{1-x^2+x^4}=\frac{2x-4x^3}{1-x^2+x^4},\\ r=2:&\quad \frac{4x^3(1-2x^4)}{1-x^4+x^8}=\frac{4x^3-8x^7}{1-x^4+x^8}, \end{aligned}$$

which is exactly the series we were asked to establish.

Another route. The identity $(1+y+y^2)(1-y+y^2)=1+y^2+y^4$ gives a telescoping product for $1+x+x^2$ directly, just like the one in (i): $1+x+x^2=\left(1+x^{2^{n+1}}+x^{2^{n+2}}\right)/\prod_{r=0}^{n}\left(1-x^{2^r}+x^{2^{r+1}}\right)$, and the numerator tends to 1.

Checks

- Put $x=0$: both sides of each series equal 1.
- At $x=\frac{1}{2}$ the left-hand side of (ii) is $\frac{2}{7/4}=\frac{8}{7}\approx 1.142857$. The first four terms of the right-hand side sum to 1.142369, and the error shrinks very fast because the powers x^{2^r} do.
- We checked the telescoped product exactly with fractions, and both series and the log identities numerically at 1000 random x in $(-0.95,0.95)$, using 30 terms.

Common slips

- Losing a factor in the chain rule: the derivative of x^{2^r} is $2^r x^{2^r-1}$, not $2^r x^{2^r}$.
- Getting the overall sign wrong in (ii): the derivative of $1-y+y^2$ is $-1+2y$, and it's this minus that cancels the minus in front of the sum.

Beyond STEP

The product $\prod(1 + x^{2^r})$ says that every positive integer can be written in exactly one way as a sum of distinct powers of 2, which is binary notation expressed as a generating function. "Ignoring convergence" hides real questions: when may an infinite product be logged, and when may a series be differentiated term by term? A first analysis course answers these with uniform convergence, and on $|x| \leq c < 1$ everything here is uniformly convergent. See Abbott, *Understanding Analysis*, for a clear treatment.

WORKED SOLUTION

STEP III 2012, Question 3

Pure • Tangent Curves and Quartics

STEP III 2012, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q3

The key idea

Substituting $y = 4 - x^2$ into the second equation gives a quartic in x whose roots are the x-coordinates of the meeting points. "Touching" means a *repeated* root. Instead of treating m and k as the unknowns, label everything by the x-coordinate a of the touching point: m, k and all the other meeting points then turn out to be simple functions of a. That tells us exactly what each of the four sketches must look like, and part (ii) is the special case $m = 12$.

Solution

Setting up: where can the curves touch?

Call the curves $C_1\colon y = 4 - x^2$ and $C_2\colon mx = k - y^2$. C_1 is the familiar downward parabola through $(\pm 2, 0)$ and $(0, 4)$. C_2 is $x = (k - y^2)/m$: a parabola symmetric about the x-axis, opening to the left (as $m > 0$), with vertex $(k/m, 0)$. It meets the y-axis at $(0, \pm\sqrt{k})$ when $k > 0$ and not at all when $k < 0$.

Every point of C_1 is $(x, 4 - x^2)$, so it lies on C_2 exactly when

$$F(x) = x^4 - 8x^2 + mx + 16 - k = 0,$$

since $(4 - x^2)^2 = x^4 - 8x^2 + 16$.

? **Ask yourself** Why does "touch" mean that F has a repeated root?

At a common point, C_1 has direction $(1, -2x)$ and C_2 (the level curve of $mx + y^2$) has normal $(m, 2y)$. The curves touch when the direction is perpendicular to the normal: $m - 4xy = 0$, that is $m - 4x(4 - x^2) = 0$. But $F'(x) = 4x^3 - 16x + m$ is exactly this expression. So the curves touch at $x = a$ precisely when $F(a) = F'(a) = 0$, a repeated root.

From $F'(a) = 0$ we get $m = 4a(4 - a^2)$. Substituting into $F(a) = 0$,

$$k = a^4 - 8a^2 + 4a^2(4 - a^2) + 16 = -3a^4 + 8a^2 + 16 = (4 - a^2)(3a^2 + 4),$$

and so

$$\frac{k}{m} = \frac{3a^2 + 4}{4a}.$$

Since $m > 0$ we need $0 < a < 2$ or $a < -2$. Dividing out the double factor,

$$F(x) = (x - a)^2\big(x^2 + 2ax + 3a^2 - 8\big),$$

as we can check by expanding: the x^3 terms cancel, the x^2 coefficient is $3a^2 - 8 - 4a^2 + a^2 = -8$, the x coefficient is $2a^3 - 2a(3a^2 - 8) = 4a(4 - a^2) = m$ and the constant is $a^2(3a^2 - 8) = 16 - k$. The quadratic factor has discriminant $8(4 - a^2)$, so there are other meeting points only when $0 < a < 2$. They are at

$$x_\pm = -a \pm \sqrt{8 - 2a^2}.$$

Now we can sort the four cases by a:

- $k < 0$ if and only if $a^2 > 4$, so $a < -2$.
- For $0 < a < 2$, $k - 16 = a^2(8 - 3a^2)$, so $k > 16$ if and only if $a < \sqrt{8/3}$.
- For $0 < a < 2$, $\frac{k}{m} < 2$ if and only if $3a^2 - 8a + 4 < 0$, i.e. $(3a - 2)(a - 2) < 0$, i.e. $a > \frac{2}{3}$.

So (a) is $a < -2$, (c) is $0 < a < \frac{2}{3}$, (d) is $\frac{2}{3} < a < \sqrt{8/3}$ and (b) is $\sqrt{8/3} < a < 2$. (No touching configuration has $0 < k < 16$ with $k/m > 2$, which is why that case isn't listed.)

Part (i)

In every case the axis intercepts are $(\pm 2, 0)$ and $(0, 4)$ for C_1, and $(k/m, 0)$, plus $(0, \pm\sqrt{k})$ when $k > 0$, for C_2.

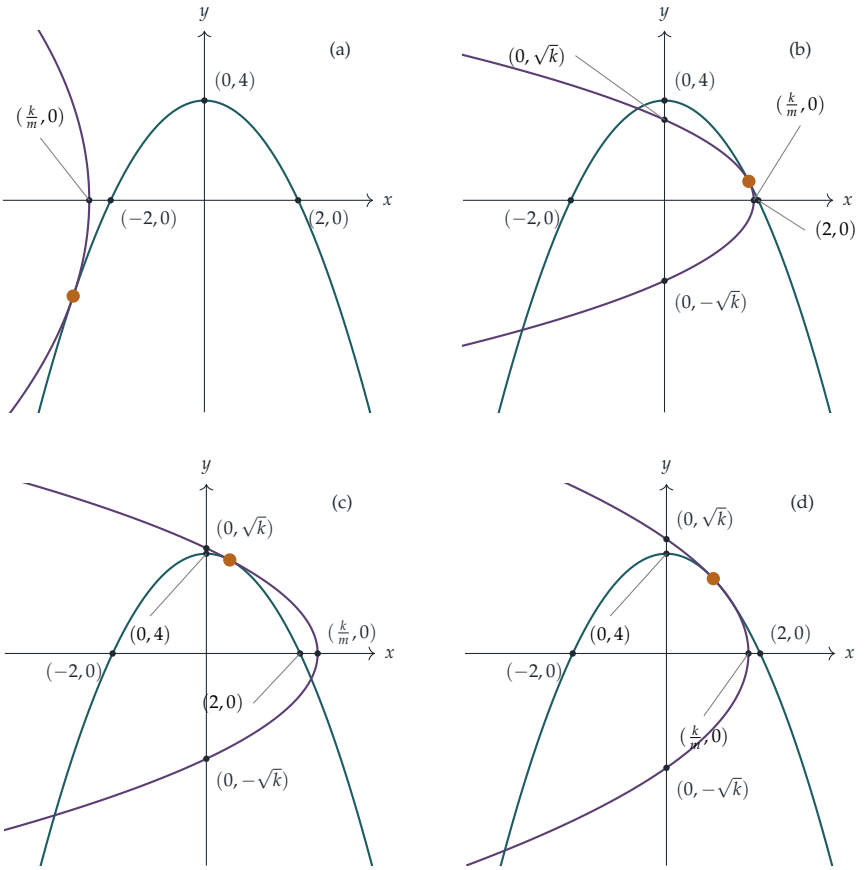
(a) Here $a < -2$, so the curves touch at $(a, 4 - a^2)$ in the third quadrant, and they don't meet anywhere else. The vertex is at $k/m = -(3a^2 + 4)/(4|a|)$, which is less than -2 because $(3|a| - 2)(|a| - 2) > 0$. So C_2 lies entirely to the left of $(-2, 0)$, and its lower arm rests on the left branch of C_1.

(b) Now $\sqrt{8/3} < a < 2$, so the touch is in the first quadrant, close to $(2, 0)$. The vertex has $\sqrt{3} \leq k/m < 2$ (by AM–GM, $3a^2 + 4 \geq 4\sqrt{3}\,a$), and $\sqrt{k} < 4$, so the vertex and both y-intercepts of C_2 lie under C_1. The upper arm touches C_1 from below and then crosses it at x_+, where $-2 < x_+ < 0$ (a point in the second quadrant). The lower arm crosses C_1 at $x_- < -2$, in the third quadrant.

(c) Here $0 < a < \frac{2}{3}$. The vertex is to the right of $(2, 0)$ and $\sqrt{k} > 4$, so C_2 wraps round the top of C_1 and touches it from outside in the first quadrant. Then $x_+ > 2$, because $\sqrt{8 - 2a^2} > 2 + a \iff (3a - 2)(a + 2) < 0$, and $x_- < -2$. So the lower arm crosses C_1 in the fourth quadrant and again in the third.

(d) Here $\frac{2}{3} < a < \sqrt{8/3}$. The vertex is between 0 and 2, under C_1, but $\sqrt{k} > 4$, so the upper arm passes above $(0,4)$. It touches C_1 in the first quadrant and also crosses it there, at $x_+ \in (0,2)$ (since $x_+ > 0 \iff a^2 < \frac{8}{3}$). The lower arm crosses at $x_- < -2$.

In every case $x_- < -2$ for $0 < a < 2$, because $a + \sqrt{8 - 2a^2} > 2$ (square $\sqrt{8-2a^2} > 2 - a$). The sketches use $a = -2.8,\ 1.8,\ 0.5,\ 1$ respectively. C_1 is teal, C_2 is plum and the touching point is amber.



Part (ii)

With $m = 12$, substituting $y = 4 - x^2$ into $12x = k - y^2$ gives $12x = k - 16 + 8x^2 - x^4$, that is

$$x^4 - 8x^2 + 12x + 16 - k = 0,$$

as required.

At the touching point $x = a$ this quartic has a repeated root (see the box above), so its derivative also vanishes there:

$$4a^3 - 16a + 12 = 0 \quad \Longrightarrow \quad a^3 - 4a + 3 = 0.$$

Since $a = 1$ is a root, $a^3 - 4a + 3 = (a-1)(a^2 + a - 3)$, and so

$$a = 1 \qquad \text{or} \qquad a = \frac{-1 \pm \sqrt{13}}{2}.$$

To find k, put $x = a$ in the quartic and reduce using $a^3 = 4a - 3$, so that $a^4 = 4a^2 - 3a$:

$$k = a^4 - 8a^2 + 12a + 16 = (4a^2 - 3a) - 8a^2 + 12a + 16 = -4a^2 + 9a + 16.$$

Now the three cases:

- $a = 1$: $k = 21$, so $k > 16$ and $k/m = \frac{7}{4} < 2$. This is case **(d)**.
- $a = \frac{1}{2}(-1 + \sqrt{13}) \approx 1.303$: here $a^2 = 3 - a$, so $k = -4(3-a) + 9a + 16 = 13a + 4 = \frac{1}{2}(13\sqrt{13} - 5) \approx 20.94$. This exceeds 16 because $13\sqrt{13} > 37$ (square: $2197 > 1369$), and it is below 24 because $13\sqrt{13} < 53$ ($2197 < 2809$). So $k > 16$ and $k/m < 2$, which is case **(d)** again.
- $a = \frac{1}{2}(-1 - \sqrt{13}) \approx -2.303$: $k = 13a + 4 = -\frac{1}{2}(5 + 13\sqrt{13}) \approx -25.94 < 0$. This is case **(a)**.

So only sketches (a) and (d) arise. This agrees with the classification by a above: 1 and 1.303 lie in $(\frac{2}{3}, \sqrt{8/3}) \approx (0.667, 1.633)$, and $-2.303 < -2$. Note that $a^3 - 4a + 3 = 0$ is just $m = 4a(4 - a^2)$ with $m = 12$.

The two (d) pictures differ slightly. For $a = 1$ the other meeting points are $x = -1 \pm \sqrt{6}$; the crossing at $x \approx 1.45$ is to the *right* of the touch, and the upper arm touches C_1 from above (as in the sketch). For $a \approx 1.303$ the crossing is at $x \approx 0.84$, to the left of the touch, and the upper arm touches from below.

Checks

- With $m = 12$, $k = 21$ the quartic factorises as $(x-1)^2(x^2 + 2x - 5)$. Expanding gives $x^4 - 8x^2 + 12x - 5$, which is the quartic with $16 - k = -5$.
- At $(1, 3)$ the gradient of C_1 is -2, and implicit differentiation of $12x = 21 - y^2$ gives $\mathrm{d}y/\mathrm{d}x = -6/y = -2$. So the curves really do touch there.
- Numerically, for random a in $(0, 2)$ and $a < -2$, the formulas $m = 4a(4 - a^2)$ and $k = (4 - a^2)(3a^2 + 4)$ make $F(a)$ and $F'(a)$ vanish, and the case boundaries fall at $a = \frac{2}{3}$ and $a = \sqrt{8/3}$. For $a \approx 1.3028$ a root search finds the other meetings at $x \approx 0.8433$ and $x \approx -3.4488$, as $x_\pm$ predicts.

Common slips

- Requiring only one root of the quartic. "Touch exactly once" allows other crossing points: in case (d) with $m = 12$ there are two.
- Drawing C_2 opening to the right. With $m > 0$, $x = (k - y^2)/m$ decreases as $|y|$ grows.

- In (a), forgetting that C_2 has no y-intercepts, or placing its vertex between -2 and 0. It must be to the left of -2.

Beyond STEP

Labelling the configurations by the touching point a is a first glimpse of *discriminants* and *envelopes*. The set of (m,k) for which F has a repeated root is the curve $\big(4a(4-a^2),\,(4-a^2)(3a^2+4)\big)$, and crossing it changes the number of real meeting points by two. This is the beginning of the algebraic geometry of plane curves, and of the singularity ("catastrophe") theory used to study how solutions change with parameters. Siklos's *Advanced Problems in Mathematics* has many more STEP-style curve problems in the same spirit.

WORKED SOLUTION

STEP III 2012, Question 4

Pure • Summing Infinite Series

STEP III 2012, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q4

The key idea

Everything rests on two standard series: $\sum x^n/n! = \mathrm{e}^x$ and $\sum x^n/n = -\ln(1-x)$. In part (i) the trick is to write each polynomial in n so that it cancels against $n!$, which means using products like $n(n-1)$ and $n(n-1)(n-2)$ rather than powers of n. In part (ii), partial fractions turn the awkward rational factor into pieces of the form $1/(n+k)$, and each of those is a shifted logarithm series.

Solution

Part (i)

Two facts do all the work. First, $\sum_{n=0}^{\infty} 1/n! = \mathrm{e}$, so

$$\sum_{n=1}^{\infty} \frac{1}{n!} = \mathrm{e} - 1.$$

Second, for any whole number $k \geq 1$, the term $n(n-1)\cdots(n-k+1)/n!$ is zero when $n < k$ and equals $1/(n-k)!$ when $n \geq k$. So

$$\sum_{n=1}^{\infty} \frac{n(n-1)\cdots(n-k+1)}{n!} = \sum_{m=0}^{\infty} \frac{1}{m!} = \mathrm{e} \qquad (k \geq 1). \tag{$*$}$$

All the series here converge absolutely (each is a polynomial in n over $n!$), so we may split them up and recombine them freely.

First sum. Since $n+1 = n+1$, using $(*)$ with $k=1$,

$$\sum_{n=1}^{\infty} \frac{n+1}{n!} = \mathrm{e} + (\mathrm{e}-1) = 2\mathrm{e} - 1,$$

as required.

Second sum. We write $(n+1)^2 = n^2+2n+1 = n(n-1)+3n+1$. Then $(*)$ with $k=2$ and $k=1$ gives

$$\sum_{n=1}^{\infty} \frac{(n+1)^2}{n!} = \mathrm{e} + 3\mathrm{e} + (\mathrm{e}-1) = 5\mathrm{e}-1,$$

as required.

Third sum.

Ask yourself How do we turn $(2n-1)^3$ into falling products $n(n-1)(n-2)$, $n(n-1)$, n?

Expanding gives $(2n-1)^3 = 8n^3 - 12n^2 + 6n - 1$. Next we use

$$n^3 = n(n-1)(n-2) + 3n(n-1) + n, \qquad n^2 = n(n-1) + n.$$

You can check these by expanding the right-hand sides, or by putting in $n = 0, 1, 2, 3$. Substituting,

$$\begin{aligned}(2n-1)^3 &= 8\big[n(n-1)(n-2) + 3n(n-1) + n\big] - 12\big[n(n-1)+n\big] + 6n - 1 \\ &= 8\,n(n-1)(n-2) + 12\,n(n-1) + 2n - 1.\end{aligned}$$

(A quick test: at $n=1$ both sides are 1, and at $n=2$ both are 27.) Now $(*)$ with $k = 3, 2, 1$ gives

$$\sum_{n=1}^{\infty} \frac{(2n-1)^3}{n!} = 8\mathrm{e} + 12\mathrm{e} + 2\mathrm{e} - (\mathrm{e}-1) = 21\mathrm{e} + 1.$$

Part (ii)

Partial fractions. The numerator and denominator have the same degree, so we divide first:

$$\frac{n^2+1}{(n+1)(n+2)} = 1 - \frac{3n+1}{(n+1)(n+2)}.$$

Writing $\dfrac{3n+1}{(n+1)(n+2)} = \dfrac{A}{n+1} + \dfrac{B}{n+2}$, the cover-up rule gives $A = \dfrac{-2}{1} = -2$ (at $n=-1$) and $B = \dfrac{-5}{-1} = 5$ (at $n=-2$). Hence

$$\frac{n^2+1}{(n+1)(n+2)} = 1 + \frac{2}{n+1} - \frac{5}{n+2}.$$

The three pieces. We put $x = \frac{1}{2}$ and use $-\ln(1-x) = \sum_{m=1}^{\infty} x^m/m$, which holds for $|x| < 1$. Each of the three series below converges (the terms shrink at least as fast as

2^{-n}), so splitting the sum is legitimate.

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} = 2,$$
$$\sum_{n=0}^{\infty} \frac{x^n}{n+1} = \frac{1}{x}\sum_{m=1}^{\infty} \frac{x^m}{m} = \frac{-\ln(1-x)}{x} = 2\ln 2,$$
$$\sum_{n=0}^{\infty} \frac{x^n}{n+2} = \frac{1}{x^2}\sum_{m=2}^{\infty} \frac{x^m}{m} = \frac{-\ln(1-x)-x}{x^2} = 4\left(\ln 2 - \tfrac{1}{2}\right) = 4\ln 2 - 2.$$

In the second and third lines we put $m = n+1$ and $m = n+2$. The third line is the full logarithm series with its $m = 1$ term, x, taken away.

? **Ask yourself** Why does the third piece need the extra $-x$?

The shift $m = n+2$ makes the sum start at $m = 2$, so the series is $-\ln(1-x)$ with its first term missing. Forgetting this is the easiest way to go wrong here.

Combining.

$$\sum_{n=0}^{\infty} \frac{(n^2+1)2^{-n}}{(n+1)(n+2)} = 2 + 2(2\ln 2) - 5(4\ln 2 - 2) = 12 - 16\ln 2.$$

Checks

- Partial sums up to $n = 60$ give 4.436564, 12.591409 and 58.083918 for the three sums in (i). These match $2e-1$, $5e-1$ and $21e+1$ to machine precision.
- In (ii), $12 - 16\ln 2 \approx 0.909645$, which agrees with the partial sums. The first few terms are $\frac{1}{2}$, $\frac{1}{6}$, $\frac{5}{48}$, $\frac{1}{16}$, so the total is plausibly a little under 1.
- The answer to (ii) must be positive, and it is: $16\ln 2 \approx 11.09 < 12$.

Common slips

- Using $\sum_{n=1}^{\infty} 1/n! = e$, which forgets that the $n = 0$ term is missing.
- Writing $\sum n^2/n!$ as $\sum n/(n-1)!$ and then stopping there. You still have to split $n = (n-1)+1$.
- In (ii), dropping the $-x$ when shifting the index on $\sum x^n/(n+2)$.

Beyond STEP

The falling products $n(n-1)\cdots(n-k+1)$ are called falling factorials, and writing a polynomial in terms of them uses the Stirling numbers of the second kind. The sum in

$(*)$ says that the kth factorial moment of a Poisson distribution with mean 1 is 1. This is why, for example, $\sum n^k/n!$ is e times the kth Bell number. The rearrangements here are justified by absolute convergence, a key theme in first-year analysis: see Abbott, *Understanding Analysis*. For the probability side, see Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP III 2012, Question 5

Pure • Rational Points on Conics

STEP III 2012, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q5

The key idea

The identity in (i)(b) is a rotation in disguise: the point $(\cos\theta + \sqrt{m}\sin\theta,\ \sin\theta - \sqrt{m}\cos\theta)$ is what you get by rotating the point $(1, -\sqrt{m})$ through the angle θ, so it stays on the circle of radius $\sqrt{1+m}$. If $\cos\theta$ and $\sin\theta$ are both rational (such as $\frac{3}{5}$ and $\frac{4}{5}$, from (i)(a)), the rotation turns one "nice" point into another. Everything in part (ii) is the same trick, starting from a suitable integer point, and for the hyperbola we swap the rotation for its hyperbolic cousin.

Solution

Part (i)

(a) The point $(1,0)$ is an integer rational point on $x^2 + y^2 = 1$. For a non-integer one we use the $3,4,5$ triangle: $\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = \frac{9+16}{25} = 1$, so $\left(\frac{3}{5}, \frac{4}{5}\right)$ works, and neither coordinate is an integer.

(b) The point $(1,1)$ is an integer rational point on $x^2 + y^2 = 2$.

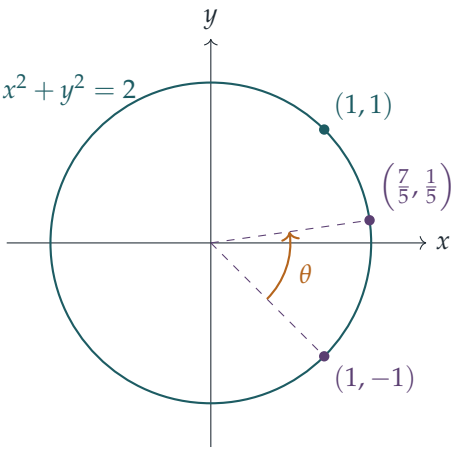
Expanding, the cross terms are $+2\sqrt{m}\sin\theta\cos\theta$ and $-2\sqrt{m}\sin\theta\cos\theta$, which cancel, so

$$\begin{aligned}(\cos\theta + \sqrt{m}\sin\theta)^2 + (\sin\theta - \sqrt{m}\cos\theta)^2 &= \cos^2\theta + m\sin^2\theta + \sin^2\theta + m\cos^2\theta \\ &= (1+m)(\cos^2\theta + \sin^2\theta) = 1 + m.\end{aligned}$$

Take $m = 1$, so the right-hand side is 2, and choose θ with $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$ (possible because $\left(\frac{3}{5}, \frac{4}{5}\right)$ lies on the unit circle). Then

$$x = \tfrac{3}{5} + \tfrac{4}{5} = \tfrac{7}{5}, \qquad y = \tfrac{4}{5} - \tfrac{3}{5} = \tfrac{1}{5}$$

satisfy $x^2 + y^2 = 2$; indeed $\frac{49+1}{25} = 2$. So $\left(\frac{7}{5}, \frac{1}{5}\right)$ is a non-integer rational point on $x^2 + y^2 = 2$.



The picture shows what happened: rotating $(1, -1)$ anticlockwise through θ, where $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$, lands on $\left(\frac{7}{5}, \frac{1}{5}\right)$.

Part (ii)

(a) Since $1^2 + (\sqrt{2})^2 = 3$, the point $(1, \sqrt{2})$, that is $p = 1$, $q = 0$, $r = 0$, $s = 1$, is an integer 2-rational point on $x^2 + y^2 = 3$.

For a non-integer one, use the identity from (i)(b) with $m = 2$, so that $1 + m = 3$, and again $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$:

$$x = \tfrac{3}{5} + \tfrac{4}{5}\sqrt{2}, \qquad y = \tfrac{4}{5} - \tfrac{3}{5}\sqrt{2}.$$

Here $p = \frac{3}{5}$, $q = \frac{4}{5}$, $r = \frac{4}{5}$, $s = -\frac{3}{5}$, none of which is an integer, and $x^2 + y^2 = 3$ by the identity. As a direct check, $x^2 = \frac{9+32+24\sqrt{2}}{25}$ and $y^2 = \frac{16+18-24\sqrt{2}}{25}$, which add to $\frac{75}{25} = 3$.

Ask yourself How can we reach 11, when the identity only gives $1 + m$ and m has to involve $\sqrt{2}$?

Scale the starting point. The identity came from rotating $(1, -\sqrt{m})$; we can just as well rotate any point $(a, -b\sqrt{2})$. The same cancellation of cross terms gives

$$(a\cos\theta + b\sqrt{2}\sin\theta)^2 + (a\sin\theta - b\sqrt{2}\cos\theta)^2 = (a^2 + 2b^2)(\cos^2\theta + \sin^2\theta) = a^2 + 2b^2.$$

So we need integers with $a^2 + 2b^2 = 11$, and $a = 3$, $b = 1$ works ($9 + 2 = 11$). In other words, we start from the integer 2-rational point $(3, \sqrt{2})$ on $x^2 + y^2 = 11$.

(b) With $a = 3$, $b = 1$, $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$:

$$x = \tfrac{9}{5} + \tfrac{4}{5}\sqrt{2}, \qquad y = \tfrac{12}{5} - \tfrac{3}{5}\sqrt{2}.$$

So $p = \frac{9}{5}$, $q = \frac{4}{5}$, $r = \frac{12}{5}$, $s = -\frac{3}{5}$, none an integer, and $x^2 + y^2 = 11$. Directly: $x^2 = \frac{81+32+72\sqrt{2}}{25}$ and $y^2 = \frac{144+18-72\sqrt{2}}{25}$, which add to $\frac{275}{25} = 11$.

(c) For the hyperbola we want a "rotation" that preserves $x^2 - y^2$ rather than $x^2 + y^2$. If $c^2 - s^2 = 1$ (think $c = \cosh t$, $s = \sinh t$), then

$$\begin{aligned}(ac + b\sqrt{2}\,s)^2 - (as + b\sqrt{2}\,c)^2 &= a^2c^2 + 2b^2s^2 - a^2s^2 - 2b^2c^2 \\ &= (a^2 - 2b^2)(c^2 - s^2) = a^2 - 2b^2,\end{aligned}$$

because the cross terms $2\sqrt{2}\,abcs$ cancel again. We need $a^2 - 2b^2 = 7$: take $a = 3$, $b = 1$ (so $(3, \sqrt{2})$ is an integer 2-rational point on the hyperbola).

Now we need rational c, s with $c^2 - s^2 = (c - s)(c + s) = 1$. Choosing $c + s = 2$ and $c - s = \frac{1}{2}$ gives $c = \frac{5}{4}$, $s = \frac{3}{4}$. Then

$$x = \tfrac{15}{4} + \tfrac{3}{4}\sqrt{2}, \qquad y = \tfrac{9}{4} + \tfrac{5}{4}\sqrt{2},$$

with $p = \frac{15}{4}$, $q = \frac{3}{4}$, $r = \frac{9}{4}$, $s = \frac{5}{4}$, none an integer. Directly: $x^2 = \frac{225+18+90\sqrt{2}}{16}$ and $y^2 = \frac{81+50+90\sqrt{2}}{16}$, so $x^2 - y^2 = \frac{112}{16} = 7$. This is a non-integer 2-rational point on $x^2 - y^2 = 7$.

Checks

Every point can be checked exactly with fractions, as done above, and numerically: for (ii)(c), $x \approx 4.81066$ and $y \approx 4.01777$, giving $x^2 - y^2 \approx 7.0000$; for (ii)(b), $x \approx 2.93137$ and $y \approx 1.55147$, giving $x^2 + y^2 \approx 11.0000$. Always finish by scanning p, q, r, s to confirm none is an integer.

Common slips

- Forgetting that 0 is an integer: a point like $\left(\frac{7}{5}\sqrt{2}, \dots\right)$ has $p = 0$, so it is not a non-integer 2-rational point.
- In (c), an innocent-looking choice can fail: $c = \frac{5}{3}$, $s = \frac{4}{3}$ also satisfies $c^2 - s^2 = 1$, but gives $x = 5 + \frac{4}{3}\sqrt{2}$, with $p = 5$ an integer. Check every coefficient.

Beyond STEP

Numbers of the form $p + q\sqrt{2}$ form a field, $\mathbb{Q}(\sqrt{2})$, and the question is really about finding points on conics over that field. The trick of rotating a known point by a rational angle is the start of a beautiful theory: once a conic has one rational point, lines through it parametrise all the others, and the rational points on $x^2 - 2y^2 = 1$ (Pell's equation) form a group under exactly the hyperbolic "rotation" used in (c). Davenport's *The Higher Arithmetic* covers Pell's equation and sums of squares, and Allenby's *Rings, Fields and Groups* introduces fields like $\mathbb{Q}(\sqrt{2})$.

WORKED SOLUTION

STEP III 2012, Question 6

Pure • Complex Root Loci

STEP III 2012, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q6

The key idea

A complex number is zero exactly when its real and imaginary parts are both zero, so one complex equation in $z = x + \mathrm{i}y$ becomes two real equations in x, y and p. The imaginary-part equation always factorises with a factor y, which splits every locus into "real roots" and "non-real roots". In each case we eliminate p to get a curve in the (x, y)-plane, and then check which points of that curve really come from a real value of p.

Solution

The first equation, $z^2 + pz + 1 = 0$

Put $z = x + \mathrm{i}y$ with x, y real. Then

$$z^2 + pz + 1 = (x^2 - y^2 + px + 1) + \mathrm{i}\,(2xy + py).$$

Since p is real, the two brackets are the real and imaginary parts, and both must vanish:

$$x^2 - y^2 + px + 1 = 0, \qquad (2x + p)y = 0,$$

as required.

From the second equation, either $p = -2x$ or $y = 0$. If $y = 0$, the first equation becomes $x^2 + px + 1 = 0$. Here $x \neq 0$, because $x = 0$ would give $1 = 0$, so we may divide by x to get $p = -(x^2 + 1)/x$. So either $p = -2x$, or $p = -(x^2 + 1)/x$ with $x \neq 0$ (and in this second case $y = 0$).

The root locus. *Case* $p = -2x$. Substituting into the real-part equation gives $x^2 - y^2 - 2x^2 + 1 = 0$, that is,

$$x^2 + y^2 = 1.$$

Conversely, take any point $x + \mathrm{i}y$ on the unit circle and set $p = -2x$, which is real. Then $(2x + p)y = 0$ and $x^2 - y^2 + px + 1 = 1 - x^2 - y^2 = 0$, so the point is a root. Hence the whole unit circle is in the locus.

Case $y = 0$, $p = -(x^2+1)/x$. For every real $x \neq 0$ this p is real, and x is a root of $x^2 + px + 1 = 0$. The point $x = 0$ is never a root, since $0^2 + p \cdot 0 + 1 = 1$.

So the root locus is the real axis with the origin missing, together with the unit circle $|z| = 1$. (The points ± 1 lie on both.)

The second equation, $pz^2 + z + 1 = 0$

With $z = x + iy$, the real and imaginary parts give

$$p(x^2 - y^2) + x + 1 = 0, \qquad (2px + 1)y = 0.$$

The origin is never a root, since $z = 0$ gives $1 = 0$.

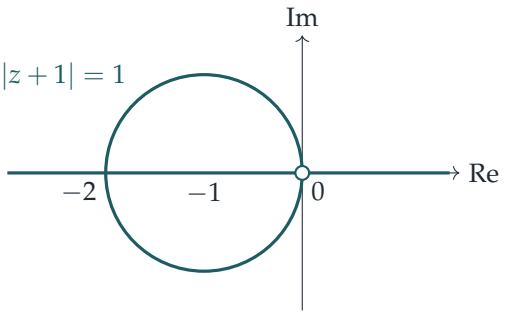
Case $y = 0$. Then $px^2 + x + 1 = 0$ with $x \neq 0$, so $p = -(x+1)/x^2$. This is real for every real $x \neq 0$, so every point of the real axis except the origin is in the locus. (When $x = -1$ we get $p = 0$, and the equation is the linear one $z + 1 = 0$, whose root is indeed -1.)

Case $2px + 1 = 0$. Then $x \neq 0$ and $p = -1/(2x)$. Substituting into the real-part equation and multiplying by $-2x$:

$$(x^2 - y^2) - 2x^2 - 2x = 0 \iff x^2 + y^2 + 2x = 0 \iff (x+1)^2 + y^2 = 1.$$

Conversely, for any point on this circle other than the origin we have $x \neq 0$, so $p = -1/(2x)$ is real, and reversing the steps shows both equations hold.

So the root locus is the real axis together with the circle $|z + 1| = 1$ (centre -1, radius 1), with the origin, which lies on both, removed.



The open circle marks the missing point $z = 0$.

Ask yourself Why does this locus look like the first one turned inside out?

If z is a root of $pz^2 + z + 1 = 0$ then $w = 1/z$ satisfies $w^2 + w + p = 0$. The map $z \mapsto 1/z$ sends the real axis (without 0) to itself and the line $\operatorname{Re} w = -\frac{1}{2}$, where the non-real roots of $w^2 + w + p = 0$ live, to the circle $|z + 1| = 1$ through the origin. The origin is missing because it would correspond to $w = \infty$.

The third equation, $pz^2 + p^2z + 2 = 0$

If $p = 0$ the equation reads $2 = 0$, which has no roots, so $p \neq 0$ throughout. The real and imaginary parts give

$$p(x^2 - y^2) + p^2x + 2 = 0, \qquad p(2x + p)y = 0,$$

and since $p \neq 0$ the second becomes $(2x + p)y = 0$.

Case $p = -2x$. Then $x \neq 0$, because $p \neq 0$. Substituting,

$$-2x(x^2 - y^2) + 4x^3 + 2 = 0 \quad \Longleftrightarrow \quad x(x^2 + y^2) = -1.$$

Conversely, any point on this curve has $x \neq 0$, so $p = -2x$ is a non-zero real number, and reversing the steps shows it is a root. Since $x^2 + y^2 > 0$, the curve has $x < 0$. Writing it as

$$y^2 = -\frac{1}{x} - x^2,$$

we need $-1/x \geq x^2$; multiplying by $x < 0$ reverses the inequality to $-1 \leq x^3$, so $-1 \leq x < 0$. The curve passes through -1 (where $y = 0$), is symmetric in the real axis, and $y^2 \to \infty$ as $x \to 0^-$, so the imaginary axis is an asymptote. Also $\mathrm{d}(y^2)/\mathrm{d}x = x^{-2} - 2x > 0$ for $x < 0$, so $|y|$ increases steadily as x goes from -1 to 0. In polar form the curve is $r^3 \cos\theta = -1$.

Case $y = 0$. Now a real number x is in the locus when there is a real p with $xp^2 + x^2p + 2 = 0$. For $x = 0$ this says $2 = 0$, so 0 is not in the locus. For $x \neq 0$ it is a quadratic in p, with a real solution exactly when its discriminant is non-negative:

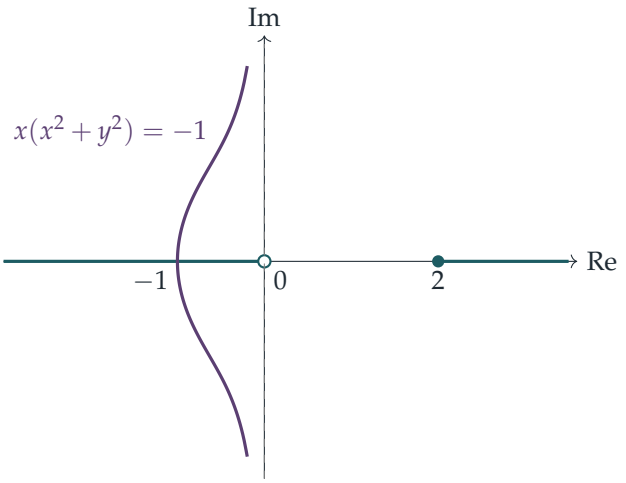
$$x^4 - 8x = x(x^3 - 8) \geq 0.$$

This holds when $x < 0$ (both factors negative) or when $x \geq 2$ (both factors non-negative), and fails for $0 < x < 2$. Any real p found this way is automatically non-zero, since $p = 0$ would give $2 = 0$. For example $x = 2$ needs $2p^2 + 4p + 2 = 0$, so $p = -1$, and indeed $-z^2 + z + 2 = -(z - 2)(z + 1)$.

Ask yourself Why solve for p here rather than for x?

We want to know which points x are attainable, so x is given and p is the unknown. The equation is quadratic in p, and the existence of a real root is exactly a discriminant condition on x.

So the root locus is the negative real axis, the ray $x \geq 2$ of the real axis, and the curve $x(x^2 + y^2) = -1$, which runs from -1 out towards the imaginary axis in both directions.



The curve meets the real axis only at -1, with a vertical tangent there, and approaches the imaginary axis as $|y| \to \infty$.

Checks

- For $z^2 + pz + 1 = 0$ the product of the roots is 1, so non-real roots are conjugates $z, \bar{z}$ with $|z|^2 = 1$: the unit circle again.
- In the third equation, $p = -1$ gives roots 2 and -1, the boundary point of the ray and the vertex of the curve. Taking $p = 1$ gives $z^2 + z + 2 = 0$, with roots $-\frac{1}{2} \pm \frac{\sqrt{7}}{2}\mathrm{i}$, and $x(x^2 + y^2) = -\frac{1}{2} \cdot 2 = -1$, as it should.
- Numerically, we solved all three equations for 10^5 random values of p and confirmed every root lies on the stated locus. We also confirmed that random points of each locus are attained with the stated p, and a fine scan of p found no real root of the third equation in $(0, 2)$; the smallest positive one was 2.0.

Common slips

- Forgetting to check the converse: eliminating p gives a curve that *contains* the locus, and you still have to show each point is reached by a real p. In the third equation, real points in $(0, 2)$ satisfy $y = 0$ but are never reached.
- Dividing by x or p without saying why it is non-zero. This is exactly where the missing origin comes from.
- Treating $p = 0$ carelessly: in the second equation it gives the genuine root -1, but in the third it gives no roots at all.

Beyond STEP

Root loci are a standard tool in control engineering: as a gain parameter varies, the poles of a system move along curves, and stability depends on whether they cross into the right half-plane. The question also hints at the geometry of the map $z \mapsto 1/z$, which sends lines and circles to lines and circles. This is the start of the theory of Möbius transformations, beautifully illustrated in Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP III 2012, Question 7

Pure • Coupled Differential Equations

STEP III 2012, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q7

The key idea

Two coupled first-order equations become one second-order equation once we eliminate a variable: differentiate the $\dot{y}$ equation, then use the first equation itself to write z in terms of y and $\dot{y}$. After that, each part is just choosing the constants A and B. In part (iii) the infinite sum is two geometric series, and matching the result with part (ii) pins down c. Physically, z_2 is the steady state reached by giving the same injection once every unit of time.

Solution

The general solution

Differentiating $\dot{y} = -2y + 2z$ and then substituting the second equation for $\dot{z}$,

$$\ddot{y} = -2\dot{y} + 2\dot{z} = -2\dot{y} + 2(-\dot{y} - 3z) = -4\dot{y} - 6z.$$

The first equation gives z in terms of y:

$$z = y + \tfrac{1}{2}\dot{y}, \qquad \text{so} \qquad 6z = 6y + 3\dot{y}.$$

Substituting,

$$\ddot{y} + 7\dot{y} + 6y = 0.$$

The auxiliary equation $m^2 + 7m + 6 = (m+1)(m+6) = 0$ has roots $m = -1$ and $m = -6$, so

$$y = Ae^{-t} + Be^{-6t}$$

for arbitrary constants A and B. Then z follows from $z = y + \frac{1}{2}\dot{y}$:

$$z = Ae^{-t}\left(1 - \tfrac{1}{2}\right) + Be^{-6t}(1 - 3) = \tfrac{1}{2}Ae^{-t} - 2Be^{-6t},$$

as required.

Ask yourself We used both equations to get the equation for y, but only the first to find z. Does the pair we found really satisfy the second equation?

It does. With $z = y + \frac{1}{2}\dot{y}$ we have $\dot{z} = \dot{y} + \frac{1}{2}\ddot{y} = \dot{y} + \frac{1}{2}(-7\dot{y} - 6y) = -\frac{5}{2}\dot{y} - 3y$. Also $-\dot{y} - 3z = -\dot{y} - 3y - \frac{3}{2}\dot{y} = -\frac{5}{2}\dot{y} - 3y$, which is the same. So every solution of the system has this form, and every pair of this form is a solution.

Part (i)

At $t = 0$ we have $y(0) = A + B$ and $z(0) = \frac{1}{2}A - 2B$. The conditions give

$$\tfrac{1}{2}A - 2B = 0, \qquad A + B = 5,$$

so $A = 4B$, and then $5B = 5$. Hence $B = 1$ and $A = 4$:

$$y = 4\mathrm{e}^{-t} + \mathrm{e}^{-6t}, \qquad z_1(t) = 2\mathrm{e}^{-t} - 2\mathrm{e}^{-6t}.$$

The drug starts in the blood, the amount in the brain rises from zero, and then it decays as it is absorbed.

Part (ii)

Write $a = \frac{1}{2}A$ and $b = 2B$, so that $z = a\mathrm{e}^{-t} - b\mathrm{e}^{-6t}$. The conditions are

$$a - b = c, \qquad a\mathrm{e}^{-1} - b\mathrm{e}^{-6} = c.$$

Subtracting the second from the first gives $a(1 - \mathrm{e}^{-1}) = b(1 - \mathrm{e}^{-6})$. Call this common value k, so that

$$a = \frac{k}{1 - \mathrm{e}^{-1}}, \qquad b = \frac{k}{1 - \mathrm{e}^{-6}}, \qquad \text{and} \qquad c = a - b = kC,$$

where

$$C = \frac{1}{1 - \mathrm{e}^{-1}} - \frac{1}{1 - \mathrm{e}^{-6}}.$$

Since $\mathrm{e}^{-1} > \mathrm{e}^{-6}$, we have $1 - \mathrm{e}^{-1} < 1 - \mathrm{e}^{-6}$, so $C > 0$. In particular $C \neq 0$, and we may divide by it: $k = c/C$. (Equivalently, the two linear equations have determinant $\mathrm{e}^{-1} - \mathrm{e}^{-6} \neq 0$, so the solution is unique.) Therefore

$$z_2(t) = \frac{c}{C}\left(\frac{\mathrm{e}^{-t}}{1 - \mathrm{e}^{-1}} - \frac{\mathrm{e}^{-6t}}{1 - \mathrm{e}^{-6}}\right),$$
$$y_2(t) = \frac{c}{C}\left(\frac{2\mathrm{e}^{-t}}{1 - \mathrm{e}^{-1}} + \frac{\mathrm{e}^{-6t}}{2(1 - \mathrm{e}^{-6})}\right),$$

where y_2 comes from $A = 2a$ and $B = \frac{1}{2}b$.

Part (iii)

Put $m = -n$, so that m runs over $0, 1, 2, \ldots$ and $t - n = t + m$. Using part (i),

$$\sum_{n=-\infty}^{0} z_1(t-n) = \sum_{m=0}^{\infty} \left(2\mathrm{e}^{-t}(\mathrm{e}^{-1})^m - 2\mathrm{e}^{-6t}(\mathrm{e}^{-6})^m\right).$$

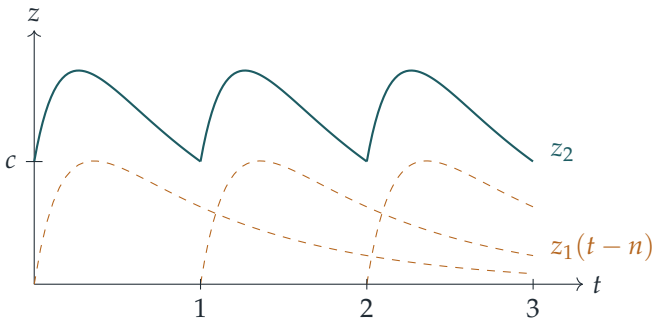
Both pieces are geometric series with ratios e^{-1} and e^{-6}, which lie strictly between 0 and 1, so each converges and we may sum them separately:

$$\sum_{n=-\infty}^{0} z_1(t-n) = \frac{2\mathrm{e}^{-t}}{1-\mathrm{e}^{-1}} - \frac{2\mathrm{e}^{-6t}}{1-\mathrm{e}^{-6}}.$$

This is exactly the bracket in $z_2(t)$, multiplied by 2. So the identity holds for $0 \le t \le 1$ precisely when $c/C = 2$, that is, when

$$c = 2C = \frac{2}{1-\mathrm{e}^{-1}} - \frac{2}{1-\mathrm{e}^{-6}} \approx 1.159.$$

(For any other c the two sides differ by a non-zero multiple of the same non-zero function, so this value is the only one.)



The picture shows the meaning. A dose of 5 injected into the blood at time n produces the brain curve $z_1(t-n)$ (dashed). If a dose is given at every integer time, from the distant past, the brain quantity is the sum of all these responses. That sum repeats every unit of time (solid curve), and on $[0, 1]$ it is z_2 with this value of c.

Checks

- In part (i), $z_1(0) = 2 - 2 = 0$ and $y(0) = 4 + 1 = 5$.
- In part (iii), $z_2(0)$ and $z_2(1)$ should both equal c. In the sum, the $t = 1$ series is the $t = 0$ series with its first term removed, and that term is $z_1(0) = 0$. This confirms $z(0) = z(1)$ without any algebra.
- With $c = 2C$, $y_2(0) - y_2(1) = 4 + 1 = 5$: over each period the blood loses exactly one dose, which the next injection replaces.
- Numerically, $c \approx 1.15898$. Summing 200 terms of $z_1(t+m)$ at $t = 0.37$ gives $1.967693\ldots$, matching $z_2(0.37)$ to 12 decimal places.

Common slips

- Forgetting that z is fixed by y through $z = y + \frac{1}{2}\dot{y}$, and solving a second equation for z with new, independent constants.
- Getting the direction of the shift wrong in part (iii): $n \leq 0$ means $t - n \geq t$, so only doses from the past contribute.

Beyond STEP

This is a linear compartment model, written as $\dot{\mathbf{x}} = M\mathbf{x}$ with a 2×2 matrix M. The decay rates 1 and 6 are minus the eigenvalues of M, which is the matrix-exponential view of linear systems that you meet in a first-year course. Part (iii) is a simple case of superposition: the response to repeated impulses is a sum of shifted copies of one impulse response. This idea leads to Green's functions and convolution. Boyce and DiPrima, *Elementary Differential Equations*, covers systems of linear equations and impulse responses.

WORKED SOLUTION

STEP III 2012, Question 8

Pure • Fibonacci Telescoping Series

STEP III 2012, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q8

The key idea

The expression in (ii) changes sign every time n goes up by one, and part (i) is a nudge to look for exactly that kind of pattern. Once we know its value for odd n, the arctan identity in (iii) is just the $\tan(A-B)$ formula in disguise. Rearranged, that identity writes each term of the series as a *difference* of two arctans, so the sum telescopes.

Solution

The first few terms are

$$F_0, F_1, \ldots, F_8 = 0,\ 1,\ 1,\ 2,\ 3,\ 5,\ 8,\ 13,\ 21.$$

Part (i)

Using these values,

$$F_0F_3 - F_1F_2 = 0\cdot 2 - 1\cdot 1 = -1, \qquad F_2F_5 - F_3F_4 = 1\cdot 5 - 2\cdot 3 = -1,$$

so the two expressions are equal, as required.

Part (ii)

Let $G_n = F_nF_{n+3} - F_{n+1}F_{n+2}$. Part (i) says $G_0 = G_2$, which suggests comparing G_{n+1} with G_n. Using the recurrence in the form $F_{n+4} = F_{n+3} + F_{n+2}$,

$$\begin{aligned} G_{n+1} &= F_{n+1}F_{n+4} - F_{n+2}F_{n+3} \\ &= F_{n+1}F_{n+3} + F_{n+1}F_{n+2} - F_{n+2}F_{n+3} \\ &= F_{n+1}F_{n+2} - F_{n+3}\left(F_{n+2} - F_{n+1}\right) \\ &= F_{n+1}F_{n+2} - F_{n+3}F_n \\ &= -G_n, \end{aligned}$$

where in the fourth line we used the recurrence again, $F_{n+2} - F_{n+1} = F_n$.

So G_n changes sign at each step. Since $G_0 = -1$ from part (i), it follows (by induction on n) that $G_n = (-1)^{n+1}$. The two cases are

$$F_n F_{n+3} - F_{n+1}F_{n+2} = \begin{cases} -1 & \text{if } n \text{ is even,} \\ 1 & \text{if } n \text{ is odd.} \end{cases}$$

For example, $n = 1$ gives $1 \cdot 3 - 1 \cdot 2 = 1$, and $n = 4$ gives $3 \cdot 13 - 5 \cdot 8 = -1$.

Part (iii)

Fix $r \geq 1$ and let

$$\alpha = \arctan \frac{1}{F_{2r}}, \qquad \beta = \arctan \frac{1}{F_{2r+1}}.$$

All the F_k with $k \geq 1$ are positive integers, so these are well defined. By the compound-angle formula,

$$\tan(\alpha - \beta) = \frac{\dfrac{1}{F_{2r}} - \dfrac{1}{F_{2r+1}}}{1 + \dfrac{1}{F_{2r}F_{2r+1}}} = \frac{F_{2r+1} - F_{2r}}{F_{2r}F_{2r+1} + 1} = \frac{F_{2r-1}}{F_{2r}F_{2r+1} + 1},$$

using $F_{2r+1} - F_{2r} = F_{2r-1}$. Now apply part (ii) with the *odd* value $n = 2r - 1$:

$$F_{2r-1}F_{2r+2} - F_{2r}F_{2r+1} = 1, \qquad \text{so} \qquad F_{2r}F_{2r+1} + 1 = F_{2r-1}F_{2r+2}.$$

Since $F_{2r-1} \geq 1$, it is non-zero and cancels:

$$\tan(\alpha - \beta) = \frac{F_{2r-1}}{F_{2r-1}F_{2r+2}} = \frac{1}{F_{2r+2}}.$$

Ask yourself Equal tangents don't always mean equal angles. Why may we conclude $\alpha - \beta = \arctan(1/F_{2r+2})$?

Because tan is one-to-one on $(-\frac{\pi}{2}, \frac{\pi}{2})$, it is enough to check that both angles lie in that interval. Here $0 < \beta \leq \alpha \leq \frac{\pi}{4}$, because $1 \leq F_{2r} \leq F_{2r+1}$, so $0 \leq \alpha - \beta < \frac{\pi}{4}$. Also $\arctan(1/F_{2r+2})$ lies in $(0, \frac{\pi}{2})$ by definition. Hence

$$\arctan \frac{1}{F_{2r}} = \arctan \frac{1}{F_{2r+1}} + \arctan \frac{1}{F_{2r+2}} \qquad (r = 1, 2, 3, \ldots),$$

which is the required identity.

The sum. Rearranged, the identity reads

$$\arctan \frac{1}{F_{2r+1}} = \arctan \frac{1}{F_{2r}} - \arctan \frac{1}{F_{2r+2}},$$

so the partial sums telescope:

$$\begin{aligned}\sum_{r=1}^{N} \arctan\frac{1}{F_{2r+1}} &= \left(\arctan\frac{1}{F_2} - \arctan\frac{1}{F_4}\right) + \left(\arctan\frac{1}{F_4} - \arctan\frac{1}{F_6}\right) + \cdots \\ &\quad + \left(\arctan\frac{1}{F_{2N}} - \arctan\frac{1}{F_{2N+2}}\right) \\ &= \arctan\frac{1}{F_2} - \arctan\frac{1}{F_{2N+2}} = \frac{\pi}{4} - \arctan\frac{1}{F_{2N+2}},\end{aligned}$$

since $F_2 = 1$. For $n \geq 2$ we have $F_{n+1} = F_n + F_{n-1} \geq F_n + 1$, so $F_n \to \infty$. Then $1/F_{2N+2} \to 0$, and since arctan is continuous with $\arctan 0 = 0$, letting $N \to \infty$ gives

$$\sum_{r=1}^{\infty} \arctan\frac{1}{F_{2r+1}} = \frac{\pi}{4}.$$

Checks

- $r = 1$: the identity says $\arctan 1 = \arctan\frac{1}{2} + \arctan\frac{1}{3}$. This is the classic $\frac{\pi}{4}$ identity, since tan of the right-hand side is $\left(\frac{1}{2} + \frac{1}{3}\right)/\left(1 - \frac{1}{6}\right) = 1$.
- The first three terms $\arctan\frac{1}{2} + \arctan\frac{1}{5} + \arctan\frac{1}{13} \approx 0.4636 + 0.1974 + 0.0768 = 0.7378$, which matches $\frac{\pi}{4} - \arctan\frac{1}{21} \approx 0.7378$.
- Numerically, 89 terms of the series give $0.785398163\ldots$, agreeing with $\frac{\pi}{4}$ to machine precision. The formula of part (ii) also holds exactly for every $n < 150$.

Common slips

- Using the even case of (ii) in part (iii). The indices there are $2r - 1$, $2r$, $2r + 1$, $2r + 2$, so $n = 2r - 1$ is odd and the value is $+1$.
- Going from equal tangents to equal angles without checking that both angles lie in $(-\frac{\pi}{2}, \frac{\pi}{2})$.
- Forgetting the first term of the telescoped sum: it is $\arctan(1/F_2) = \frac{\pi}{4}$, not $\arctan(1/F_1)$.

Beyond STEP

Part (ii) is a cousin of Cassini's identity $F_{n-1}F_{n+1} - F_n^2 = (-1)^n$. Both are best understood through the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, whose nth power has the Fibonacci numbers as entries, so that taking determinants gives Cassini's identity at once. Telescoping series, and the care needed to pass from partial sums to a limit, are a staple of a first analysis course. Abbott's *Understanding Analysis* is a good place to read about them, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with recurrences.

WORKED SOLUTION

STEP III 2012, Question 9

Mechanics • Rotating Pulley Dynamics

STEP III 2012, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

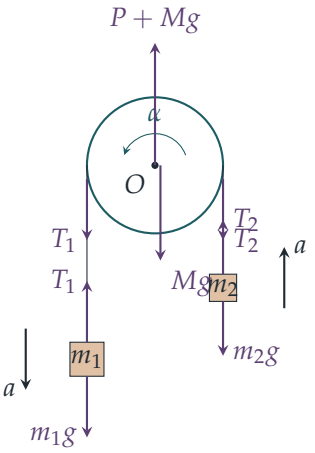
step.maths.org/questions/12-s3-q9

The key idea

The measured force on the axle is the clue. The centre of mass of the pulley doesn't move, so the axle must hold up the pulley's weight and also balance the two string tensions. That makes P the sum of the tensions, and from that the acceleration of the particles follows. Once we know the acceleration, the pulley's rotational equation gives the moment of inertia. A friction couple changes that one equation and nothing else.

Solution

Model. The particles are point masses and the string is light and inextensible. The string doesn't slip on the pulley, so its speed equals the rim speed of the disc. We take the system to have been released from rest, so the heavier particle m_1 descends. Let a be the acceleration of m_1 downwards, which is also the acceleration of m_2 upwards. The pulley's angular acceleration is then $\alpha = a/r$. The tension is T_1 in the string on the m_1 side and T_2 on the m_2 side (the two differ because the pulley has inertia). Let I be the moment of inertia about the axle.



Newton's second law for each particle gives

$$m_1 g - T_1 = m_1 a, \qquad T_2 - m_2 g = m_2 a,$$

so

$$T_1 = m_1(g - a), \qquad T_2 = m_2(g + a).$$

The axle force. The centre of mass of the pulley is at O, which is fixed, so the net force on the pulley is zero. The forces acting on it are its weight Mg, the two tensions (both pulling down, since the string hangs vertically on each side) and the force from the axle. A friction couple (which we add in part (ii)) has no resultant force, so it doesn't enter here. The vertical force on the axle is therefore $Mg + T_1 + T_2$, and comparing with the measurement gives

$$P = T_1 + T_2 = (m_1 + m_2)g - (m_1 - m_2)a,$$

so

$$a = \frac{(m_1 + m_2)g - P}{m_1 - m_2}. \tag{1}$$

This uses only the measurement and the particles' equations, so it holds in both parts.

Part (i)

With no friction at the axle, only the tensions have moments about O. Taking moments about the axle,

$$(T_1 - T_2)r = I\alpha = \frac{Ia}{r}, \qquad \text{so} \qquad I = \frac{(T_1 - T_2)r^2}{a}.$$

Now $T_1 - T_2 = (m_1 - m_2)g - (m_1 + m_2)a$, so

$$I = r^2\left(\frac{(m_1 - m_2)g}{a} - (m_1 + m_2)\right).$$

Substituting (1) for a:

$$\begin{aligned} I &= r^2\left(\frac{(m_1-m_2)^2 g}{(m_1+m_2)g-P} - (m_1+m_2)\right) \\ &= r^2\,\frac{(m_1-m_2)^2 g - (m_1+m_2)^2 g + (m_1+m_2)P}{(m_1+m_2)g-P} \\ &= \frac{\big((m_1+m_2)P - 4m_1m_2 g\big)r^2}{(m_1+m_2)g-P}, \end{aligned}$$

using $(m_1+m_2)^2-(m_1-m_2)^2 = 4m_1m_2$. This is $(*)$, as required.

Part (ii)

Call the new calculated value I'. The friction couple C opposes the rotation, and the rotation is in the sense driven by T_1, so the moment equation becomes

$$(T_1-T_2)r - C = \frac{I'a}{r}.$$

The acceleration is still given by (1), because the same measured P is used and the axle-force argument doesn't involve C. So T_1-T_2 and a are exactly as in part (i), and

$$I' = \frac{(T_1-T_2)r^2}{a} - \frac{Cr}{a} = (*) - \frac{Cr}{a}.$$

? **Ask yourself** How do we know the sign of a?

The system starts from rest and the pulley turns with m_1 going down, so m_1 must have a positive downward acceleration. (If a were zero or negative, m_1 would never start moving downwards.) So $a>0$. Since $C>0$ and $r>0$, we get $Cr/a>0$, and **the value calculated with friction is smaller than** $(*)$. This makes sense: part of the net torque from the tensions goes into overcoming friction, so less inertia is needed to explain the same observed acceleration.

The bound on C. Substitute the tensions into the moment equation:

$$\big((m_1-m_2)g - (m_1+m_2)a\big)r - C = \frac{I'a}{r},$$

which rearranges to

$$(m_1-m_2)rg - C = a\left(\frac{I'}{r} + (m_1+m_2)r\right).$$

On the right, $a>0$, $I'\geq 0$ (a moment of inertia can't be negative) and $(m_1+m_2)r>0$, so the right-hand side is strictly positive. Hence

$$C < (m_1-m_2)rg,$$

as required. In words, the friction couple must be less than the couple $(m_1-m_2)gr$ from the out-of-balance weight. If it weren't, the system wouldn't start to move at all.

Checks

- *Light pulley:* if $I = 0$ and $C = 0$ then $a = \frac{(m_1 - m_2)g}{m_1 + m_2}$ (the familiar Atwood result) and $P = \frac{4m_1 m_2 g}{m_1 + m_2}$. The numerator of $(*)$ then vanishes, as it should.
- *Stationary pulley:* as $a \to 0$ we get $P \to (m_1 + m_2)g$, the denominator of $(*)$ tends to zero and $(*)$ blows up. An infinitely heavy pulley doesn't accelerate.
- *Numerically:* with $g = 9.8$, $m_1 = 3$, $m_2 = 1$, $r = 0.5$, $I = 0.7$ and $C = 2$ (SI units), the true motion has $a \approx 2.29\ \mathrm{m\,s^{-2}}$ and $P \approx 34.6$ N. Formula $(*)$ gives $I \approx 1.14$, which is too big by exactly $Cr/a \approx 0.44$, while the friction model gives back 0.7. Here $C = 2 < (m_1 - m_2)rg = 9.8$. We tested 10^4 random cases the same way.

Common slips

- Taking the tension to be the same on both sides. That holds only for a light or frictionless pulley, and here it would make I disappear from the problem.
- Forgetting the pulley's own weight in the axle force. That is why the measurement is written as $P + Mg$.
- Thinking the friction couple changes the axle force. A couple has zero resultant, so $P = T_1 + T_2$ in both parts.

Beyond STEP

This is a small example of *inferring a hidden parameter from a measurement*. The answer depends on the model you assume, and leaving out friction biases the estimate in a predictable direction. In first-year mechanics, rigid bodies are treated with the full apparatus of linear and angular momentum, and the moment of inertia becomes a tensor for bodies rotating about general axes. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers fixed-axis rotation and Atwood machines with massive pulleys in depth.

WORKED SOLUTION

STEP III 2012, Question 10

Mechanics • Circular Motion

STEP III 2012, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

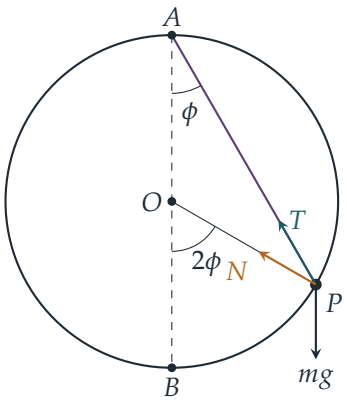
step.maths.org/questions/12-s3-q10

The key idea

Everything is geometry of the isosceles triangle formed by the centre of the hoop, the top of the hoop and the ring. Once we see that the string has length $2a\cos\phi$ and that the ring's radius makes angle 2ϕ with the downward vertical, the problem is one energy equation (for λ and the speed) and one radial equation of motion (for the reaction).

Solution

We model the ring as a particle moving on a smooth circle, with a light elastic string obeying Hooke's law. Let O be the centre of the hoop, A its highest point, B its lowest point and P the ring, and write $c = \cos\phi$.



Geometry

Triangle OAP has $OA = OP = a$, so it is isosceles and the angles at A and P are both ϕ. Hence

$$AP = 2a\cos\phi, \qquad \angle AOP = \pi - 2\phi, \qquad \angle BOP = 2\phi.$$

So the depth of P below O is $a\cos 2\phi$, and its height above B is $a(1-\cos 2\phi) = 2a\sin^2\phi$. The extension of the string (while taut) is $x = 2a\cos\phi - a = a(2c-1)$, and the string goes slack when $2c-1=0$, that is, at $\phi = \frac{\pi}{3}$.

The modulus of elasticity

The hoop is smooth and the reaction is perpendicular to the motion, so mechanical energy is conserved. At B ($\phi = 0$) the ring is at rest with extension a, so the elastic energy is $\lambda a^2/(2a) = \frac{1}{2}\lambda a$. At $\phi = \frac{\pi}{3}$ it is at rest again, the string has just become slack (no elastic energy), and it has risen $2a\sin^2\frac{\pi}{3} = \frac{3}{2}a$. Therefore

$$\tfrac{1}{2}\lambda a = mg\cdot\tfrac{3}{2}a \quad\Longrightarrow\quad \lambda = 3mg.$$

The speed

At a general angle ϕ with the string taut, energy conservation from B gives

$$\begin{aligned}\tfrac{1}{2}mv^2 &= \tfrac{1}{2}\lambda a - \frac{\lambda a^2(2c-1)^2}{2a} - 2mga\sin^2\phi\\ &= \tfrac{3}{2}mga\left[1-(2c-1)^2\right] - 2mga(1-c^2)\\ &= \tfrac{3}{2}mga(4c-4c^2) - 2mga + 2mgac^2\\ &= mga(-4c^2+6c-2),\end{aligned}$$

so

$$\frac{mv^2}{a} = mg(-8c^2+12c-4) = -4mg(2c-1)(c-1).$$

This vanishes at $c=1$ and $c=\frac{1}{2}$ and is positive in between, consistent with the ring moving between $\phi = -\frac{\pi}{3}$ and $\phi = \frac{\pi}{3}$: after coming to rest it slides back through B to the symmetric point on the other side, and so on. Throughout, $\frac{1}{2}\le c\le 1$ and the string is taut (just slack at the extreme points), with tension

$$T = \frac{\lambda x}{a} = 3mg(2c-1).$$

The reaction

Let N be the component of the reaction towards O, as drawn (it may turn out negative). We resolve towards the centre along PO:

- the tension acts along PA, which makes angle ϕ with PO (the base angle of the isosceles triangle), giving $T\cos\phi$ towards O;
- OP makes angle 2ϕ with the downward vertical, so the weight has component $mg\cos 2\phi$ directed *away* from O.

? **Ask yourself** Why is the weight's component outward rather than inward here?

For $\phi < \frac{\pi}{4}$ the ring is below O and gravity pulls it away from the centre; the single expression $-mg\cos 2\phi$ (towards O) covers both cases, because $\cos 2\phi$ changes sign when P passes the level of O.

Newton's second law towards the centre gives

$$N + T\cos\phi - mg\cos 2\phi = \frac{mv^2}{a}.$$

Substituting T, mv^2/a and $\cos 2\phi = 2c^2 - 1$:

$$\begin{aligned} N &= mg(-8c^2 + 12c - 4) - 3mg(2c-1)c + mg(2c^2 - 1) \\ &= mg(-8c^2 + 12c - 4 - 6c^2 + 3c + 2c^2 - 1) \\ &= -mg(12c^2 - 15c + 5). \end{aligned}$$

Hence the magnitude of the reaction is

$$R = (12\cos^2\phi - 15\cos\phi + 5)\,mg,$$

as required, provided the bracket is never negative, which we now show. (The minus sign tells us the hoop pushes the ring *outwards*, away from O, throughout.)

R is never zero

The quadratic $12c^2 - 15c + 5$ has discriminant $15^2 - 4\cdot 12\cdot 5 = 225 - 240 = -15 < 0$ and positive leading coefficient, so it is strictly positive for every real c. Completing the square,

$$12c^2 - 15c + 5 = 12\left(c - \tfrac{5}{8}\right)^2 + \tfrac{5}{16} \geq \tfrac{5}{16},$$

so $R \geq \frac{5}{16}mg > 0$ throughout the motion, with the minimum when $\cos\phi = \frac{5}{8}$ (which lies in $[\frac{1}{2}, 1]$, so it is actually attained).

Checks

- At B ($c = 1$): the ring is at rest, $T = 3mg$ upwards, weight mg downwards, so the hoop must push down with $2mg$. The formula gives $(12 - 15 + 5)mg = 2mg$.
- At the slack point ($c = \frac{1}{2}$): no tension and no speed, P is $\frac{a}{2}$ above O, so the radial reaction balances the inward weight component $mg\cos\frac{\pi}{3} = \frac{1}{2}mg$. The formula gives $(3 - 7.5 + 5)mg = \frac{1}{2}mg$.
- We integrated the equation of motion numerically (RK4, starting just off B): the ring stops at angle $2.094 \approx \frac{2\pi}{3}$ from B, the computed reaction matches the formula to within 10^{-7}, and its minimum is $0.3125\,mg = \frac{5}{16}mg$.

Common slips

- Taking the ring's angular position to be ϕ rather than 2ϕ: the angle at the circumference is half the angle at the centre.

- Using the full string length $2a\cos\phi$ instead of the extension $2a\cos\phi - a$ in Hooke's law or the elastic energy.
- Assuming the reaction points inwards and then being puzzled by a negative answer; the question asks only for its magnitude.

Beyond STEP

The ring has one degree of freedom, and the cleanest way to handle such systems is through an energy (or Lagrangian) formulation in a single coordinate, with constraint forces like R recovered afterwards from Newton's law. That split, between equations of motion from energy and constraint forces from resolving, is exactly how first-year mechanics courses treat beads on wires. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers circular motion with constraints and energy methods in this spirit.

WORKED SOLUTION

STEP III 2012, Question 11

Mechanics • Falling Folded Rope

STEP III 2012, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q11

The key idea

Everything follows from the geometry of the loop. Once we know how long the stationary side and the moving side are when the particle has fallen x, we can write down the potential energy lost (using the centre of mass of each straight section) and the kinetic energy gained (only the particle and the moving side have speed). Energy conservation then gives v^2 as a function of x, and the acceleration comes from differentiating $\frac{1}{2}v^2$ with respect to x, since $v\,\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}v}{\mathrm{d}t}$.

Solution

The model and the geometry

We treat the rope as a uniform, inextensible, perfectly flexible line of mass per unit length

$$\rho = \frac{2M}{2L} = \frac{M}{L},$$

the particle as a point mass, and we take the bend at the bottom of the loop to have negligible size. Measure depths downwards from P.

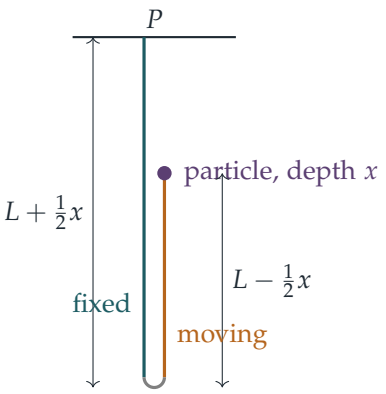
When the particle has fallen a distance x, the rope consists of two vertical sections joined at the bottom of the loop, at depth d say:

- the *fixed* section, from P down to the bottom, which is at rest and has length d;
- the *moving* section, from the particle (depth x) down to the bottom, which has length $d - x$.

The total length is $2L$, so $d + (d - x) = 2L$, giving

$$d = L + \tfrac{1}{2}x, \qquad \text{moving length } = L - \tfrac{1}{2}x.$$

(At the start, $x = 0$ and both sections have length L.)



Potential energy lost

Each straight section is uniform, so its centre of mass is at its midpoint. With depth h, a mass μ has potential energy $-\mu g h$. Write $D(x)$ for the sum of mass $\times$ depth over the whole system:

$$\begin{aligned} D(x) &= mx + \rho d \cdot \frac{d}{2} + \rho(d-x) \cdot \frac{x+d}{2} \\ &= mx + \frac{\rho}{2}(2d^2 - x^2) \\ &= mx + \rho\left(L^2 + Lx + \tfrac{1}{4}x^2 - \tfrac{1}{2}x^2\right) = mx + \rho\left(L^2 + Lx - \tfrac{1}{4}x^2\right). \end{aligned}$$

At the start $D(0) = \rho L^2$, so the potential energy lost is

$$g\big(D(x) - D(0)\big) = gx\left(m + \rho L - \tfrac{1}{4}\rho x\right) = \frac{gx}{L}\left(mL + ML - \tfrac{1}{4}Mx\right),$$

using $\rho L = M$.

Kinetic energy gained

The fixed section is at rest. The particle and every point of the moving section have speed v (as we are told to assume), so the moving mass is $m + \rho\left(L - \frac{1}{2}x\right)$ and

$$\text{KE} = \tfrac{1}{2}\left(m + M - \frac{Mx}{2L}\right)v^2 = \frac{1}{2L}\left(mL + ML - \tfrac{1}{2}Mx\right)v^2.$$

The system starts from rest, so energy conservation gives KE = PE lost:

$$\frac{1}{2L}\left(mL + ML - \tfrac{1}{2}Mx\right)v^2 = \frac{gx}{L}\left(mL + ML - \tfrac{1}{4}Mx\right),$$

that is,

$$v^2 = \frac{2gx\left(mL + ML - \frac{1}{4}Mx\right)}{mL + ML - \frac{1}{2}Mx},$$

as required. The denominator is never zero for $x < 2L$, since then $ML - \frac{1}{2}Mx > 0$, so $mL + ML - \frac{1}{2}Mx > mL \geq 0$.

The acceleration

For brevity write

$$A = mL + ML - \tfrac{1}{4}Mx, \qquad B = mL + ML - \tfrac{1}{2}Mx,$$

so that $\tfrac{1}{2}v^2 = gxA/B$, with $A' = -\tfrac{1}{4}M$ and $B' = -\tfrac{1}{2}M$ (differentiating with respect to x).

Ask yourself How do we get an acceleration from v^2 as a function of x?

By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}x}\left(\tfrac{1}{2}v^2\right) = v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}x}{\mathrm{d}t}\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}v}{\mathrm{d}t}$, which is the acceleration. This is valid while the particle is moving down, so x increases with t.

By the product and quotient rules,

$$\begin{aligned}\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{gxA}{B}\right) &= g\left[\frac{A}{B} + x\,\frac{A'B - AB'}{B^2}\right] \\ &= g\left[\frac{A}{B} + x\,\frac{-\tfrac{1}{4}MB + \tfrac{1}{2}MA}{B^2}\right] \\ &= g\left[\frac{A - \tfrac{1}{4}Mx}{B} + \frac{MxA}{2B^2}\right].\end{aligned}$$

The key simplification is that $A - \tfrac{1}{4}Mx = mL + ML - \tfrac{1}{2}Mx = B$, so the first fraction is exactly 1. Hence

$$\frac{\mathrm{d}v}{\mathrm{d}t} = g + \frac{Mgx\left(mL + ML - \tfrac{1}{4}Mx\right)}{2\left(mL + ML - \tfrac{1}{2}Mx\right)^2},$$

as required.

Acceleration exceeds g

After release, $0 < x < 2L$. Then

- $B = mL + M\left(L - \tfrac{1}{2}x\right) > 0$, as shown above, so $B^2 > 0$;
- $A = B + \tfrac{1}{4}Mx > B > 0$;
- $M > 0$, $g > 0$ and $x > 0$.

So the extra term $\dfrac{MgxA}{2B^2}$ is strictly positive, and the acceleration is strictly greater than g for every x in $0 < x < 2L$. (At the instant of release, $x = 0$ and the acceleration is exactly g.)

Checks

- If $M = 0$ (a light rope) both formulae reduce to free fall: $v^2 = 2gx$ and acceleration g.

- Take $m = 0.3$, $M = L = 1$ and $x = 1$, with $g = 9.8$: then $A = 1.05$, $B = 0.8$, so $v^2 = 25.725$ and the acceleration is $9.8 + 9.8 \times 1.05/1.28 \approx 17.84\,\mathrm{m\,s^{-2}}$, well above g. Differentiating $\frac{1}{2}v^2$ numerically at this point gives the same value, and stepping the equation of motion forward from rest reproduces the v^2 formula.
- Units: mL, ML and Mx all have the units of mass $\times$ length, so the fractions are dimensionless and v^2 has units of gx.

Common slips

- Getting the moving length wrong: the bottom of the loop moves down by $\frac{1}{2}x$, not x, because the rope taken up by the falling side is shared between both sides.
- Giving the fixed section kinetic energy, or forgetting that it also changes its potential energy (it gets longer, so its centre of mass drops).
- Differentiating v^2 instead of $\frac{1}{2}v^2$, which gives twice the acceleration.

Beyond STEP

This is the classic "falling folded chain" problem, a system whose mass in motion changes with time. The energy-conserving model used here predicts that the end accelerates faster than g, and if $m = 0$ the speed grows without bound as $x \to 2L$, which is the idea behind the crack of a whip. Whether energy really is conserved at the bend has been debated in the research literature, and experiments with real chains show accelerations greater than g, as here. Variable-mass systems and the careful use of energy versus momentum are treated in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2012, Question 12

Probability and Statistics • Geometric Probability

STEP III 2012, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q12

The key idea

Splitting the triangle into three smaller triangles that share the vertex P shows that the three distances always add up to the height. For the minimum, the clean way in is the tail probability $\mathrm{P}(X > x)$. The event that every distance exceeds x is a smaller triangle similar to the original, and areas of similar figures scale as the square of the length ratio. In three dimensions the same idea works with volumes, which scale as the cube.

Solution

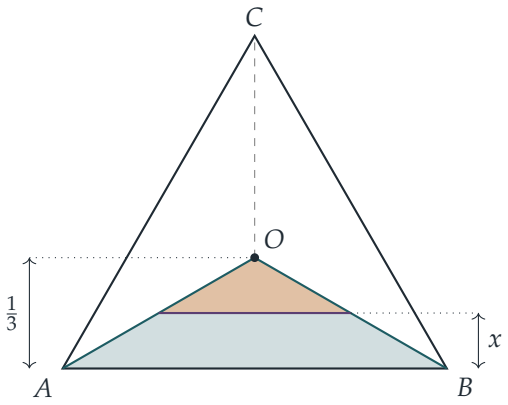
Part (i)

Let the side of the triangle be s. Since the height is 1, the area of ABC is $\frac{1}{2}s$. Joining P to the three vertices splits ABC into the triangles PAB, PBC and PCA. Their bases are the sides, each of length s, and their heights are x_1, x_2 and x_3. Adding the areas,

$$\tfrac{1}{2}sx_1 + \tfrac{1}{2}sx_2 + \tfrac{1}{2}sx_3 = \tfrac{1}{2}s, \qquad \text{so} \qquad x_1 + x_2 + x_3 = 1,$$

as required.

The region where X_1 is smallest. The points equidistant from AB and CA lie on the bisector of angle A. On the B side of this bisector, P is closer to AB than to CA. In the same way, P is closer to AB than to BC exactly when it lies on the A side of the bisector of angle B. In an equilateral triangle the angle bisectors are also the medians, and they meet at the centre O. So $X_1 = \min(X_1, X_2, X_3)$ exactly when P lies in the triangle ABO (shaded below). The distance from O to AB is a third of the height, $\frac{1}{3}$.



The density of X. The three triangles ABO, BCO and CAO are congruent, so each has probability $\frac{1}{3}$. (Ties between distances happen only on the bisectors, which have zero area.) By symmetry, X has the same distribution as X_1 conditioned on P lying in ABO. In that case $X = X_1$, the distance from P to AB.

Ask yourself Inside triangle ABO, which points have $X_1 > x$?

They are the points above the line parallel to AB at distance x from it, which is the small amber triangle with apex O. It is similar to ABO, with height $\frac{1}{3} - x$ against $\frac{1}{3}$, so its area is a fraction $\left((\frac{1}{3} - x)/\frac{1}{3}\right)^2 = (1-3x)^2$ of the area of ABO. Hence, for $0 \le x \le \frac{1}{3}$,

$$\mathrm{P}(X > x) = (1-3x)^2, \qquad \mathrm{F}(x) = \mathrm{P}(X \le x) = 1 - (1-3x)^2.$$

Also X can't exceed $\frac{1}{3}$, since the three distances add to 1. Differentiating,

$$\mathrm{f}(x) = \mathrm{F}'(x) = 6(1-3x) \quad (0 \le x \le \tfrac{1}{3}),$$

and $\mathrm{f}(x) = 0$ otherwise, as required.

Expected value.

$$\mathrm{E}(X) = \int_0^{1/3} 6x(1-3x)\,\mathrm{d}x = \left[3x^2 - 6x^3\right]_0^{1/3} = \tfrac{1}{3} - \tfrac{2}{9} = \tfrac{1}{9}.$$

Part (ii)

Let the four distances from the point to the faces be $Y_1, \ldots, Y_4$, and let $Y = \min(Y_1, \ldots, Y_4)$. Each face has the same area S, and the volume is $\frac{1}{3}S \cdot 1$. Joining the point to the four vertices splits the tetrahedron into four smaller tetrahedra. Each has a face as its base and one of the Y_i as its height, so

$$\tfrac{1}{3}S(Y_1 + Y_2 + Y_3 + Y_4) = \tfrac{1}{3}S, \qquad \text{so} \qquad Y_1 + Y_2 + Y_3 + Y_4 = 1.$$

The centroid is at distance $\frac{1}{4}$ from every face, so the inscribed sphere has radius $\frac{1}{4}$.

Ask yourself What does the set $\{Y > y\}$ look like now?

It is the set of points at distance more than y from all four faces. Each face moves inward by y, parallel to itself, which gives a smaller regular tetrahedron with the same centroid. The distance from the centroid to each face drops from $\frac{1}{4}$ to $\frac{1}{4} - y$, so the scale factor is $(\frac{1}{4} - y)/\frac{1}{4} = 1 - 4y$. Volumes scale as the cube of the scale factor, so for $0 \le y \le \frac{1}{4}$,

$$\mathrm{P}(Y > y) = (1 - 4y)^3.$$

(The same argument applied to the triangle in part (i) gives $(1 - 3x)^2$ directly: the inner triangle has inradius $\frac{1}{3} - x$.)

Since $Y \ge 0$, we can find $\mathrm{E}(Y)$ by integrating the tail probability:

$$\mathrm{E}(Y) = \int_0^{1/4} \mathrm{P}(Y > y)\,\mathrm{d}y = \int_0^{1/4} (1 - 4y)^3\,\mathrm{d}y = \left[-\tfrac{1}{16}(1 - 4y)^4\right]_0^{1/4} = \tfrac{1}{16}.$$

If you prefer to use the density, it is $12(1 - 4y)^2$ on $[0, \frac{1}{4}]$, and $\int_0^{1/4} 12y(1 - 4y)^2\,\mathrm{d}y = \frac{1}{16}$ as well.

So the expected distance to the nearest face is $\frac{1}{16}$.

Checks

- $\int_0^{1/3} 6(1 - 3x)\,\mathrm{d}x = 1$, so f is a genuine density.
- There's a pattern: in a triangle of height 1, $\mathrm{E}(X) = \frac{1}{9} = \frac{1}{3^2}$, and in a tetrahedron of height 1, $\mathrm{E}(Y) = \frac{1}{16} = \frac{1}{4^2}$. In one dimension, the distance from a random point of $[0, 1]$ to the nearer end has mean $\frac{1}{4} = \frac{1}{2^2}$, which fits.
- A Monte Carlo simulation with 4×10^5 random points in the triangle gave a mean of 0.1110 (against $\frac{1}{9} \approx 0.1111$), and a histogram that matches $6(1 - 3x)$. With 3×10^5 points in the tetrahedron, the mean was 0.0625 (against $\frac{1}{16} = 0.0625$).

Common slips

- Differentiating $\mathrm{P}(X > x)$ and forgetting the minus sign: the density is $-\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{P}(X > x)$.
- Forgetting that the inner triangle's height is $1 - 3x$, not $1 - x$: each of the three sides moves in by x.
- Using a squared scale factor in part (ii). Volumes scale as the cube.

Beyond STEP

Two tools from this question come up all the time in a first probability course: finding the distribution of a minimum through $\mathrm{P}(\min > x)$, and computing $\mathrm{E}(Y) = \int_0^\infty \mathrm{P}(Y >$

$y)\,\mathrm{d}y$ for a non-negative random variable. The pattern continues: in an n-dimensional simplex of height 1, the expected distance to the nearest face is $1/(n+1)^2$. Grimmett and Welsh, *Probability: An Introduction*, treats both tools carefully.

WORKED SOLUTION

STEP III 2012, Question 13

Probability and Statistics • Truncated Normal Distribution

STEP III 2012, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q13

The key idea

A conditional expectation over an interval is an integral of $z\,f(z)$ over that interval, divided by the probability of the interval. For the standard Normal, $z\,\mathrm{e}^{-z^2/2}$ is exactly the derivative of $-\mathrm{e}^{-z^2/2}$, so the integral can be done exactly. Everything after that is standardising, $X = \mu + \sigma Z$, and splitting $|X|$ according to the sign of X.

Solution

Throughout, write $\phi(z) = \dfrac{1}{\sqrt{2\pi}}\,\mathrm{e}^{-z^2/2}$ for the probability density function of Z, so that $\Phi'(z) = \phi(z)$ and $\phi'(z) = -z\,\phi(z)$.

Part (i)

The conditional density of Z given $a < Z < b$ is $\phi(z)/\mathrm{P}(a < Z < b)$ on (a, b), and $\mathrm{P}(a < Z < b) = \Phi(b) - \Phi(a)$. Hence

$$\begin{aligned}
\mathrm{E}(Z \mid a < Z < b) &= \frac{1}{\Phi(b) - \Phi(a)} \int_a^b z\,\phi(z)\,\mathrm{d}z \\
&= \frac{1}{\sqrt{2\pi}\,\big(\Phi(b) - \Phi(a)\big)} \Big[-\mathrm{e}^{-z^2/2} \Big]_a^b \\
&= \frac{\mathrm{e}^{-a^2/2} - \mathrm{e}^{-b^2/2}}{\sqrt{2\pi}\,\big(\Phi(b) - \Phi(a)\big)},
\end{aligned}$$

since $\dfrac{\mathrm{d}}{\mathrm{d}z}\big(-\mathrm{e}^{-z^2/2}\big) = z\,\mathrm{e}^{-z^2/2}$. This is the required result. It also holds with $b = \infty$ (or $a = -\infty$), reading $\mathrm{e}^{-\infty} = 0$ and $\Phi(\infty) = 1$, because the integral converges.

Part (ii)

Write $X = \mu + \sigma Z$ with $Z \sim \mathrm{N}(0,1)$ and $\sigma > 0$. Since $\sigma > 0$,

$$X > 0 \iff Z > -\mu/\sigma,$$

so conditioning on $X > 0$ is the same as conditioning on $Z > -\mu/\sigma$. Expectation is linear under conditioning too, so

$$\mathrm{E}(X \mid X > 0) = \mathrm{E}(\mu + \sigma Z \mid Z > -\mu/\sigma) = \mu + \sigma\,\mathrm{E}(Z \mid Z > -\mu/\sigma),$$

as required.

The mean of $|X|$. Put $c = \mu/\sigma$. Part (i) with $a = -c$ and $b = \infty$ gives

$$\mathrm{E}(Z \mid Z > -c) = \frac{\mathrm{e}^{-c^2/2}}{\sqrt{2\pi}\,\bigl(1 - \Phi(-c)\bigr)} = \frac{\phi(c)}{p}, \qquad \text{where } p = \mathrm{P}(X > 0) = 1 - \Phi(-c).$$

Multiplying the previous display by p,

$$\mathrm{E}(X;\, X > 0) := \mathrm{E}(X \mid X > 0)\,\mathrm{P}(X > 0) = \mu p + \sigma\phi(c).$$

> **Ask yourself** How do we get the contribution from $X < 0$ without a second integral?

The two pieces add up to the whole mean: $\mathrm{E}(X;\, X > 0) + \mathrm{E}(X;\, X < 0) = \mathrm{E}(X) = \mu$ (the event $X = 0$ has probability 0). So $\mathrm{E}(X;\, X < 0) = \mu - \mu p - \sigma\phi(c)$. On $X < 0$ we have $|X| = -X$, and so

$$\begin{aligned} m = \mathrm{E}|X| &= \mathrm{E}(X;\, X > 0) - \mathrm{E}(X;\, X < 0) \\ &= 2\bigl(\mu p + \sigma\phi(c)\bigr) - \mu \\ &= \mu(2p - 1) + 2\sigma\phi(c). \end{aligned}$$

Finally $2p - 1 = 1 - 2\Phi(-\mu/\sigma)$ and $2\phi(c) = \sqrt{2/\pi}\,\mathrm{e}^{-\mu^2/(2\sigma^2)}$, giving

$$m = \mu\bigl(1 - 2\Phi(-\mu/\sigma)\bigr) + \sigma\sqrt{2/\pi}\,\exp\bigl(-\tfrac{1}{2}\mu^2/\sigma^2\bigr),$$

as required.

The variance of $|X|$. The point is that $|X|^2 = X^2$, so the second moment needs no new work:

$$\mathrm{E}\bigl(|X|^2\bigr) = \mathrm{E}(X^2) = \mathrm{Var}(X) + \bigl(\mathrm{E}X\bigr)^2 = \sigma^2 + \mu^2.$$

Therefore

$$\mathrm{Var}|X| = \mu^2 + \sigma^2 - m^2.$$

Checks

- If $\mu = 0$ then $|X|$ is half-Normal and the formula gives $m = \sigma\sqrt{2/\pi}$, the familiar value; the variance is then $\sigma^2(1 - 2/\pi)$, which is positive.
- If $\mu \gg \sigma$ then X is almost always positive, $\Phi(-\mu/\sigma) \approx 0$ and the exponential term is tiny, so $m \approx \mu$ and $\mathrm{Var}|X| \approx \sigma^2$, as it should be.
- The formula for m is unchanged on replacing μ by $-\mu$ (use $1 - 2\Phi(-c) = -(1 - 2\Phi(c))$), matching the symmetry $|X| \sim |-X|$.
- Numerically, with $\mu = 1$, $\sigma = 2$ we get $m \approx 1.7912$ and $\mathrm{Var}|X| \approx 1.7917$, which agree with direct quadrature and with a simulation of 2×10^5 values.

Common slips

- Forgetting to divide by $\Phi(b) - \Phi(a)$ in part (i): the integral alone is $\mathrm{E}(Z;\, a < Z < b)$, not the conditional expectation.
- Writing $\mathrm{E}|X| = \mathrm{E}(X \mid X > 0)$: the conditional expectations must be weighted by $\mathrm{P}(X > 0)$ and $\mathrm{P}(X < 0)$.
- Computing $\mathrm{E}(X^2)$ by another integral, when $|X|^2 = X^2$ gives it at once.

Beyond STEP

The distribution of $|X|$ is the *folded Normal*, and the conditional distribution in part (i) is a *truncated Normal*; the ratio $\phi(c)/\big(1 - \Phi(-c)\big)$ is the inverse Mills ratio, central to models in econometrics where data are censored or selected. The trick $\phi'(z) = -z\phi(z)$ is also the start of Stein's method for Normal approximation. For more on conditional expectation, see Blitzstein and Hwang, *Introduction to Probability*.

Beyond STEP

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